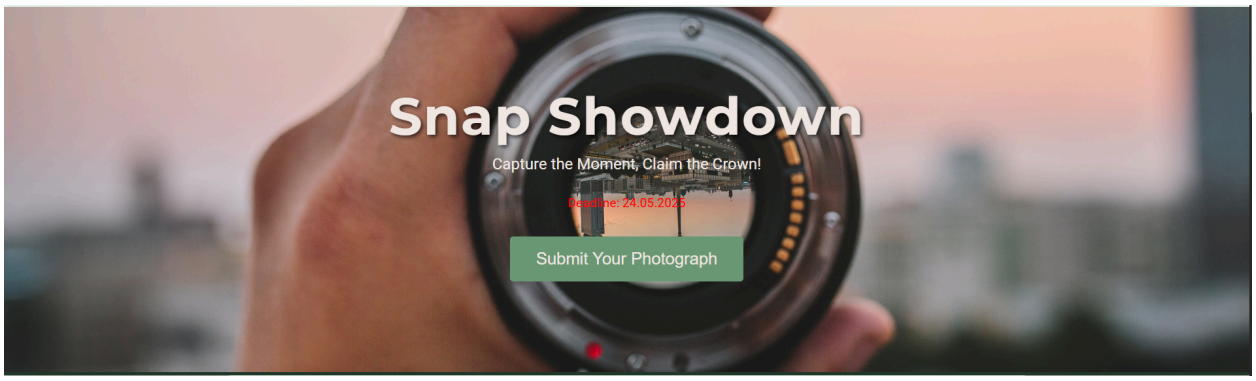
EAST WEST UNIVERSITY



Snap Showdown : A Photography Competition Entrance Website

Course

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Abstract

A Photography Competition Website is an online place where photography contests can happen. The main goal of this project is to make sure that participants, administrators, and viewers have a smooth experience. We used a planned method called Quality Function Deployment (QFD) to do this.

This report talks about the basic, normal, and special features of the system. We made sure the website has all the important things users need, and we also added extra features to make it more fun. The main functions include logging in, checking photo submissions, voting, managing comments, and ranking winners. Some extra features like connecting with social media, an event calendar, and a winner's gallery make the website even better.By using QFD, we made sure that user needs were turned into a simple and easy-to-use website for competitions.

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Chapter 1: Introduction

This chapter is a part of the "Photography Competition Entrance Website" project. It explains why this document is important and who should read it.

1.1 Purpose

This document is called the Software Requirements Specification (SRS) for the "Photography Competition Entrance Website." It lists all the things needed to build the system, including its functions, extra features, and support needs. The SRS helps everyone involved in the project understand what is required and keeps everything well-organized.

The goal of this SRS is to be a clear way for users and developers to communicate. It makes sure the system is built the right way and meets expectations. Over time, it will be updated as new ideas and feedback come in.

1.2 Intended Audience

The Software Requirements Specification (SRS) document serves as a crucial guide for all stakeholders involved in the development process. It outlines the system's functionality, design, and expectations, ensuring a clear understanding among customers, project managers, designers, developers, and testers. Each role relies on the SRS to fulfill their responsibilities effectively and contribute to the successful completion of the system.

- Customers: They will check this document to see if the system is made as they want. It helps them understand if their needs are met and if any changes are required.

- Project Managers: They will use the SRS to plan the development steps, set deadlines, and ensure the project is completed on time. This document helps them track progress and manage tasks efficiently.

- Designers: They will follow the document to create a system that is both user-friendly and visually appealing. The SRS guides them in making a design that meets user needs and provides a smooth experience.

- Developers: They will use the SRS to understand system requirements and build features accordingly. This document ensures that the system is developed correctly and functions as expected.

- Testers: They will make test plans based on this document to check if everything is working before the system is released. The SRS helps them identify potential issues and ensure the system meets all required standards.

1.3 Conclusion

This document is important for making and understanding the Photography Competition Entrance Website. It clearly lists all needs so that everyone has the same idea about the project. This helps in creating a well-planned and successful system.

Chapter 2: Inception

2.1 Introduction

Inception is the initial phase of requirements engineering. It defines how a software project begins and establishes the scope and nature of the problem to be solved. The objective of the inception phase is to identify the needs of stakeholders and resolve any conflicting requirements in the early stages of development. To lay the foundation for this project, we have considered the following key factors in the inception phase:

- Identifying Stakeholders
- Recognizing Multiple Viewpoints
- Working Towards Collaboration
- Asking the First Questions

2.1.1 Identifying Stakeholders

A stakeholder is any person or group who will be directly or indirectly affected by the system. Stakeholders include end-users who interact with the system as well as others who may influence or benefit from its operation. During inception, we identify the people who will contribute to the requirements elicitation process. The initial list expands as more stakeholders are identified.

The following stakeholders were identified for the “Snap Showdown: A Photography Competition Entrance Website”:

- **Administrator:** The person responsible for managing the website, approving or rejecting submissions, overseeing voting, moderating comments, and ensuring the smooth operation of the competition.
- **Participant (Photographers):** Users who register on the platform and submit their photographs to participate in competitions.
- **General Users (Voters/Viewers):** Visitors who explore the website, view submitted photographs, vote for their favorite entries, and engage through likes and comments.
- **Web Developers & Designers:** The team responsible for building and maintaining the website, ensuring proper functionality and a seamless user experience.

2.1.2 Recognizing Multiple Viewpoints

Different stakeholders have varying expectations and demands from the software. To ensure that the platform meets these needs, we have considered multiple viewpoints:

- **Administrator's Viewpoint:**
  - Ability to approve or reject submitted photographs
  - Moderation of comments and handling user reports
  - Managing user roles and ensuring fair competition
  - Monitoring voting activity to prevent fraud
  - Updating competition rules and deadlines
  - Managing event calendar
  - Updating leaderboard
  - Updating winner information
- **Participant's Viewpoint:**
  - Simple and intuitive photo submission process
  - Ability to track the status of submitted photographs
  - Fair voting and competition procedures
  - Receiving feedback through comments and likes
  - Visibility and recognition for their work
- **General Users (Voter's) Viewpoint:**
  - Easy access to browse photographs in the gallery
  - Transparent voting system
  - Ability to like and comment on photographs
  - User-friendly interface for seamless interaction
- **Developer's Viewpoint:**
  - Well-structured and scalable system
  - Security measures to prevent data breaches
  - Efficient database management
  - Minimal conflicts in requirements

2.1.3 Working Towards Collaboration

Since multiple stakeholders are involved, certain requirements may conflict. To finalize the essential features, collaboration is necessary. The following steps were followed to merge different viewpoints:

- Identifying common and conflicting requirements
- Categorizing the requirements based on priority
- Gathering feedback from stakeholders to prioritize essential features
- Finalizing a consensus on the requirements

Common Requirements:

- Secure user authentication and role management
- A user-friendly web interface
- A well-structured voting and commenting system
- A moderation feature for handling inappropriate content
- A centralized database for storing user data and submissions

Conflicting Requirements:

- Anonymous vs. registered voting system
- Whether to allow unlimited votes or restrict them per user
- Whether to allow editing of submitted photos after submission
- Whether the participant can delete the approved photograph from the website

Finalized Requirements:

- Secure, user-friendly registration and login system
- Transparent and fair voting system (with necessary restrictions)
- Administrator control over content moderation and approval
- A responsive and visually appealing homepage gallery
- An engaging comment and liking system for user interaction

2.1.4 Asking the First Questions

To understand the core requirements and set the direction of development, we began with a set of context-free questions focused on stakeholders, project goals, and benefits. The questions helped us clarify the problem domain and explore potential alternatives for system design.

Some key questions included:

- Who are the primary users of the system?
- What key functionalities should be prioritized in the initial release?
- How will the voting and judging system work?
- What security measures should be implemented to prevent fraud?
- What are the challenges of maintaining an interactive gallery?

These questions provided valuable insights and helped refine the system's requirements to better serve all stakeholders.

2.2 Conclusion

The inception phase allowed us to establish a fundamental understanding of the "Photography Competition Website," identify key stakeholders, define the system's objectives, and resolve early conflicts in requirements. This phase set the groundwork for future development by clarifying the scope, needs, and expectations of all involved parties. Continued communication and further analysis will ensure that the system meets its intended purpose effectively.

Chapter 3: Elicitation

3.1 Introduction

Elicitation is the process of gathering requirements from stakeholders to ensure the development of a system that meets their expectations. In this phase, various techniques are used to collect, clarify, and analyze user needs. This chapter details how requirements for the Photography Competition Website were elicited through interactions with stakeholders, research, and collaborative discussions.

3.2 Eliciting Requirements

To develop a well-functioning and user-centric Photography Competition Website, it was essential to gather requirements from multiple sources. The following techniques were used for requirement elicitation:

- **Interviews**: Discussions with potential users, including photographers, competition organizers, and website administrators, to understand their needs and expectations.
- **Surveys and Questionnaires**: Online surveys were conducted to collect feedback from photographers and users regarding essential features, such as voting systems, submission guidelines, and user interaction.
- **Observation**: Analyzing existing competition platforms which are similar helped identify industry best practices and potential improvements.
- **Prototyping**: Initial wireframes and mockups will be created and shared with stakeholders to refine and validate requirements.
- **Workshops and Brainstorming Sessions**: Meetings with developers, designers, and users helped uncover specific needs, challenges, and possible enhancements to the system.

3.3 Collaborative Requirements

Gathering Collaboration between stakeholders ensures that requirements are comprehensive and reflect the needs of all users. The following steps were taken to facilitate effective collaboration:

- **Stakeholder Meetings**: Meetings were held with administrators, competition organizers and developers to discuss project goals, scope, and potential constraints.
- **Requirement Prioritization**: Features were categorized based on their importance, feasibility, and impact on the user experience.
- **Feedback Loops**: Continuous feedback was collected through iterative discussions and prototype reviews to refine the requirements.
- **Consensus Building**: Conflicting requirements were addressed through discussions and compromise to ensure that all perspectives were considered in the final system design.

By following these elicitation techniques, a comprehensive set of requirements was developed to guide the successful implementation of the Photography Competition Website.

3.4 Quality Function Deployment

Quality Function Deployment (QFD) is a structured approach used to ensure that the needs and desires of clients are translated into specific technical requirements for the product. QFD helps teams prioritize features and align their efforts with what truly matters to users maximizing customer satisfaction from the software engineering process. With respect to our project the following requirements are identified by a QFD.

3.4.1 Normal Requirements

Normal requirements are the essential functionalities and features that the website must have to meet the basic expectations of users. These requirements focus on the core capabilities of the website and ensure it operates effectively.
Below are the normal requirements for our project.

1. Accessible via the Internet.
2. Allow valid users to sign up.
3. Allow valid users to login and logout.
4. Allow registered users to submit their photography entries for the competition.
5. Allow administrators to approve or delete photography submissions.
6. Restrict access to functionality of the system based upon user roles.
7. Keep track of the number of entries submitted for each category to manage participation.
8. Display approved photographs on the website.
9. Allow viewers to vote for their favorite photographs.
10. Allow viewers to comment on the submitted photographs.

3.4.2 Expected Requirements

Expected requirements outline the features and functionalities that users anticipate will be included in the photography competition website. These requirements enhance user experience and satisfaction by meeting common expectations.
Below are the expected requirements of our project.

1. User-Friendly Interface
2. Error free software
3. Strong authentication system
4. Allow users to change their password via email if forgotten.
5. Effective system
6. Allow administrators to change data for the user.

7. Send notification if the photograph is approved by the administration
8. Help section to explain how users can navigate the site and participate in the competition.
9. Include a frequently asked questions section to address common queries regarding the competition.
10. Allow users to create and manage personal profiles where they can track and delete their submissions and participation history.
11. Allow visitors to view approved photographs.
12. Allow users to search photographs by participant's name or photography name,

3.4.3 Exciting Requirements

Exciting requirements are features that go beyond the basic expectations of the client, adding unique value and engaging experiences for users. These elements can enhance the overall appeal of the photography competition website.
Below the exciting requirements that we find will enhance the user experience.

1. Display leaderboard showing top photographs based on votes.
2. Enable users to share their entries and the competition on social media platforms.
3. Recommend more photos that voters might like.
4. Virtual exhibitions of past winners photographs
5. Show information about awards, prizes, and recognition for winners.
6. Send notification if the photograph gets a new vote or comment.
7. An event calendar showcasing important dates related to the competition.
8. Show previous winners and photographs.

Usage Scenario

Authentication

Register:

In the “Photography Competition Website”, there will be a secure authentication system that allows all types of users such as administrators, participants, voters and viewers to access various functionalities. To access specific functions users must create an account. To create an account, users must complete a registration form, providing essential information such as their first name, last name, email address, address, phone number, password, and account type, which determines their access level. The system includes validation checks to ensure that both email addresses and

phone numbers are unique and correctly formatted. Upon successful registration, users will receive a verification code via email.

Sign In:

If any user already has an account, they can sign in to the system. To sign in, the user has to give their email id or mobile number and password. The email id or mobile number and password will be verified. If the verification is successful, the user can sign in to the system successfully. If the password is wrong, there is a retry option. If the retry count is 5, the user will be blocked for 2 minutes. After block time has finished, users can attempt to sign in to the system again. If the user forgets the password, he has an option to change password by entering their email address. New password is sent to the email address and then the user can sign in again using the new password.

Sign Out:

Users have an option to log out from the system. Users will automatically be logged out of their account if the account remains idle for more than two hours.

Submit Photography:

Participants in the photography competition can easily submit their photographs directly from their profile. After logging into their accounts, they navigate to the "Profile" section and select "Submit Photography." Here, they can enter the photography name and a brief description for each photo they wish to upload. Each participant is allowed to upload up to five photographs at a time; however, they can submit as many photos as they want over the course of the competition. After entering the required details and uploading the images, participants simply click "Submit." They receive confirmation for each successfully submitted photograph, which is then marked as "Pending Review" in their profile.

Validate Submitted Photographs:

The administrator has the capability to validate each participant's photograph to determine its authenticity. If a submission is found to be original and not sourced from elsewhere, the administrator can approve it, resulting in the photograph being displayed in the home gallery. The participant will receive a confirmation notification stating that their photo has been approved, and the approved image will appear in the participant's profile. Conversely, if a submission is deemed inauthentic, the administrator has the authority to bin the submission, removing it from consideration.

Voting Process:

Registered voters can view the submitted photographs in the home gallery and participate in the voting process. Each voter is allowed to cast a vote for their favorite photographs. After

voting, voters receive a confirmation that their vote has been recorded. The voting period is limited, and results are displayed after the voting ends.

Commenting Feature:
Users can leave comments on photographs in the home gallery. Users can edit or delete their comments within a specific time frame after posting.

Profile Management:
Users can manage their profiles by updating personal information such as their name, email, password, and profile picture. They can also view their submission history, including approved and pending photographs.

Notifications:
Users receive notifications for various activities, such as submission confirmations, approval notifications, comments on their photographs, and reminders about upcoming voting deadlines.

Winner Gallery:
The website can have a dedicated section for showcasing winner photographs that highlight the best submissions from past competitions.

Help Section:
A dedicated help and support section can be added to provide users with FAQs, contact information for support, and guidelines for participants on how to take quality photographs and submit them.

Social Media Integration:
Participants and voters can share their favorite photographs or their own submissions on social media platforms directly from the website to promote their photographs.

Event Calendar:
An event calendar showcasing important dates related to the competition, such as submission deadlines, voting periods, and announcement dates, can help users stay informed about key events.

Results Announcement:
After the voting period ends, the website can feature a results announcement section where winners are displayed along with their photographs.

Leaderboard:

This feature on the photography competition website showcases the top participants based on the number of votes received for their submissions. Accessible to all users, it displays rankings along with each participant's profile picture and total votes.

Chapter 4: Scenario Based Modelling

4.1 Introduction

This model uses different scenarios to understand and analyze potential outcomes and behaviors of a system or process. It helps in visualizing interactions and identifying requirements by simulating real-world situations and their implications.

4.2 Use Case Scenario

Level 0	Level 1	Level 2
	Authentication	Sign up
		Log in
		Log out
		Reset Password
	View	Browse Photos
		Search
	Manage Profile	Submit Photography
		Update Photography
		Delete Photography
		Receive Notifications
		Update Information
		Share Submission

Snap Showdown: A Photography Competition Entrance Website		View Submitted Photographs with votes
	Vote	Vote for photos
		Comment
	Submission	View submitted photography
		Approve submission
		Disapprove submission
	Event Information	View Event Information
		Update Event Information
		View Event Calendar
		See Leaderboard
		Update Leaderboard
		View Result Announcements
		Update Result
		Previous Winners
	About	Help
		Frequently Asked Questions

4.3 Use Case Diagram

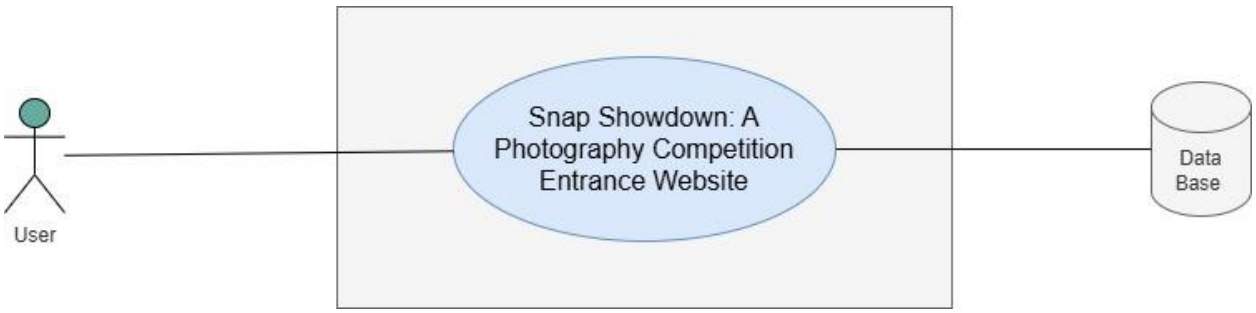


Figure 1: Level 0 Use Case Diagram

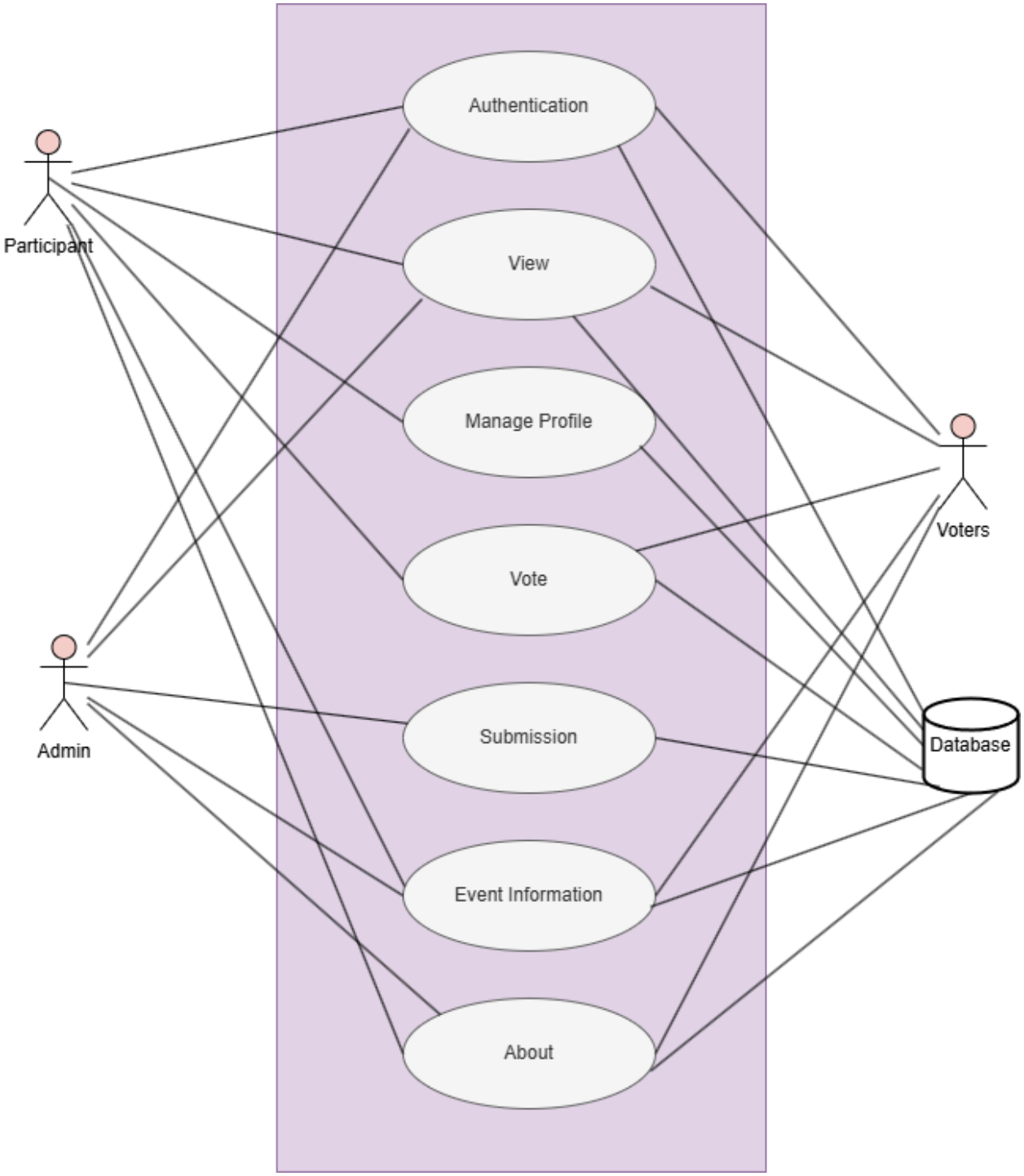


Figure 2: Level 1 Use Case Diagram

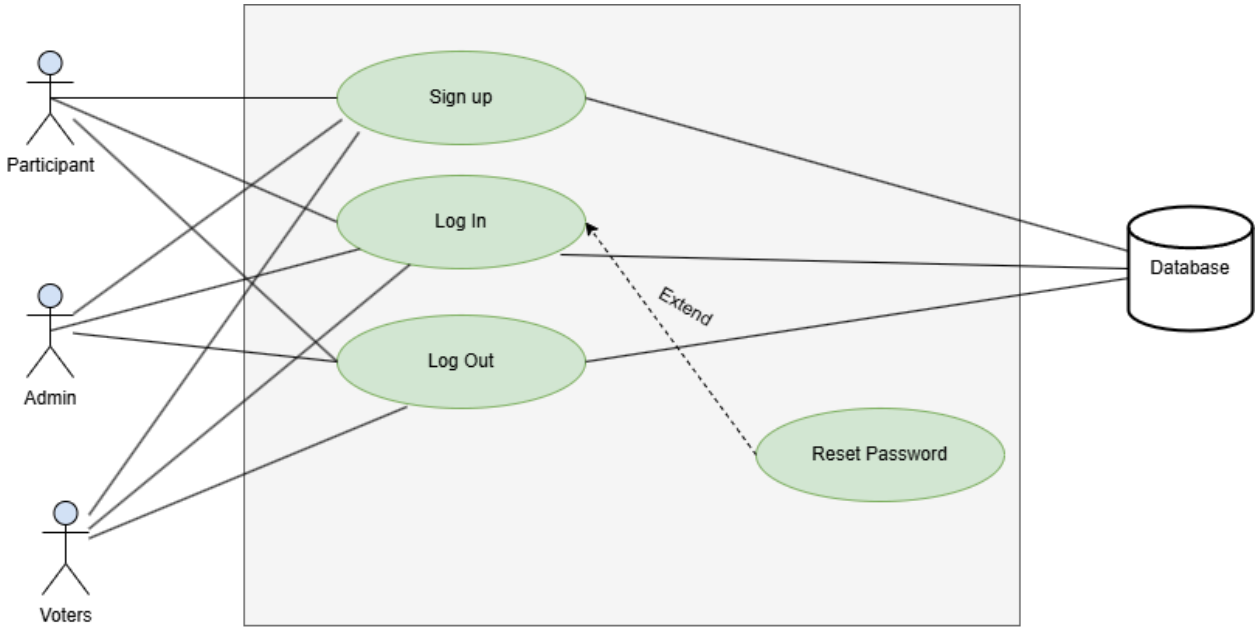


Figure 3: Level 2.1 Use Case Diagram

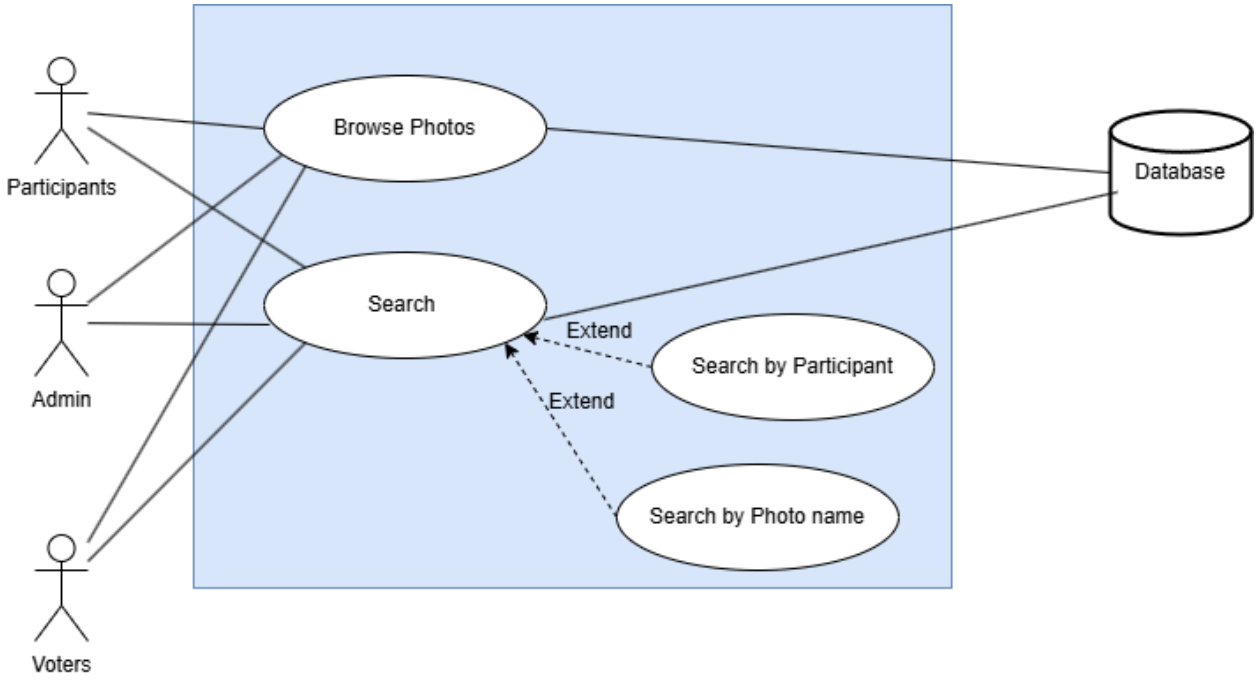


Figure 4: Level 2.2 Use Case Diagram

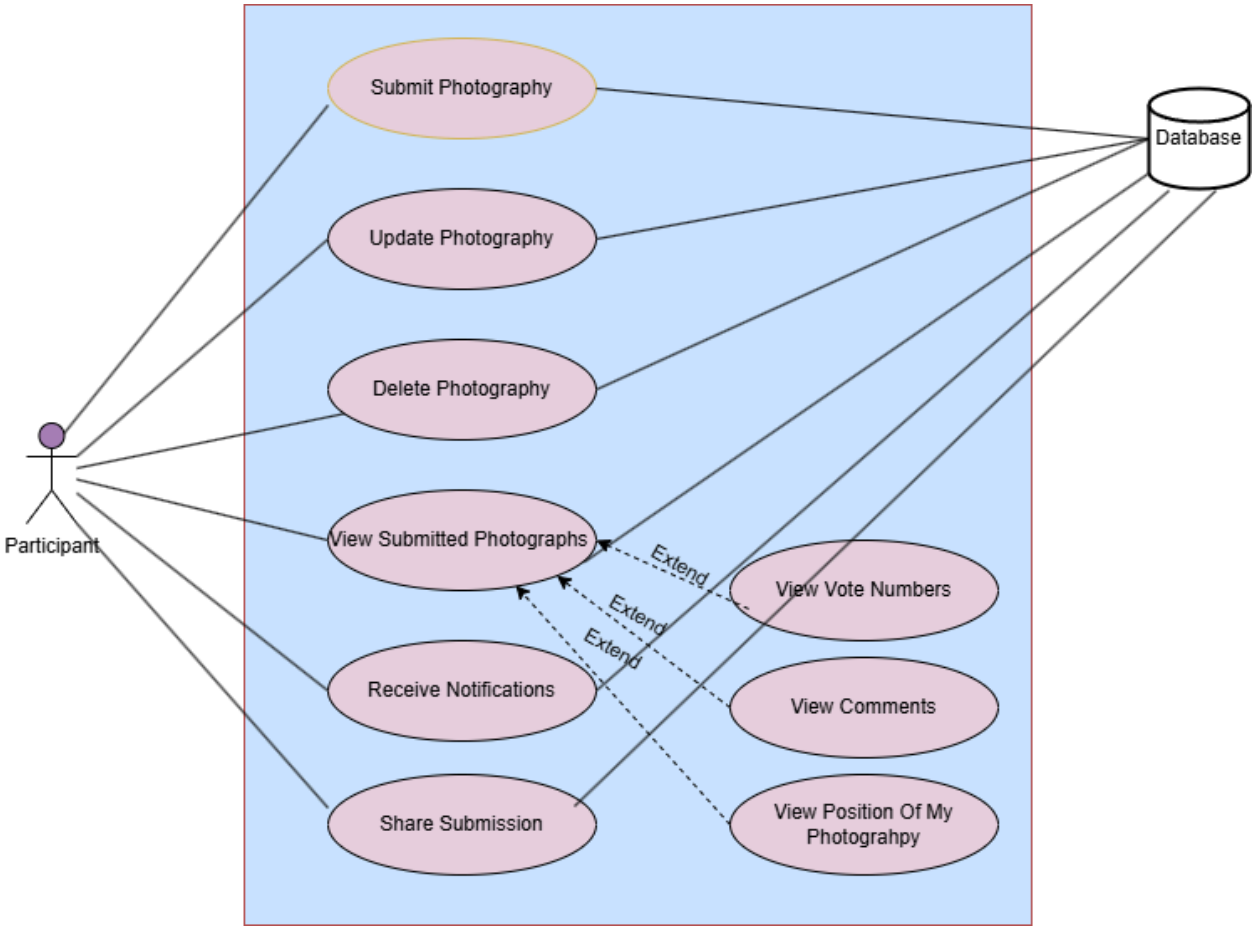


Figure 5: Level 2.3 Use Case Diagram

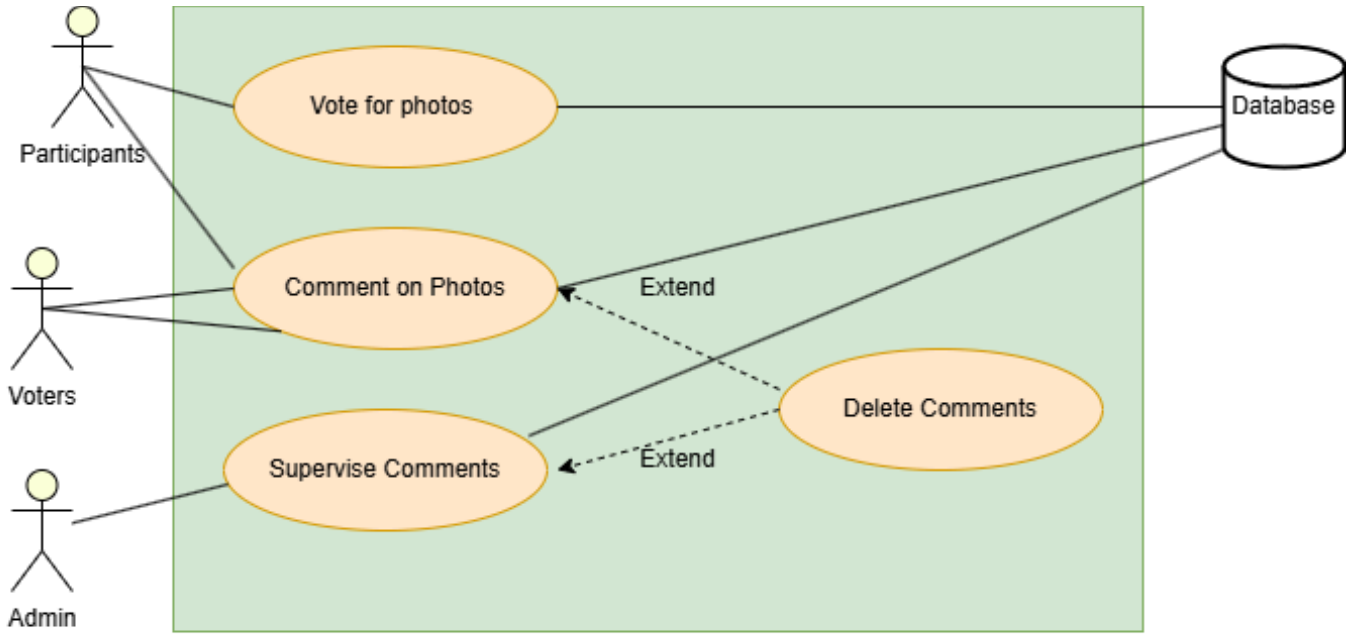


Figure 6: Level 2.4 Use Case Diagram

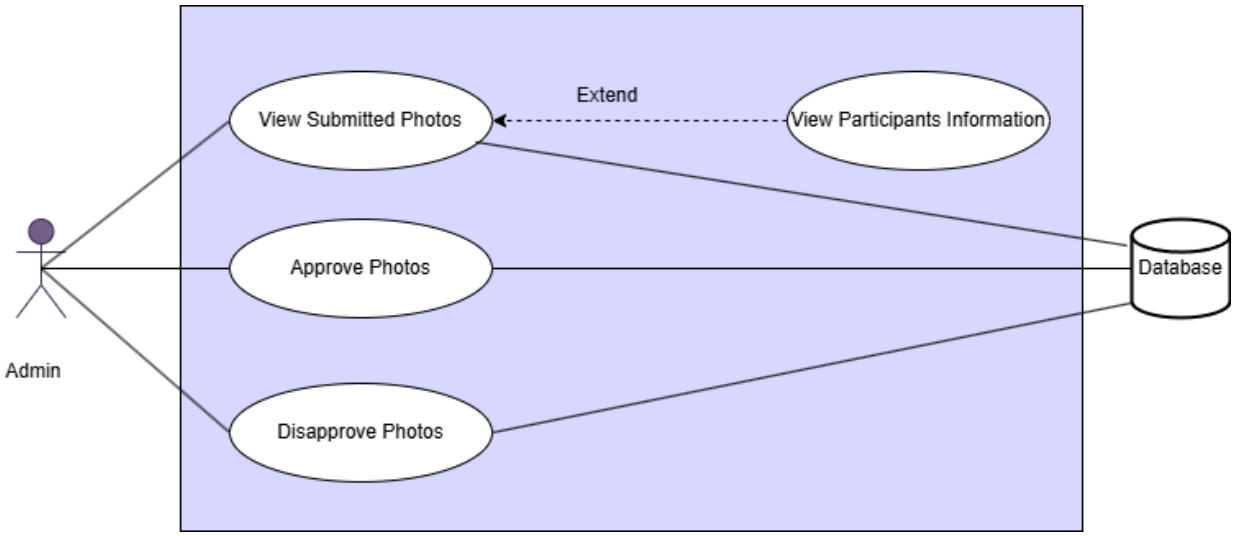


Figure 7: Level 2.5 Use Case Diagram

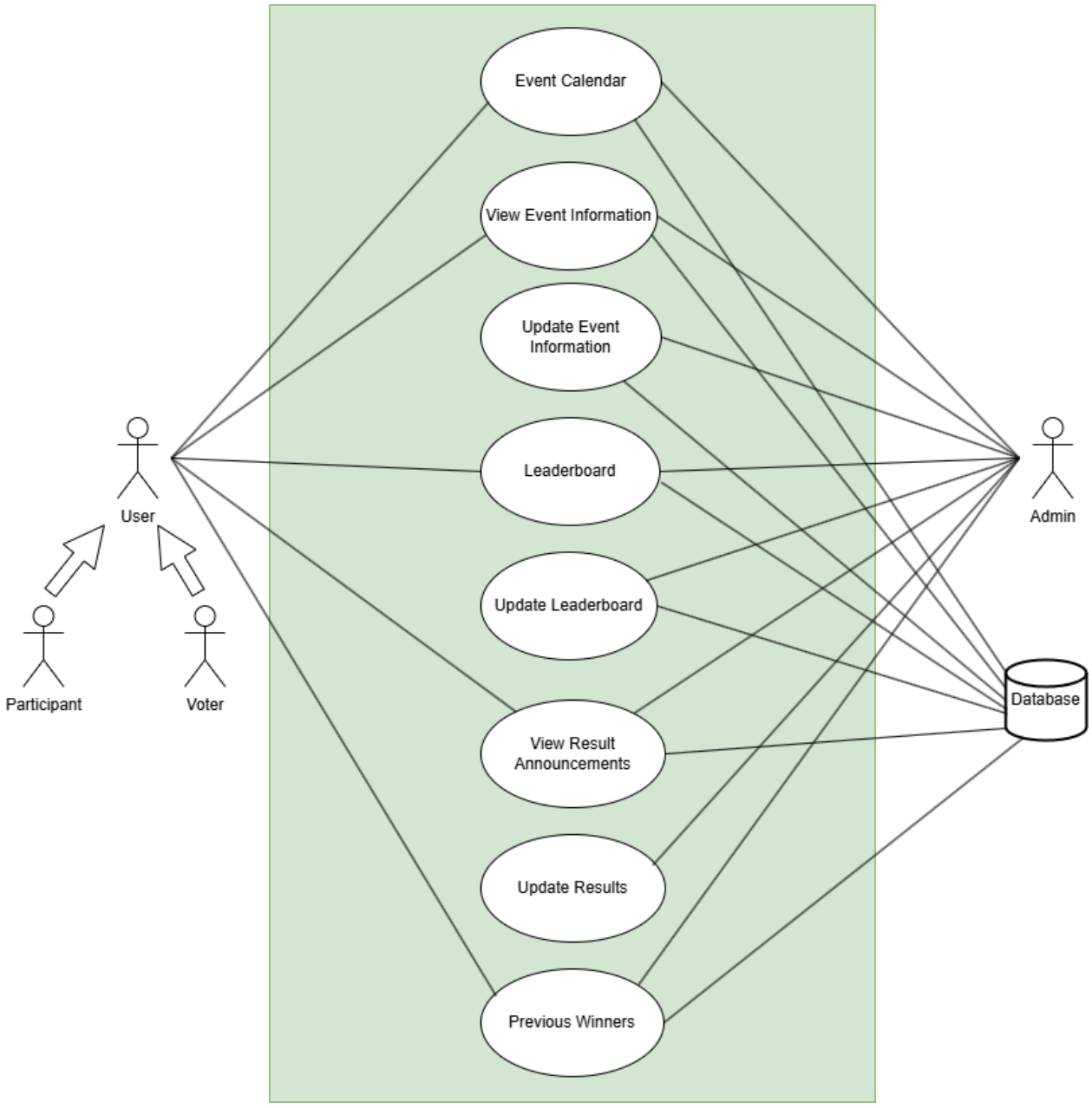


Figure 8: Level 2.6 Use Case Diagram

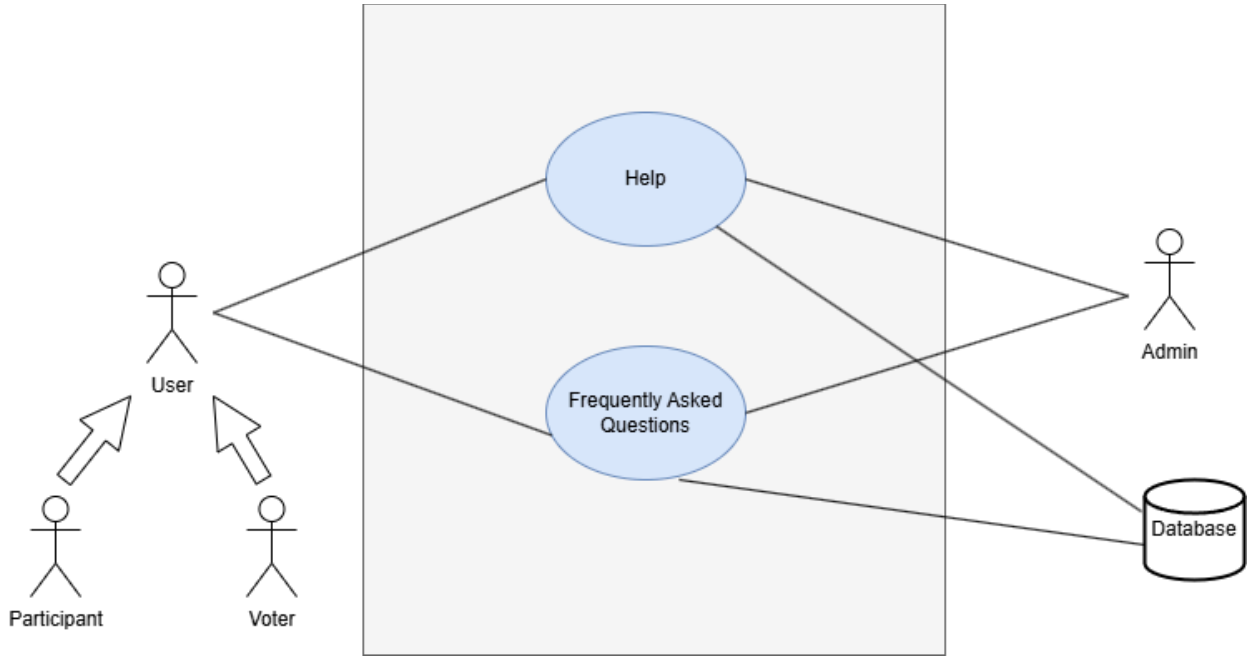


Figure 9: Level 2.7 Use Case Diagram

4.5 Activity Diagram of generated Use Cases:

Use Case 1: Sign Up
Activity Diagram:

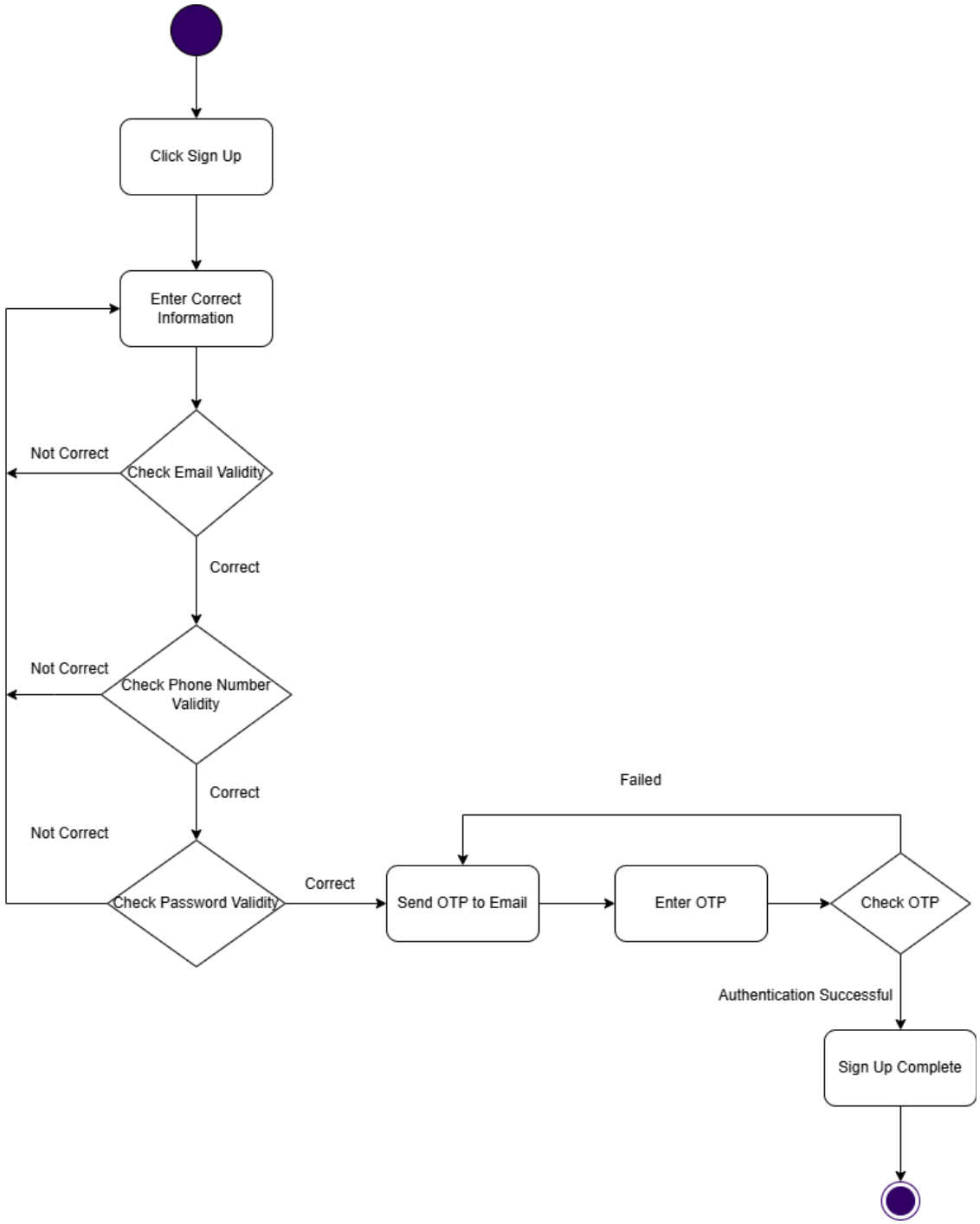


Figure 10: Activity Diagram Of Sign Up Use Case

Swimlane Diagram

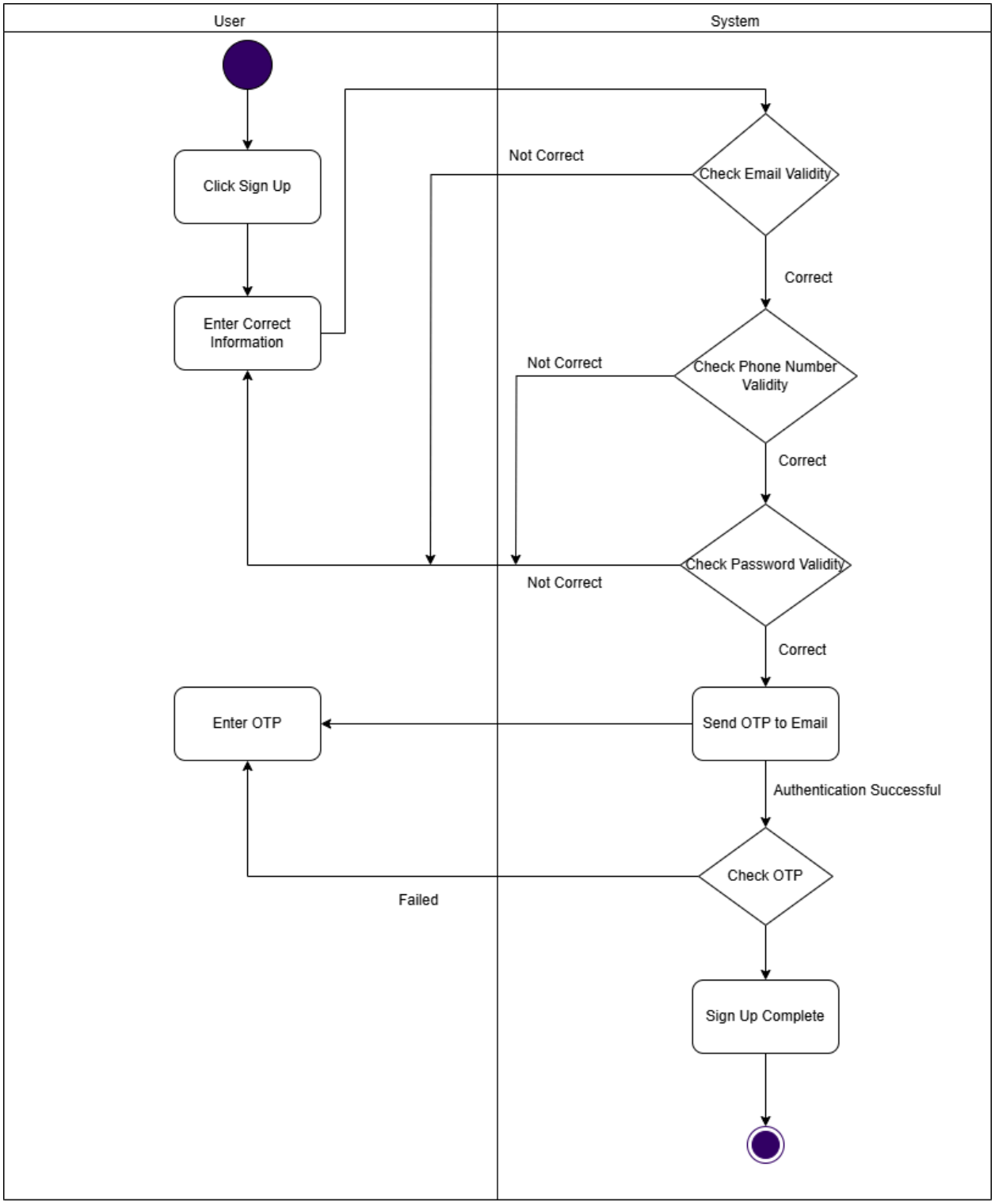


Figure 11. Swimlane Diagram of Sign Up Use Case

Use Case 2: Login
Activity Diagram

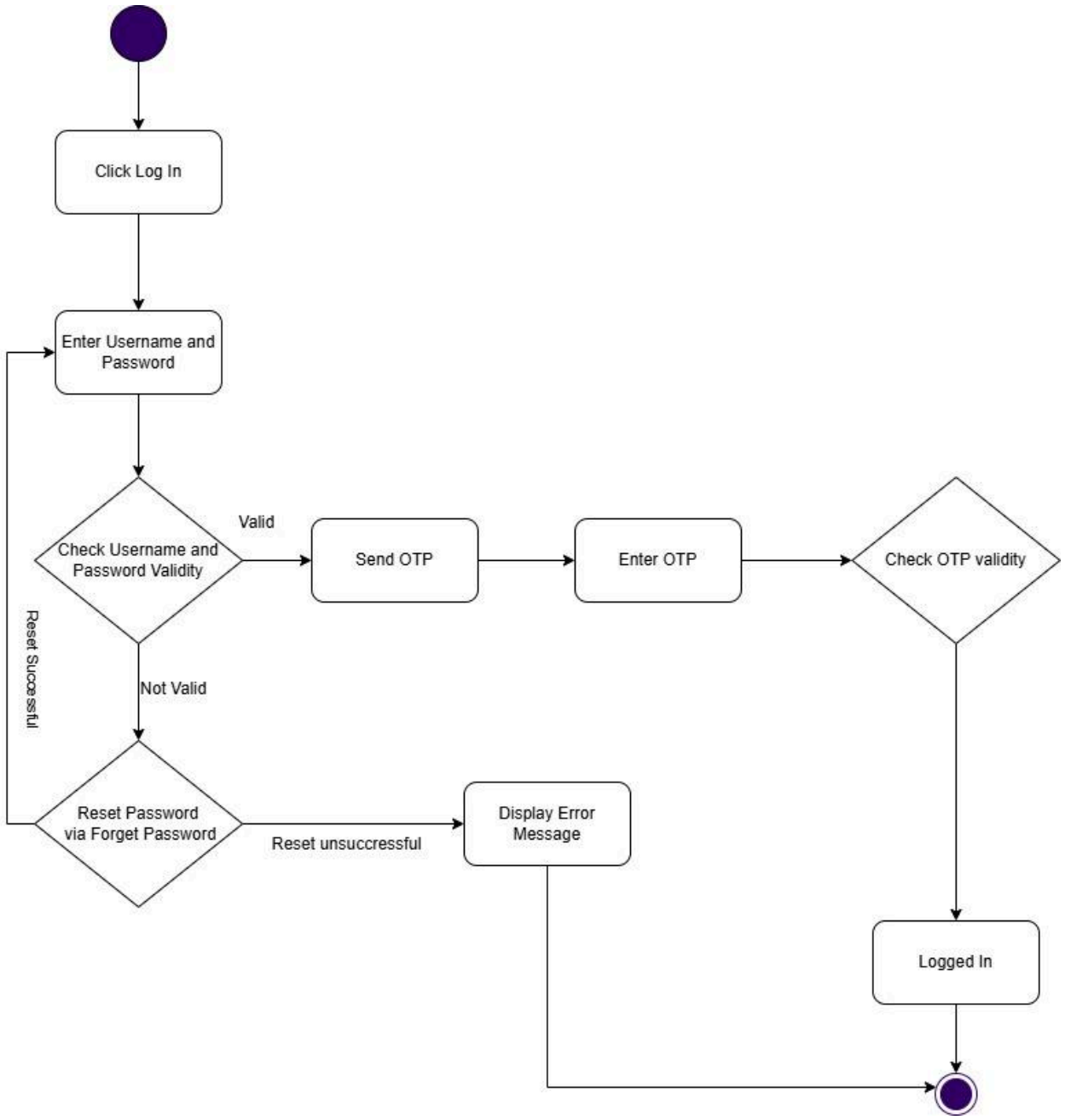


Figure 12: Activity Diagram for Login Use Case

Swimlane Diagram

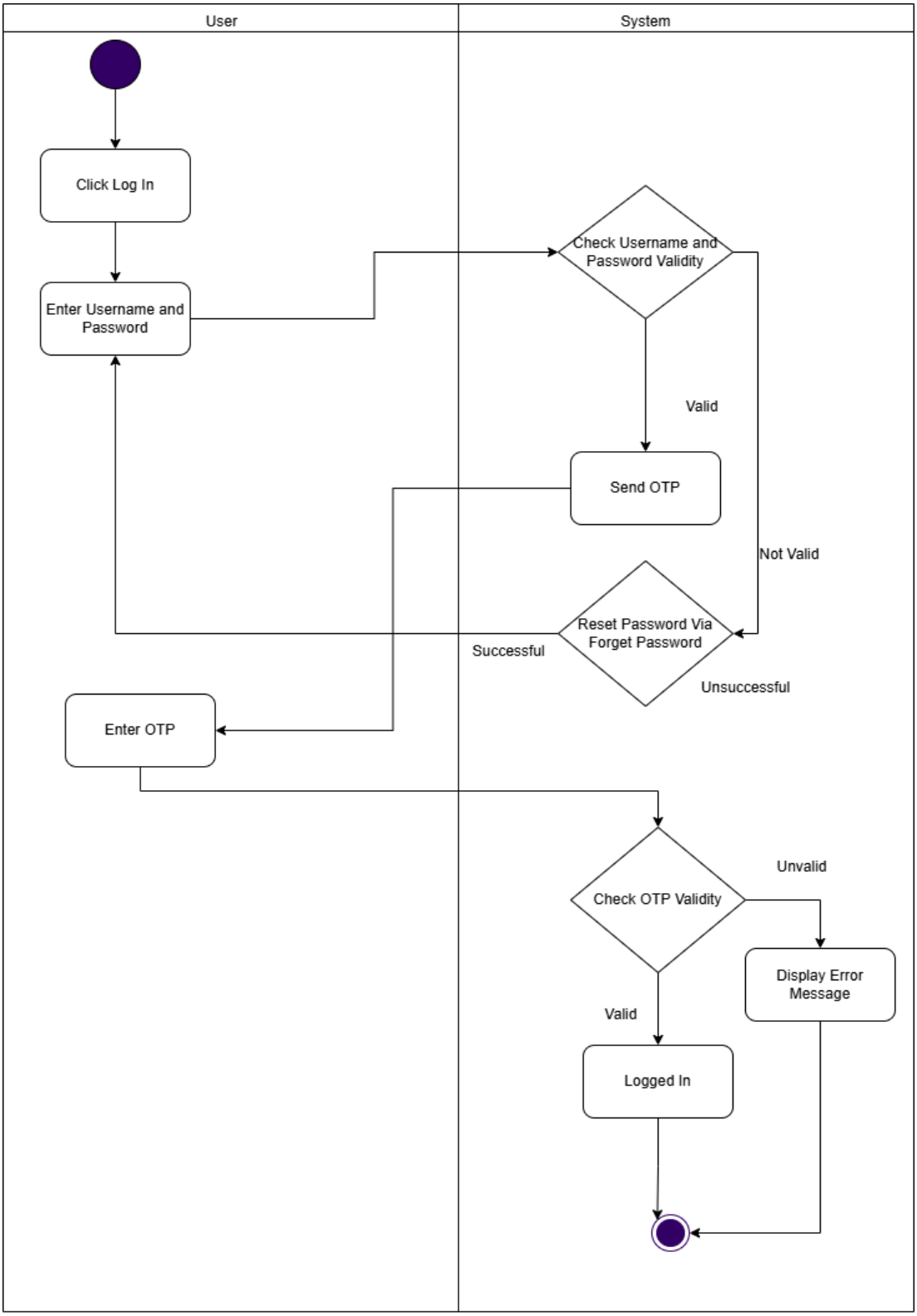


Figure 13: Swimlane Diagram for Login Use Case

Use Case 3: Log Out
Activity Diagram:

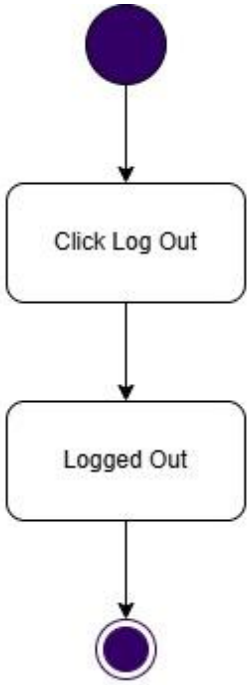


Figure 14: Activity Diagram For Log Out

Swimlane Diagram

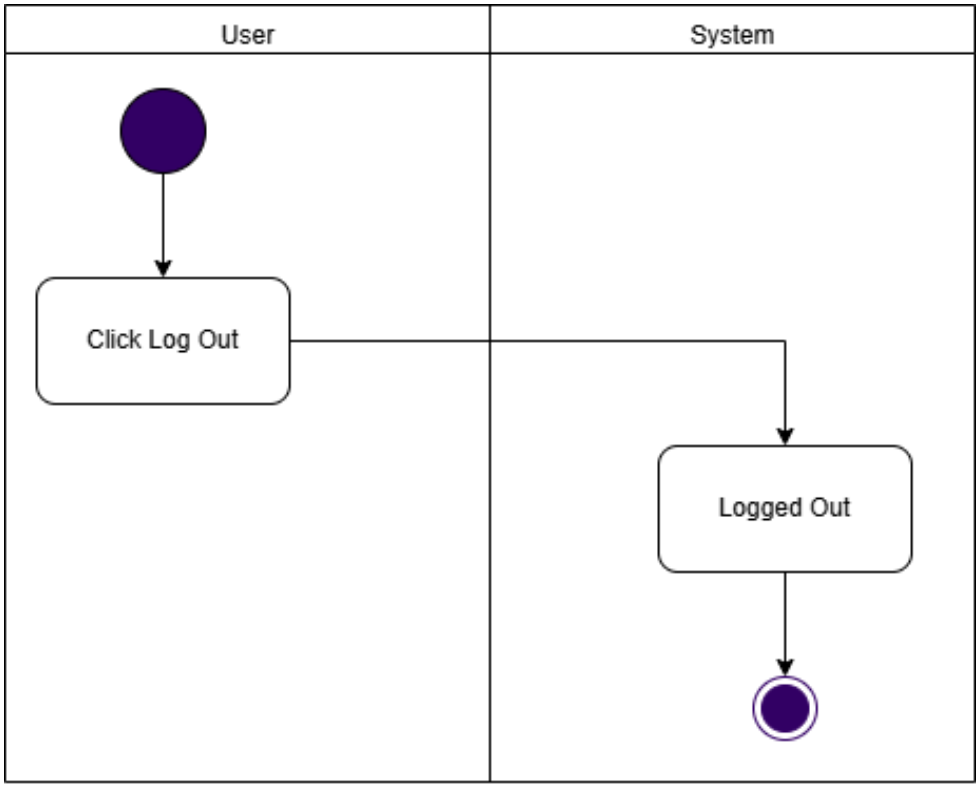


Figure 15: Swimlane Diagram For Log Out Use Case

Use Case 4: Browse Photos
Activity Diagram

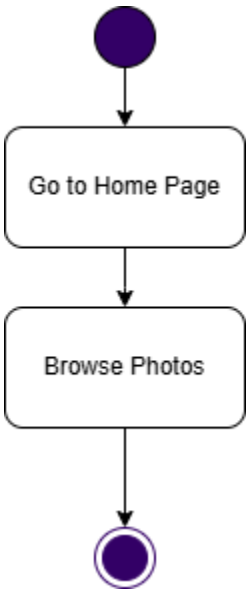


Figure 16: Activity Diagram For Browse Photos

Use Case 5: Search Photos
Activity Diagram

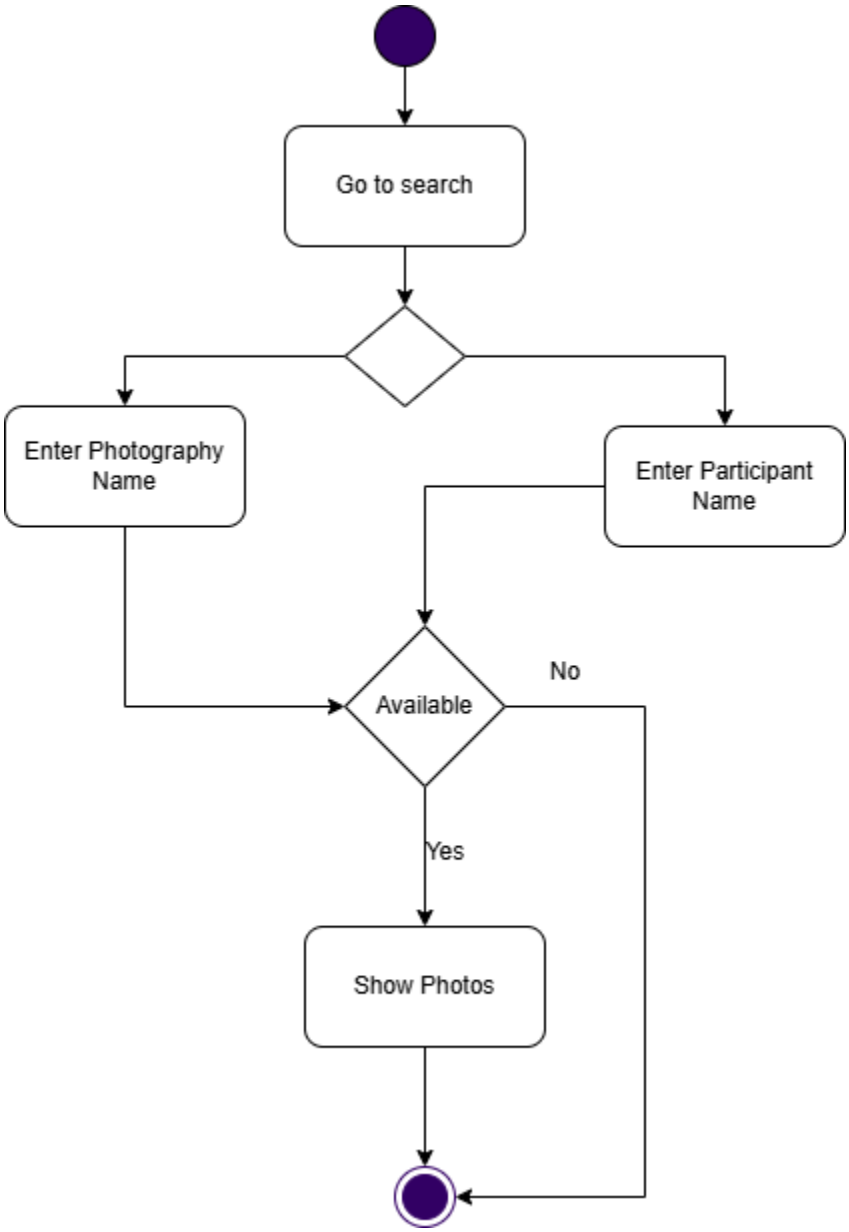


Figure 17: Activity Diagram for Search Photo

Swimlane Diagram

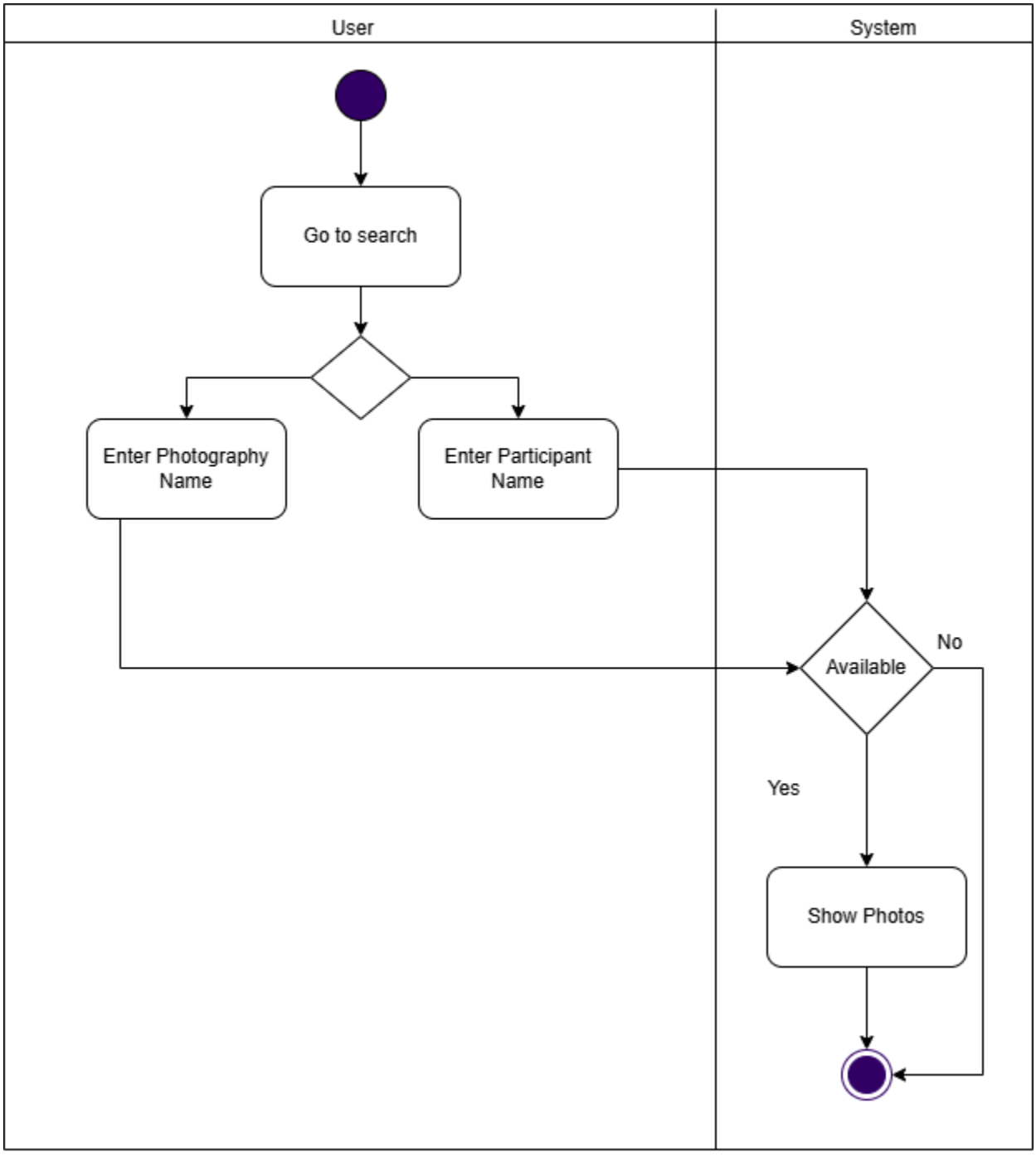


Figure 18: Swimlane Diagram For Search Photos

Use Case 6: Submit Photography
Activity Diagram

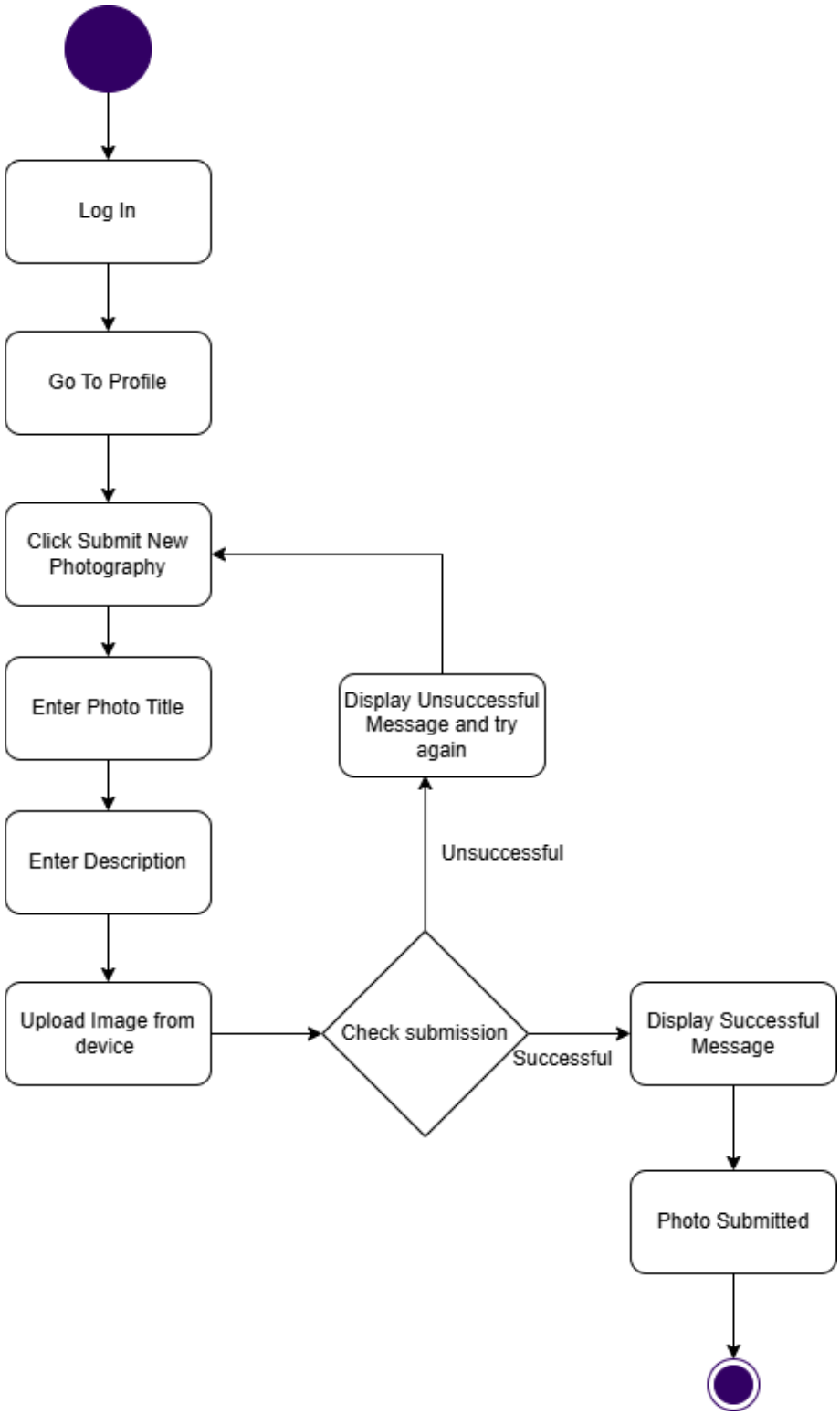


Figure 19: Activity Diagram For Submit Photography

Swimlane Diagram

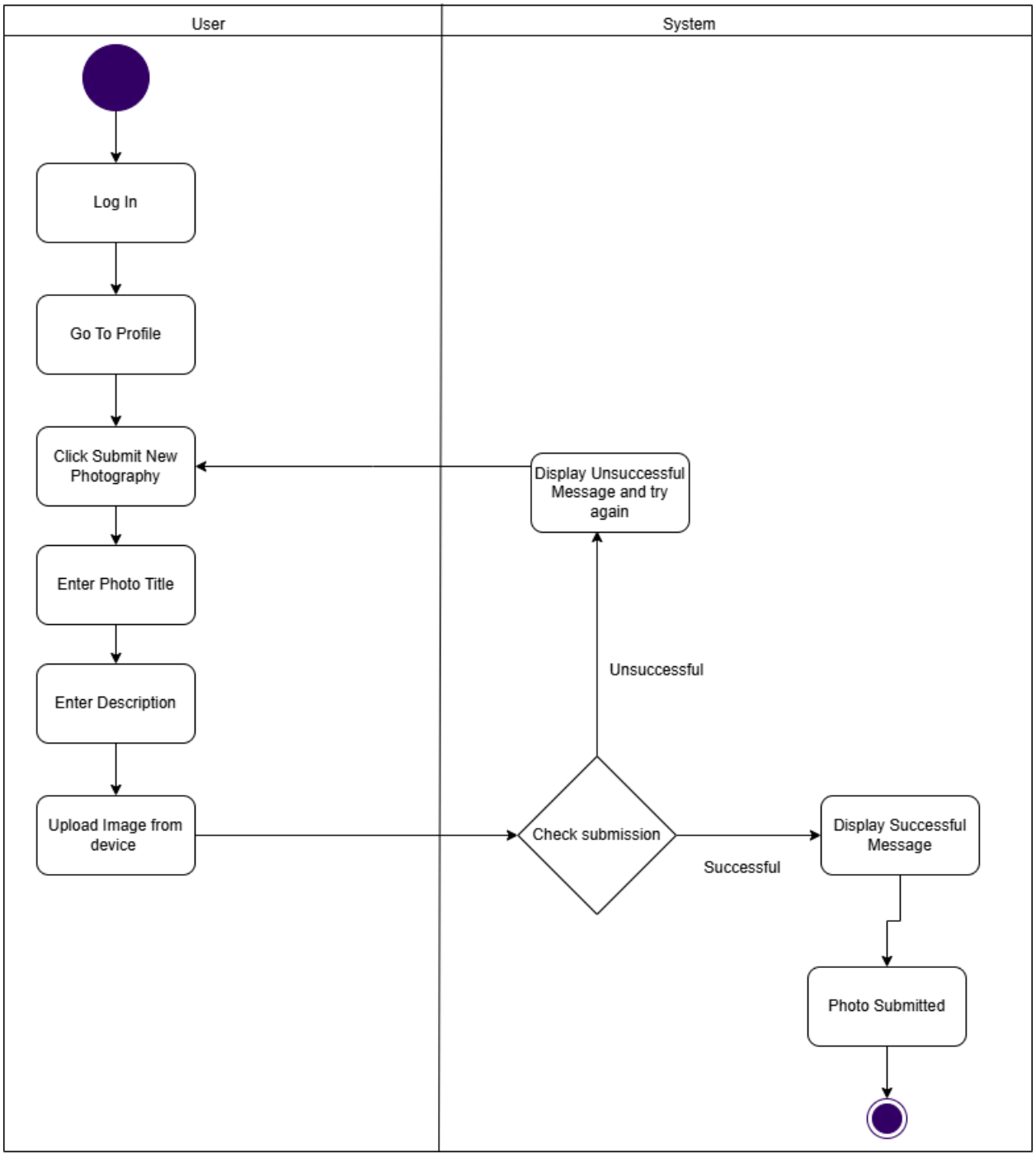


Figure 20: Swimlane Diagram for Submit Photography

Use Case 7: Update and Delete Photograph
Activity Diagram

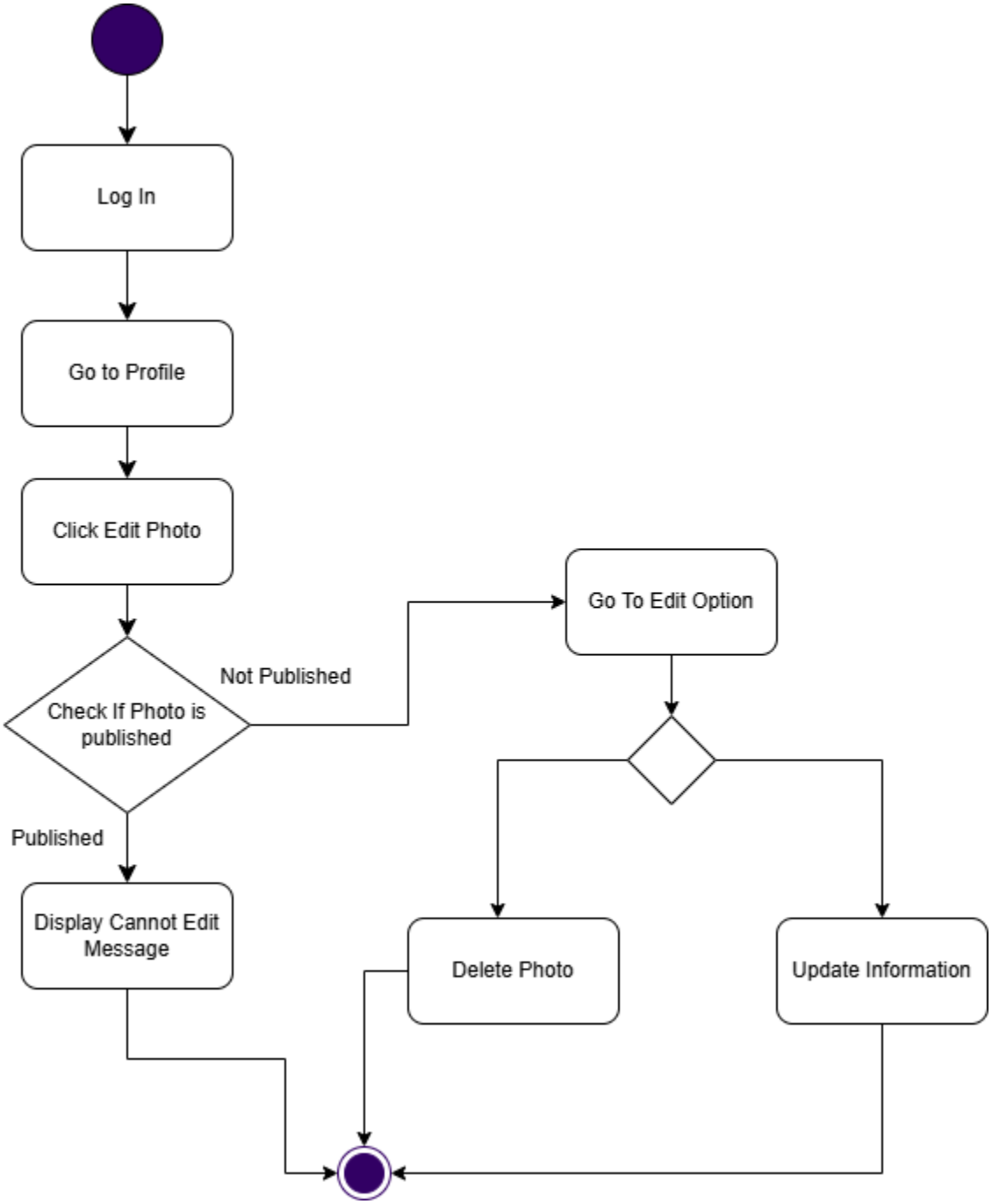


Figure 21: Activity Diagram For Update and Delete Photography

Swimlane Diagram

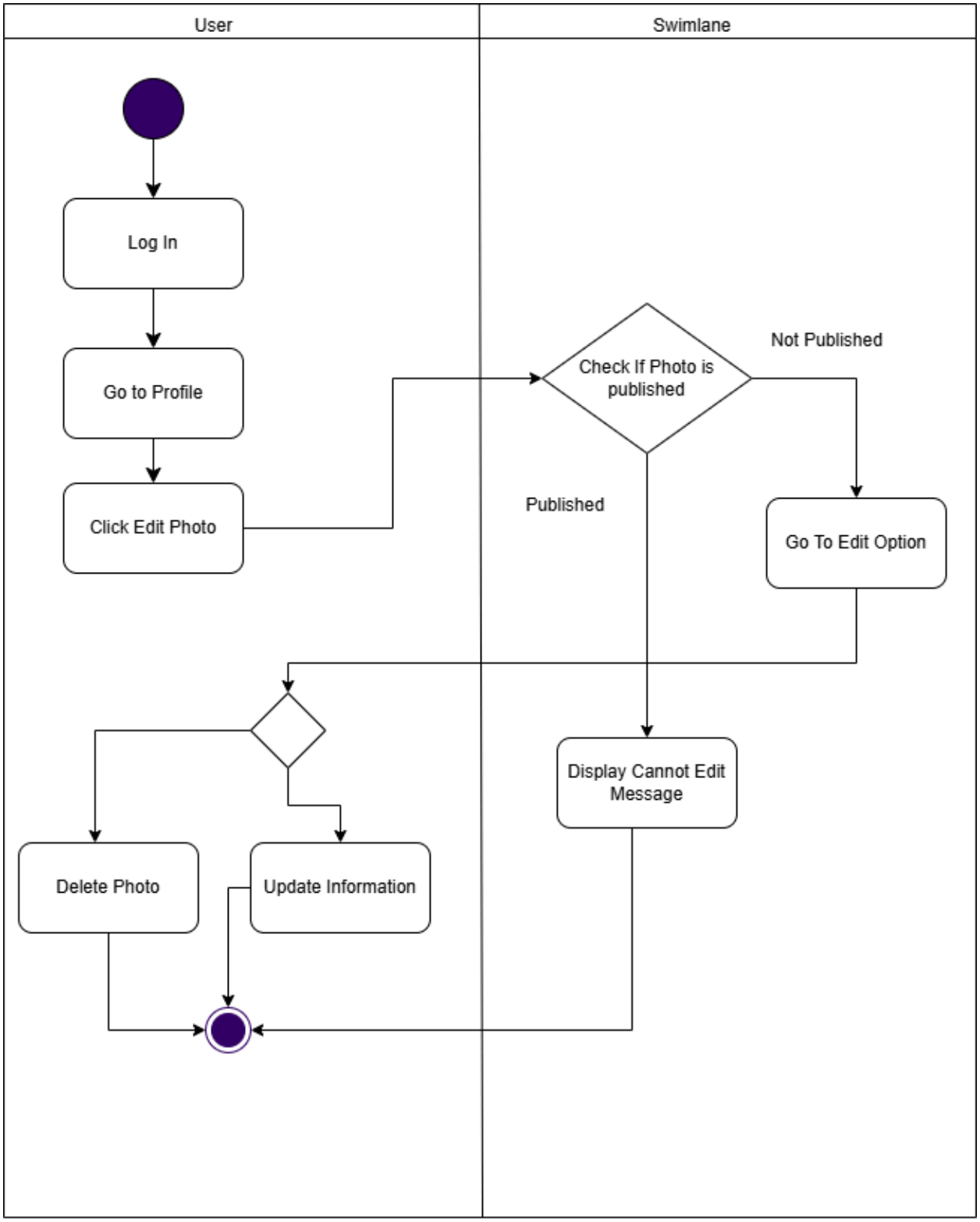


Figure 22: Swimlane Diagram For Update and Delete Photography

Use Case 8: View Submitted Photo
Activity Diagram

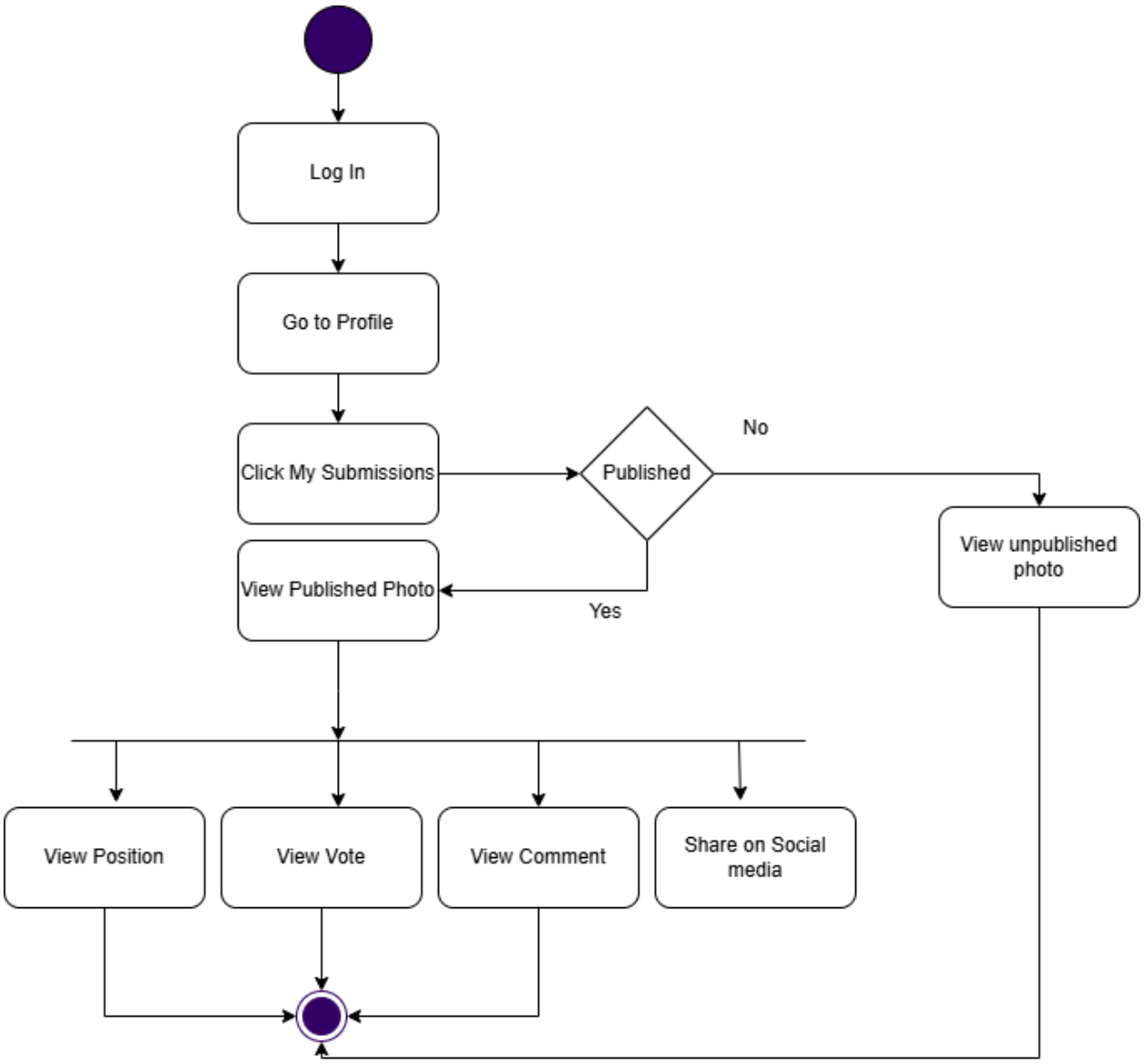


Figure 23: Activity Diagram For View Submitted Photo

Swimlane Diagram

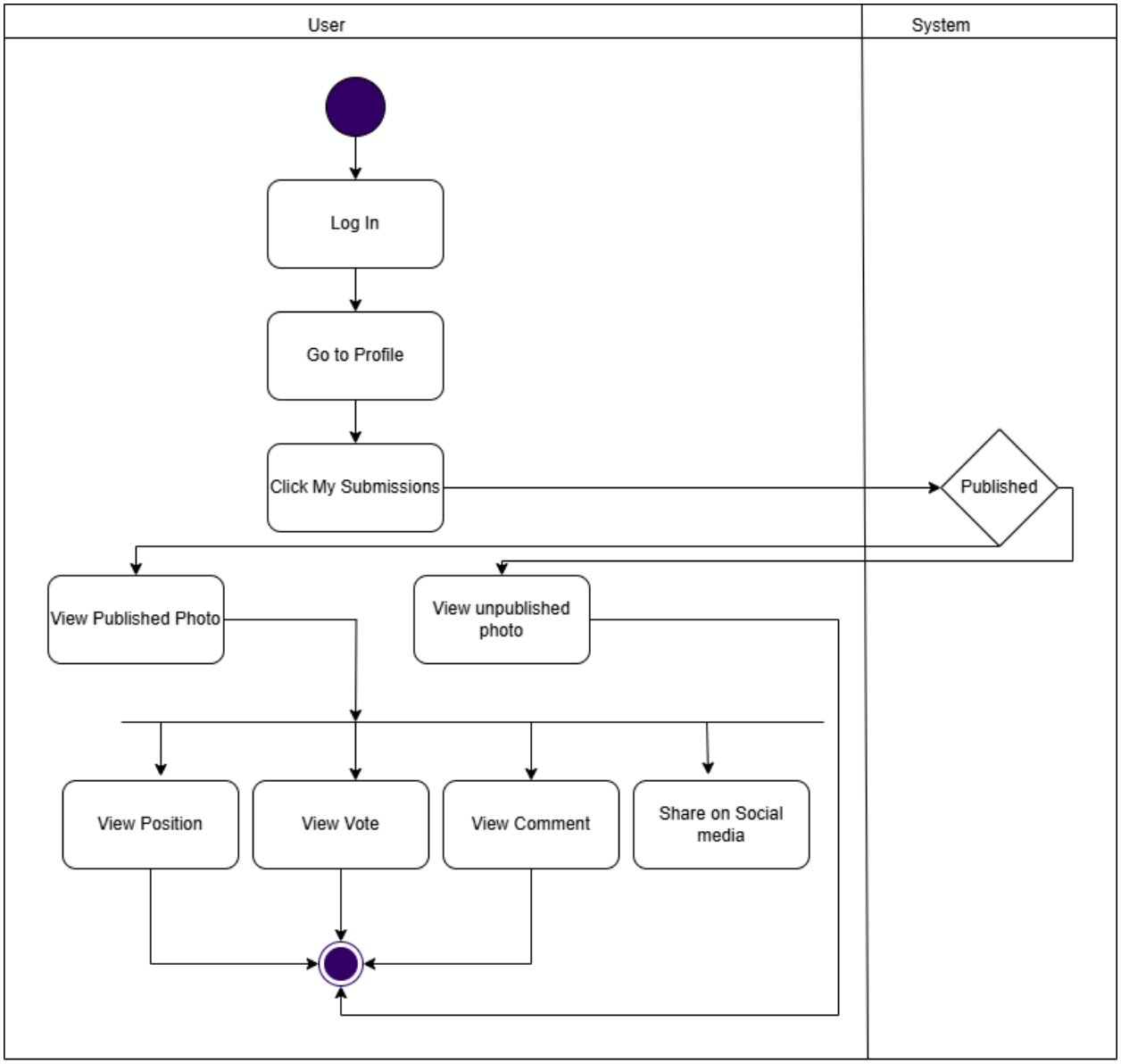


Figure 24: Swimlane Diagram For View Submitted Photo

Use Case 9: Receive Notifications
Activity Diagram

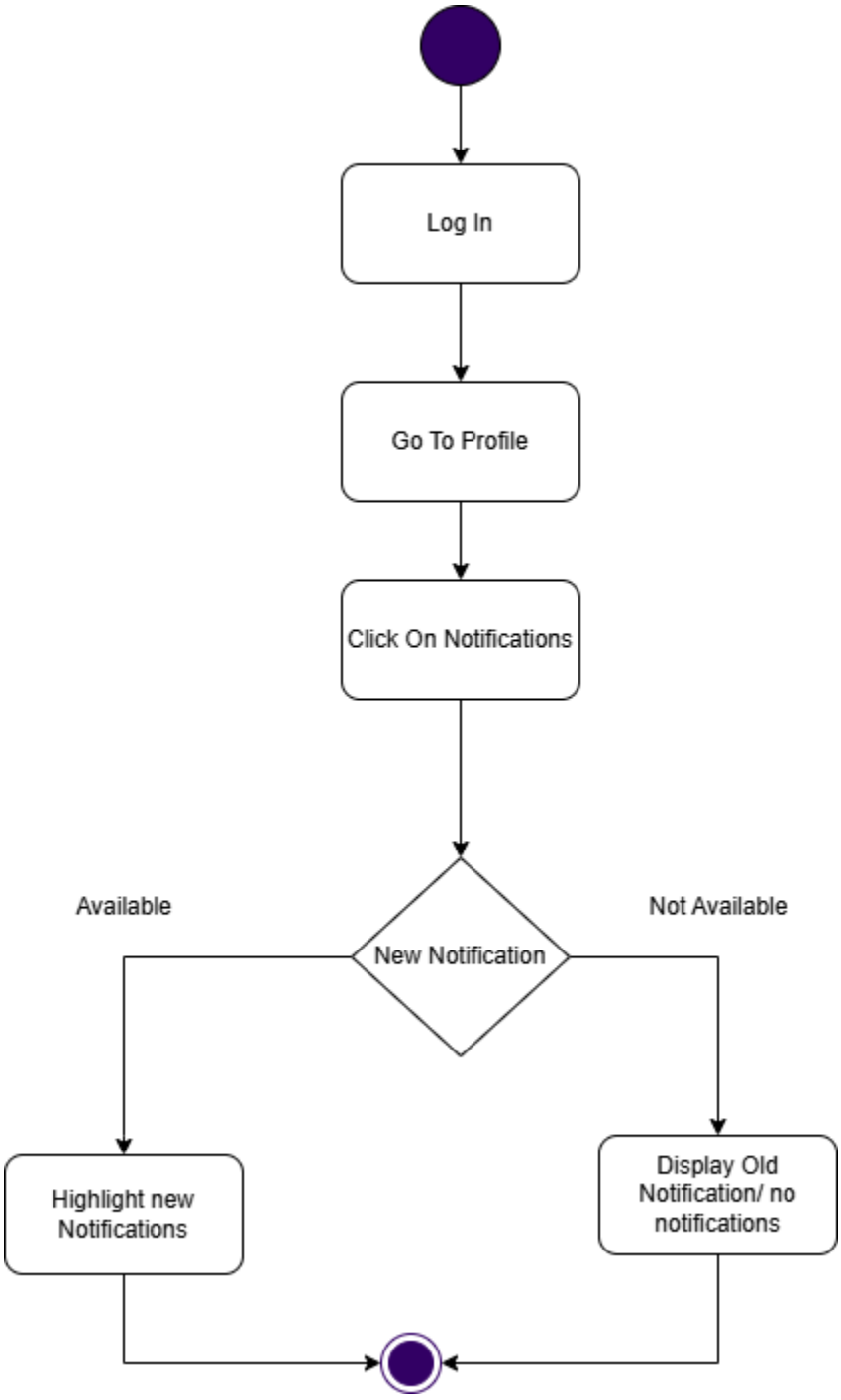


Figure 25: Activity Diagram For Receive Notifications

Swimlane Diagram

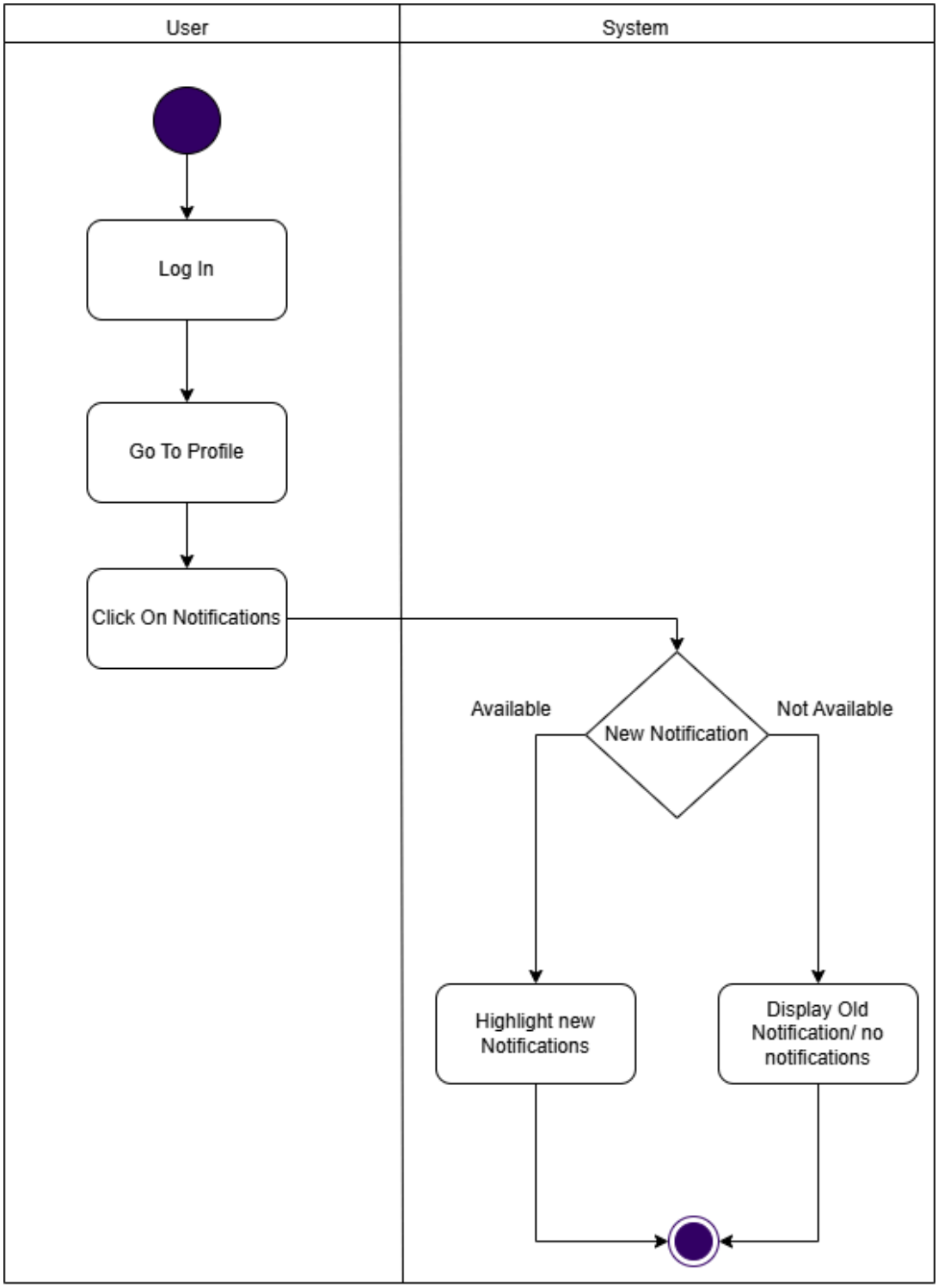


Figure 26: Swimlane Diagram For Recieve Notifications

Use Case 10: Vote And Comment on Photos
Activity Diagram

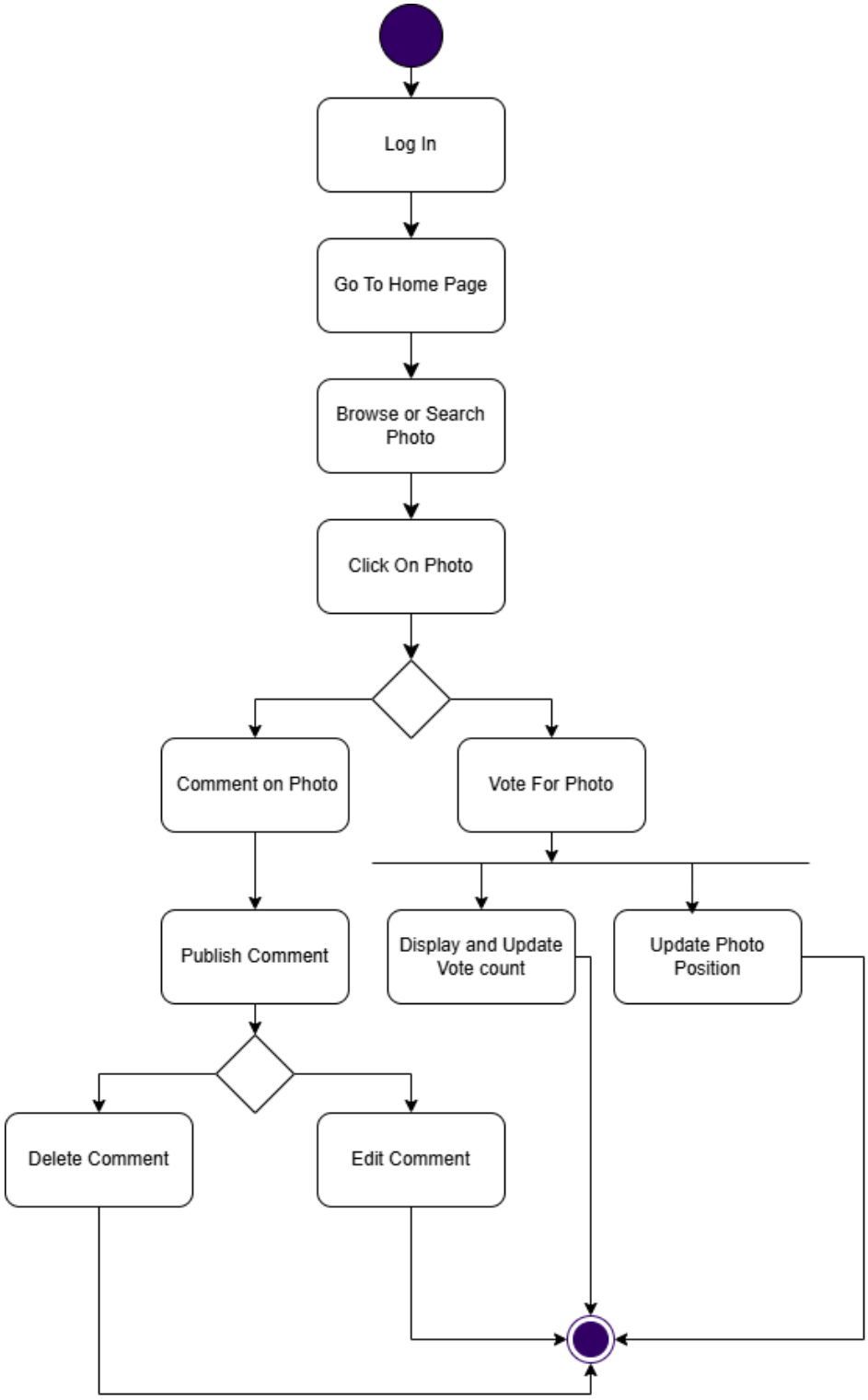


Figure 27: Activity Diagram For Vote And Comment On Photos

Swimlane Diagram

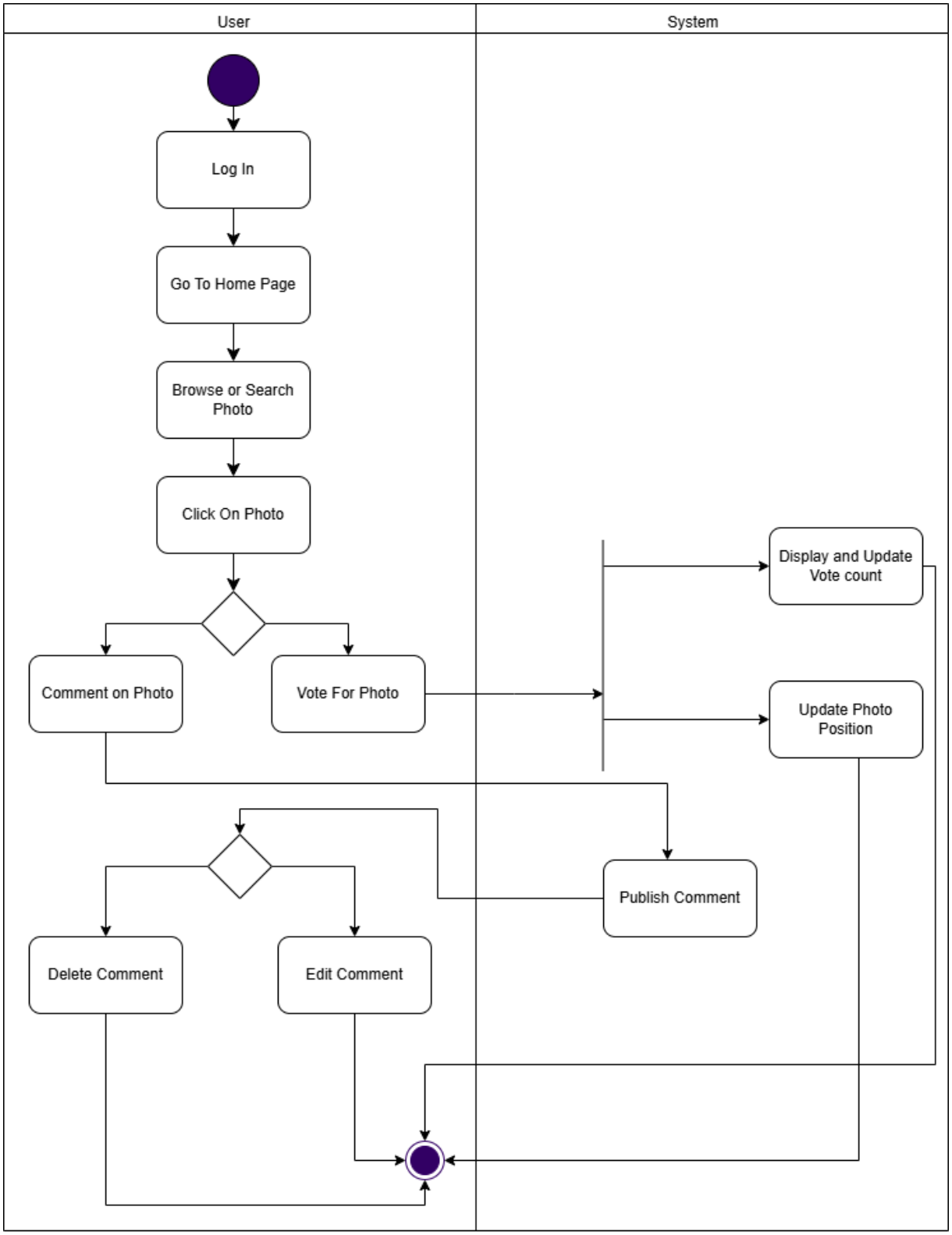


Figure 28: Swimlane Diagram For Vote And Comment On Photos

Use Case 11: Supervise Comments
Activity Diagram

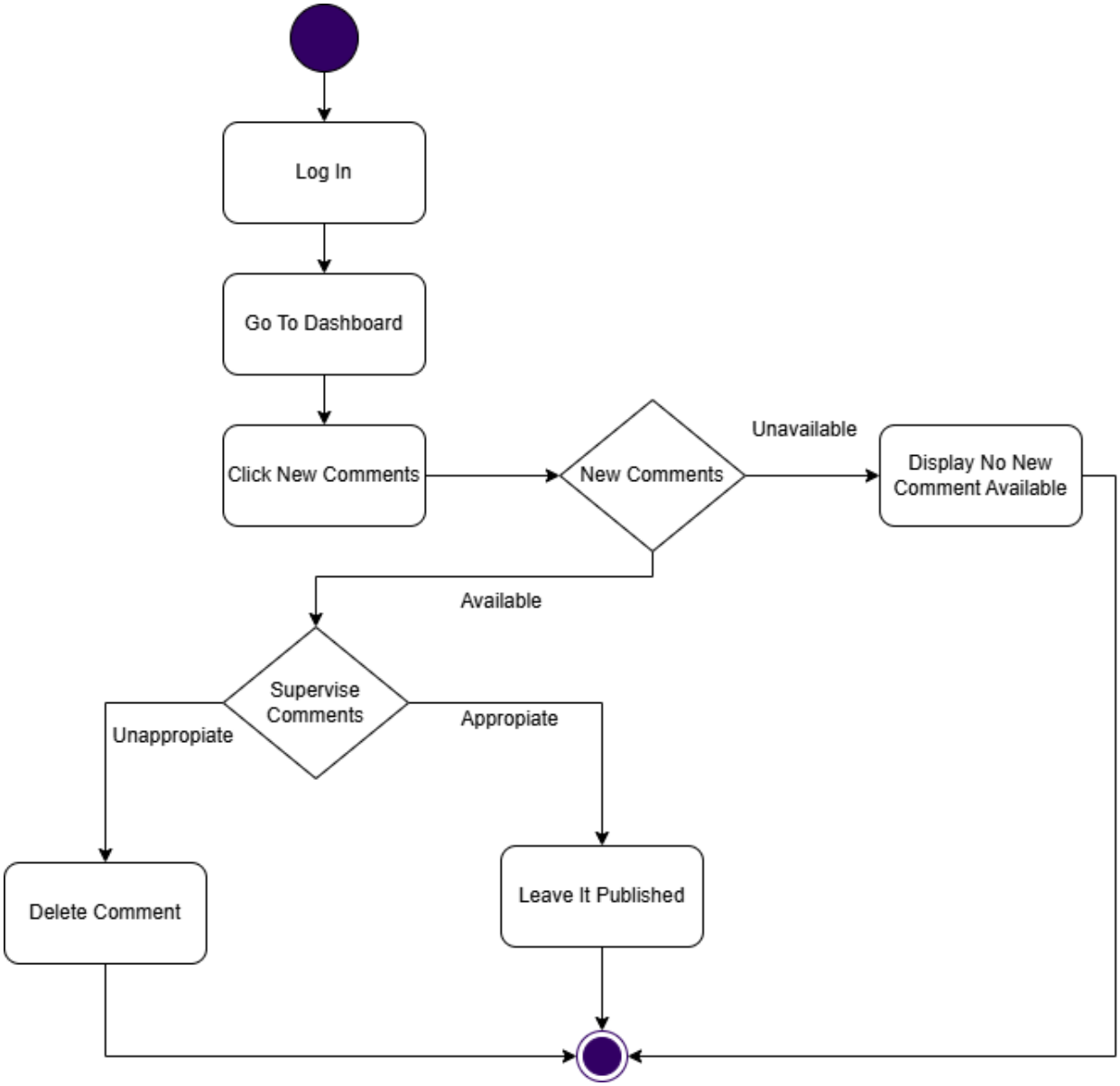


Figure 29: Activity Diagram For Supervise Comments

Swimlane Diagram

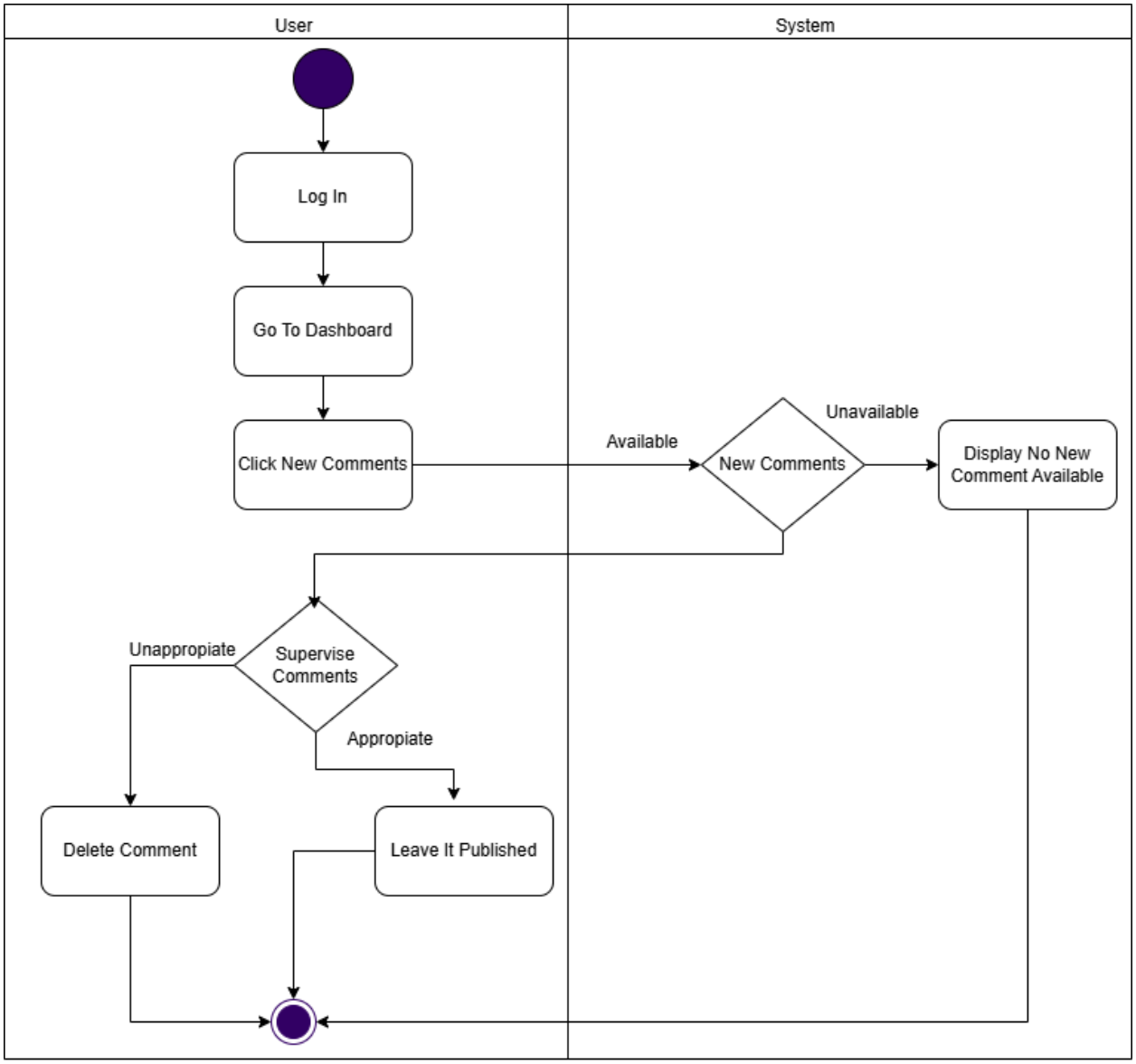


Figure 30: Swimlane For Supervise Comments

Use Case 12: Approving and Disapproving Submissions
Activity Diagram

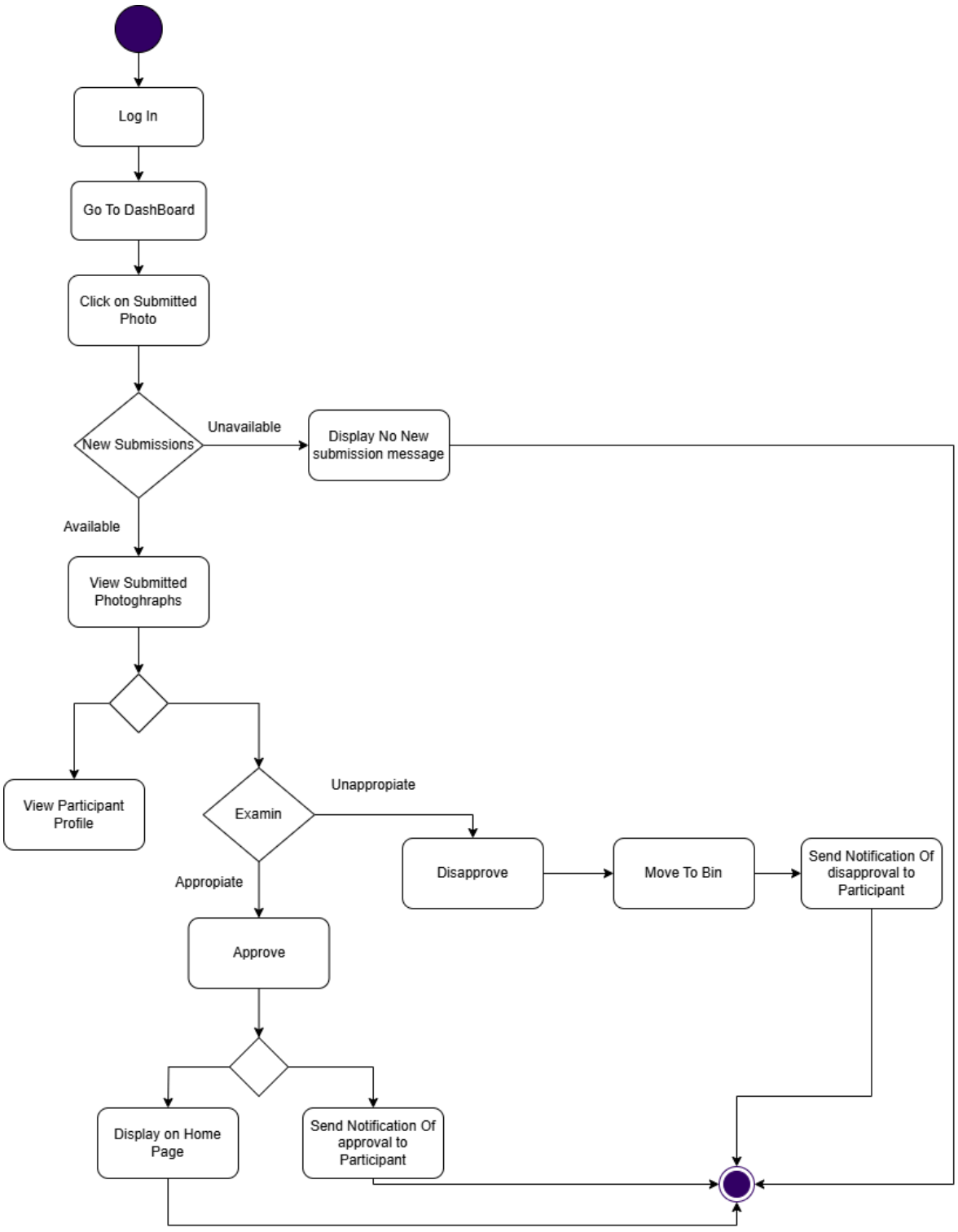


Figure 31: Activity Diagram For Approving Submission

Swimlane Diagram

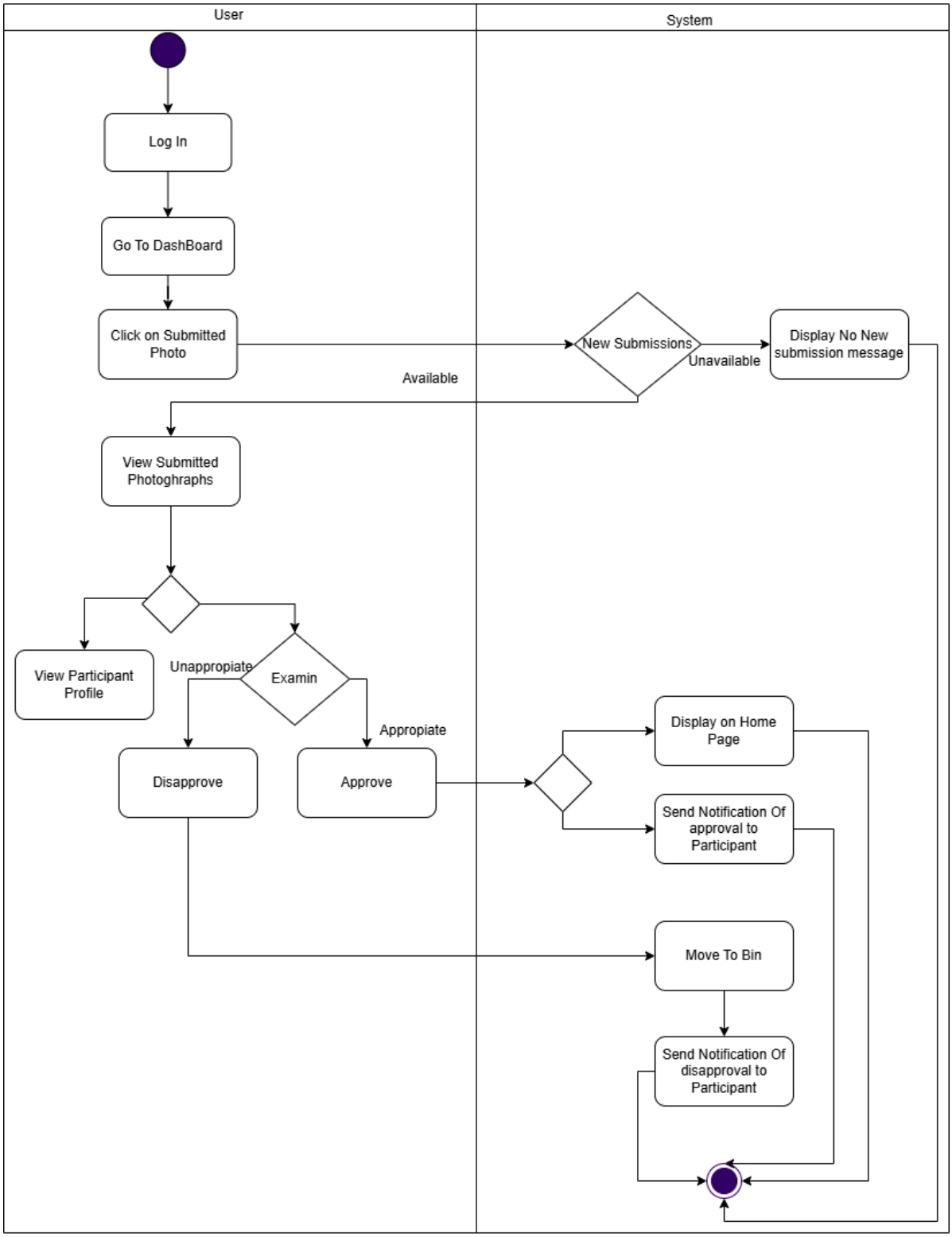


Figure 32: Swimlane Diagram For Approving Submissions

Use Case 13: View Event Information, Leaderboard, Calendar, Results
Activity Diagram

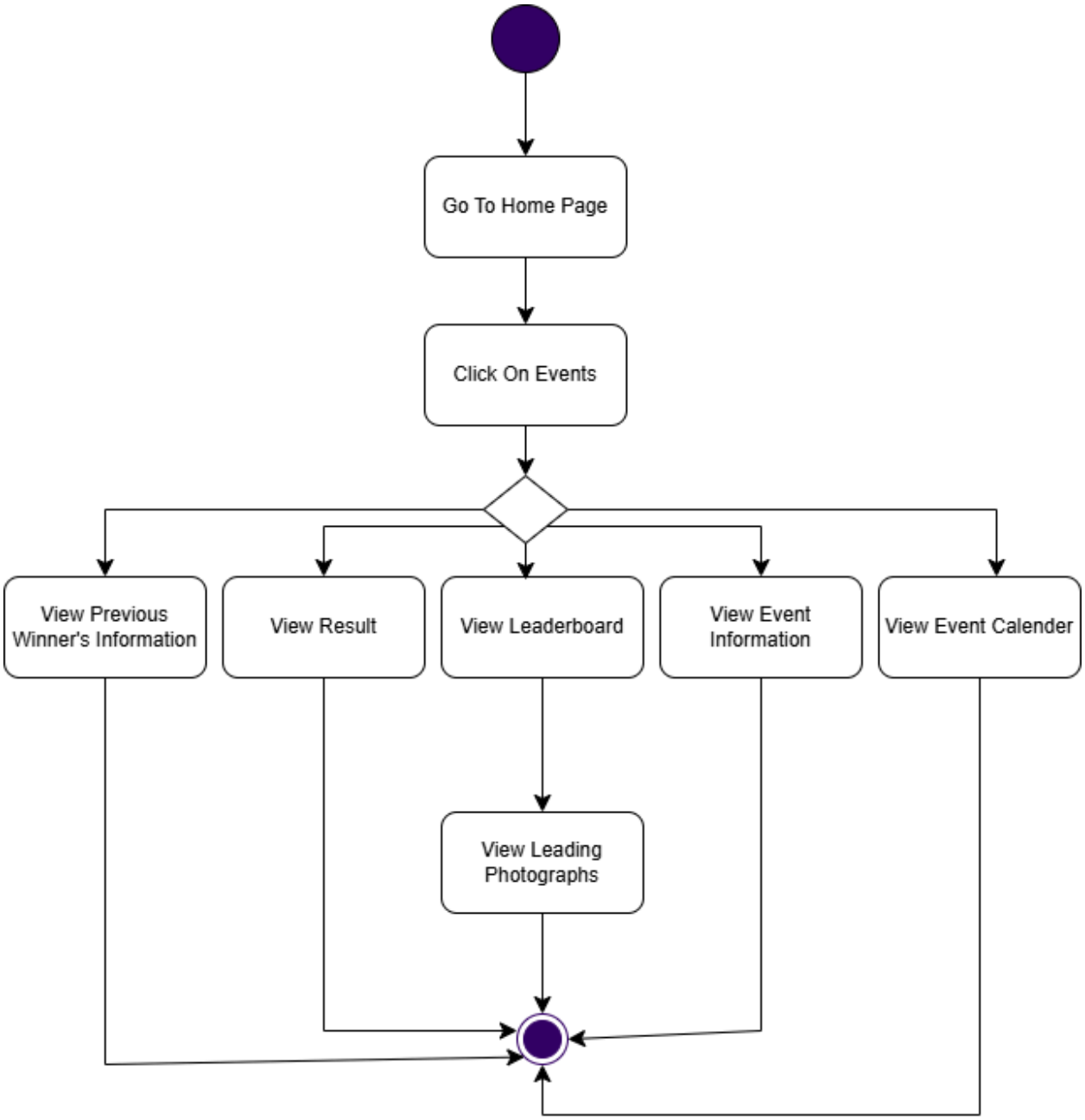


Figure 33: Activity Diagram For View Event, Information, Calendar, Leaderboard, Results

Use Case 14: Update Event, Information, Calendar, Leaderboard, Results
Activity Diagram

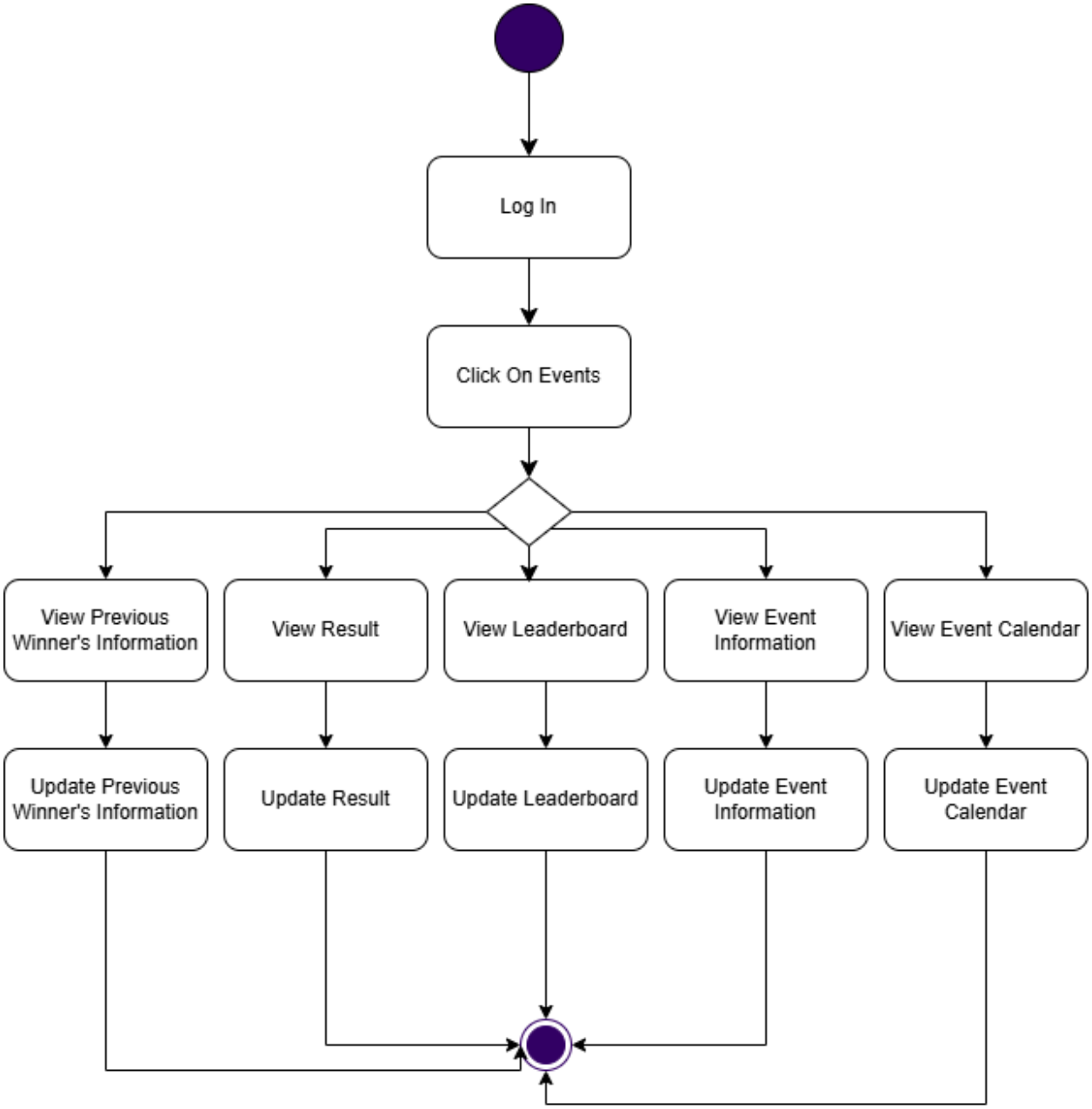


Figure 34: Activity Diagram For Update Event, Information, Calendar, Leaderboard, Results

Use Case 15: View Help and Frequently Asked Questions
Activity Diagram

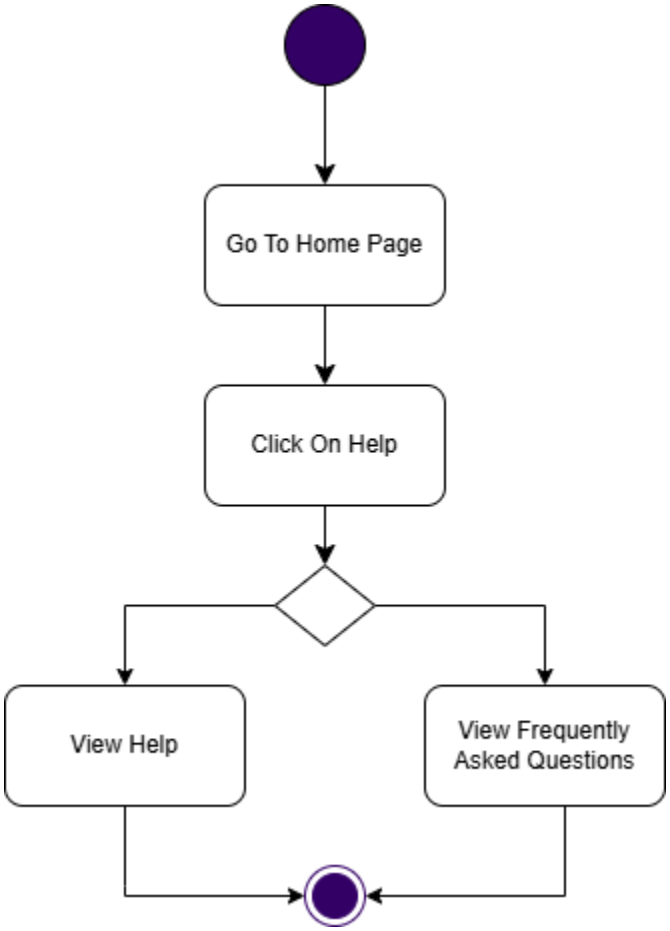


Figure 35: Activity Diagram For View Help and Frequently Asked Questions

Use Case 16: Update Help and Frequently Asked Questions
Activity Diagram

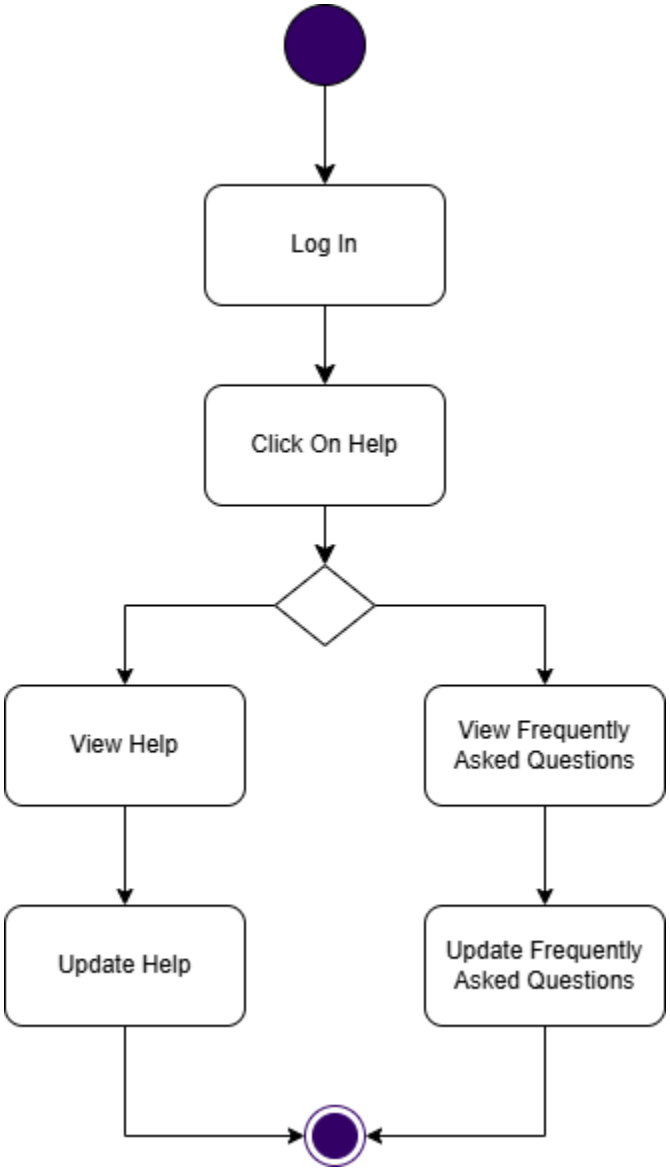


Figure 36: Update Help and Frequently Asked Questions

Chapter 5: Class Based Modeling

5.1 Class Based Modeling Concept

Class-based modeling represents the objects that the system will manipulate, the operations that will applied to the objects, relationships between the objects and the collaborations that occur between the classes that are defined

5.2 General Classification

To identify the potential classes, nouns were selected from the solution space of the story. These were then characterized in seven general classifications. The seven general characteristics are as follows:

1. External entities
2. Things
3. Events
4. Roles
5. Organizational units
6. Places
7. Structures

5.2.1 External Entities

Social Media Platforms (For sharing/authentication)
Payment Gateways (For competition entry fees/premium features)
Email Service Providers (For notifications/password resets)
Cloud Storage Services (For photo uploads/hosting)
Judges/External Reviewers (For competition judging)

5.2.2 Things

Photograph/Submission (Core entity with attributes like title, description, EXIF data)
Competition/Event (Contains rules, deadlines, categories)
Vote (User votes on photographs)
Comment (Feedback on submissions)
Prize (Competition awards)

Badge/Achievement (For user profiles)
FAQ Item (Help content)
Leaderboard Entry (Ranking data)
Notification (Alerts for users)

5.2.3 Events

Submission Deadline (Time-bound event)
Voting Period (Start/end dates)
Winner Announcement (Result publication)
User Registration (Account creation timestamp)
Photo Upload (Submission timestamp)
Judging Phase (Admin-side event)

5.2.4 Roles

Participant (Photographer submitting work)
Admin (Manages competitions, approves submissions)
Judge (Special role for evaluating entries)
Visitor (Unauthenticated user browsing photos)
Moderator (Manages comments/flags)

5.2.5 Organizational Units

Competition Categories (Portrait, Landscape)
Sponsors (Organizations funding prizes)
Support Team (Helpdesk/contact)

5.2.6 Places

User Profile Page (Virtual "place" for user data)
Gallery Page (Where photos are displayed)
Leaderboard Page (Rankings view)
Competition Landing Page (Event details)
Help Center (FAQ/contact section)
Geographic Tags (Locations in photo metadata)

5.2.7 Structures

User Account (Stores credentials/profile data)
Photo Metadata (EXIF data structure)
Voting System (Rules/constraints for votes)
Comment Thread (Nested replies structure)
Search Index (For photo discovery)
Notification System (In-app/email alerts)
Analytics Dashboard (For admins to track engagement)

5.3 Selection Criteria

The potential classes were then selected as classes by six Selection Criteria. A potential class becomes a
class when it fulfills all six characteristics.

1. Retained Information
2. Needed Services
3. Multiple Attributes
4. Common attributes
5. Common operations
6. Essential requirements

5.4 Associate Noun and Verb Identification

The nouns and the verbs associated with the potential classes are identified to find out the attributes and methods of each class.

Table 1: Associate Noun and Verbs

No	Class	Noun	Verb
1	Database	users events ,accounts	stores users, manages accounts, inserts event info
2	User	userID ,firstName . lastName ,email phoneNumber , password loginStatus	logs in, resets password, updates profile, verifies login
3	Event_Information	eventID ,title ,description ,startDate ,endDate ,rules ,prizes	displays calendar, updates leaderboard, announces results
4	Account	accNumber ,verificationCode user ,accDetails	signs up, views account info
5	Participant	participantID, submissionsCount , bio	submits photos, deletes photos
6	Voter	voterID ,votesCast , lastVoteDate	views photos, votes on submissions, comments on photos
7	Admin	adminID ,permissions, lastAccess	manages events, approves submissions, locks accounts
8	About	content , lastUpdated , author	displays help, shows FAQs
9	Activity	activityID ,timestamp , type	counts votes, stores feedback, saves photo info
10	Submit Photo	photoID ,title description,uploadDate, filePath	views submissions, updates submissions
11	Vote	voteID, value , timestamp	adds votes, posts comments
12	Manage Event	eventID,changesLog,	updates event info, approves/rejects

		lastModified	submissions
13	Notification	notificationID ,message isRead, priority	sends alerts, marks notifications as read

5.5 Attribute Selection

Table 2: Attribute Selection of Classes

No	Class	Noun
1	Database	users events accounts
2	User	userID firstName lastName email phoneNumber password loginStatus
3	Event_Information	eventID title Description startDate endDate rules prizes

4	Account	accNumber verificationCode user accDetails
5	Participant	participantID submissionsCount bio
6	Voter	voterID votesCast lastVoteDate
7	Admin	adminID Permissions lastAccess
8	About	content lastUpdated author
9	Activity	activityID timestamp type
10	Submit Photo	photoID title Description uploadDate filePath
11	Vote	voteID value timestamp
12	Manage Event	eventID changesLog lastModified
13	Notification	notificationID message isRead priority

5.6 Methods Identification

Table 3: Methods of Class

No	Class	Methods
1	Database	+insertInfo()
2	User	+verifylogin() +resetPassword() +updateProfile()
3	Event_Information	+viewCalendar() +seeLeaderBoard() +getResult()
4	Account	+signUp() +acc_Info()
5	Participant	+submitPhoto() +deletephoto()
6	Voter	+viewphoto() +voting()
7	Admin	+manage() +approveSubmission()

8	About	+displayHelp() +displayFAQ()
9	Activity	+count() +storeVote() +saveInfo()
10	Submit Photo	+viewSub() +updareSub()
11	Vote	+addvote() +commnet()
12	Manage Event	+updateinfo() +approval()
13	Notification	+sendNotification() +markAsRead()

5.6 Class Cards

Table 4: Class Card of Database

Class Name	Database
Attributes	User Account Event_Information
Methods	insertInfo()
Responsibilities	Collaborators
Data storage	All Classes
Information retrieval	User, Account

Table 5: Class Card of User

Class Name	User
Attributes	UserID Password LoginStatus Email
Methods	Verifylogin() resetPassword() updateProfile()
Responsibilities	Collaborators
Authentication	Database
Profile management	Account

Table 6: Class Card of Event_Information

Class Name	Event_Information
Attributes	Calendar Leaderboard Result
Methods	ViewCalendar() SeeLeaderBoard() GetResult()
Responsibilities	Collaborators
Event scheduling	Admin
Results display	Leaderboard

Table 7: Class Card of Account

Class Name	Account
Attributes	Acc_no Acc_details
Methods	SignUp() Acc_Info()
Responsibilities	Collaborators
. Account creation	Database
Profile linking	User

Table 8: Class Card of Participant

Class Name	Participant
Attributes	P_name
Methods	SubmitPhoto() deletephoto()
Responsibilities	Collaborators
Photo submission	Submit Photo
Content management	User

Table 9: Class Card of Voter

Class Name	Voter
Attributes	V_name
Methods	viewphoto() voting()
Responsibilities	Collaborators
Photo interaction	Vote
Voting system	Leaderboard

Table 10: Class Card of Admin

Class Name	Admin
Attributes	Admin_id
Methods	Manage() ApproveSubmission()
Responsibilities	Collaborators
Content moderation	Submit Photo
System management	Database

Table 11: Class Card of About

Class Name	About
Attributes	content title question answer
Methods	displayHelp() displayFAQ()
Responsibilities	Collaborators
User support	User
Information display	None

Table 12: Class Card of Activity

Class Name	Activity
Attributes	Photo feedback
Methods	Count() StoreVote() SaveInfo()
Responsibilities	Collaborators
Activity tracking	Vote
Data recording	Database

Table 13: Class Card of Submit Photo

Class Name	Submit Photo
Attributes	Sub_no delete
Methods	ViewSub() UpdareSub()
Responsibilities	Collaborators
Photo handling	Participant
Submission management	Admin

Table 14: Class Card of Vote

Class Name	Vote
Attributes	Vote_no
Methods	Addvote() commnet()
Responsibilities	Collaborators
Voting mechanism	Voter
Feedback system	Activity

Table 15: Class Card of Manage Event

Class Name	Manage Event
Attributes	info
Methods	Updateinfo() Approval()
Responsibilities	Collaborators
Event updates	Event_Information
Content approval	Admin

Table 16: Class Card of Notification

Class Name	Notification
Attributes	N_no message date_Sent is_Read
Methods	sendNotification() markAsRead()
Responsibilities	Collaborators
Alert system	User
Status tracking	Database

5.7 UML Class Diagram

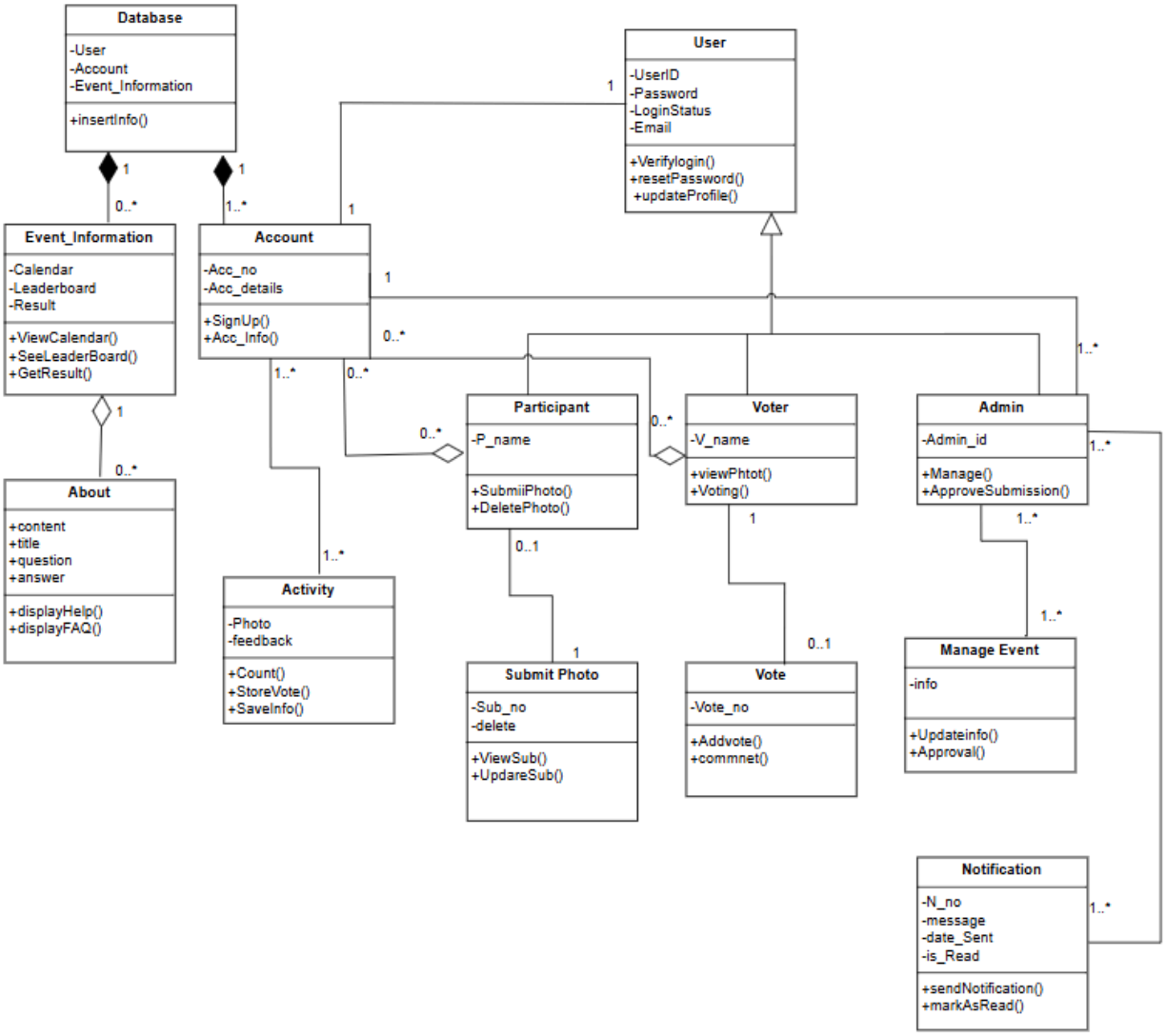


Figure 37:UML Class Diagram of Snap Showdown A Photography Competition Entrance Website

Chapter 6: Flow-Oriented Model

6.1 Introduction

Although data flow-oriented modeling is perceived as an outdated technique by some software engineers, it continues to be one of the most widely used requirements analysis notations in use today. It provides additional insight into system requirements and flow.

6.2 Data Flow Diagram (DFD)

The DFD takes an input-process-output view of a system. In the figures, data objects are represented by labeled arrows and transformations are represented by circles.

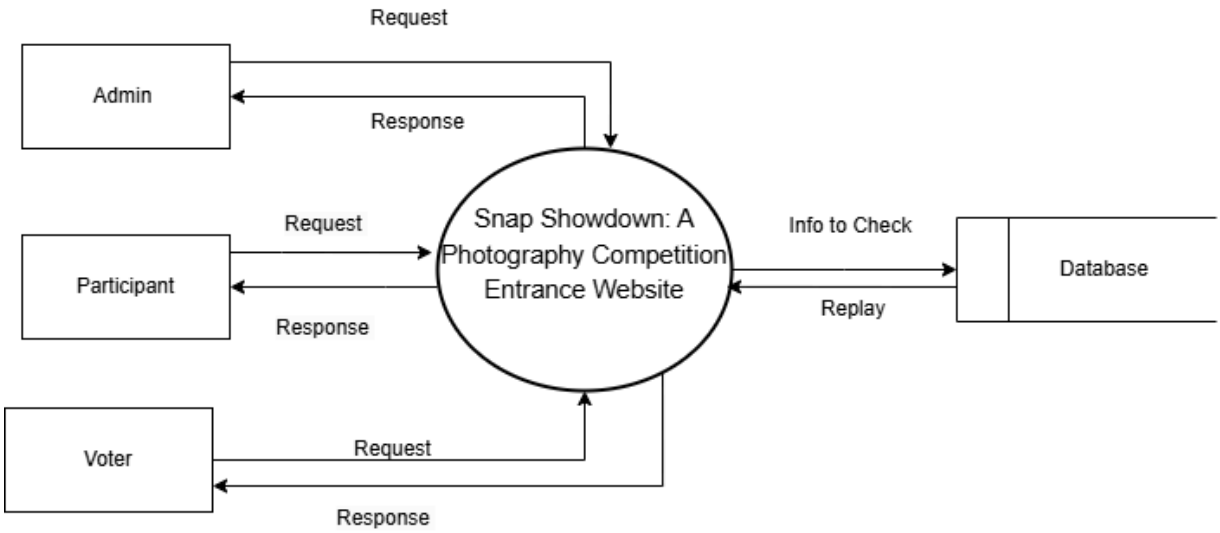


Figure 38: Level 0 Data Flow Diagram

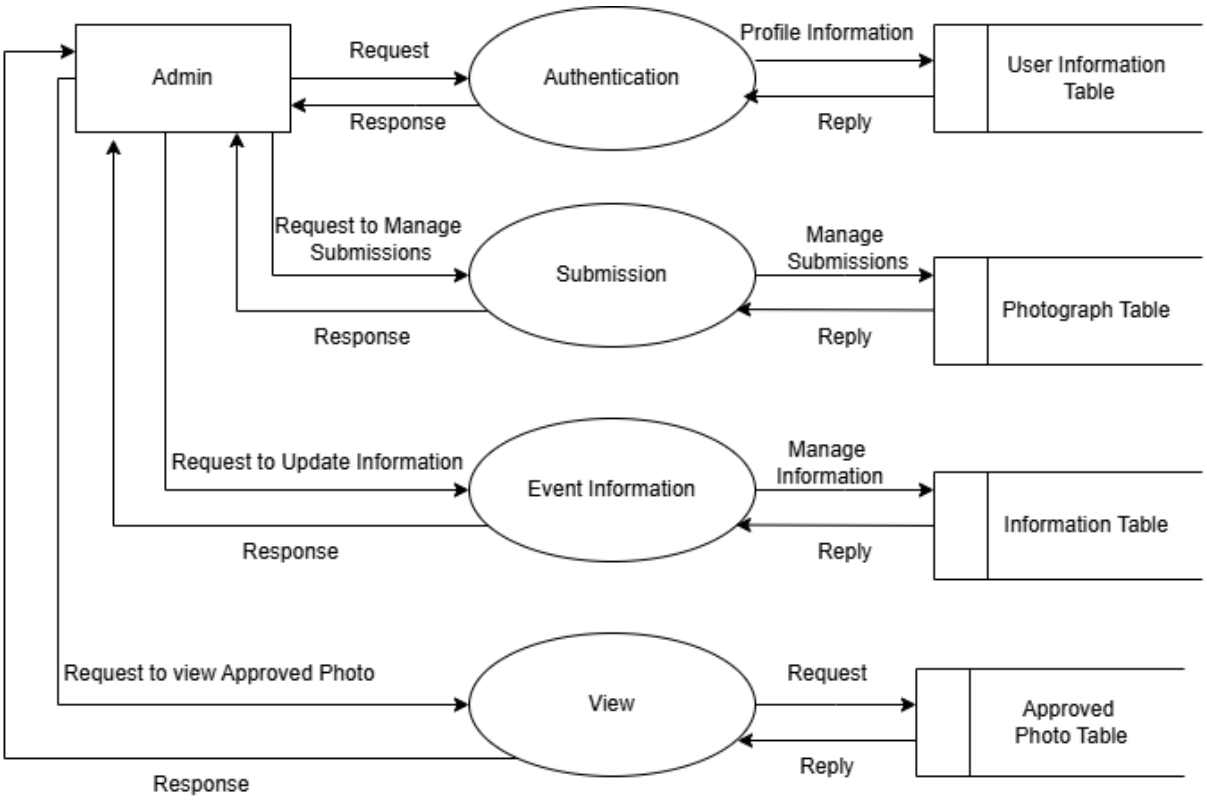


Figure 39: Level 1.1 Data Flow Diagram

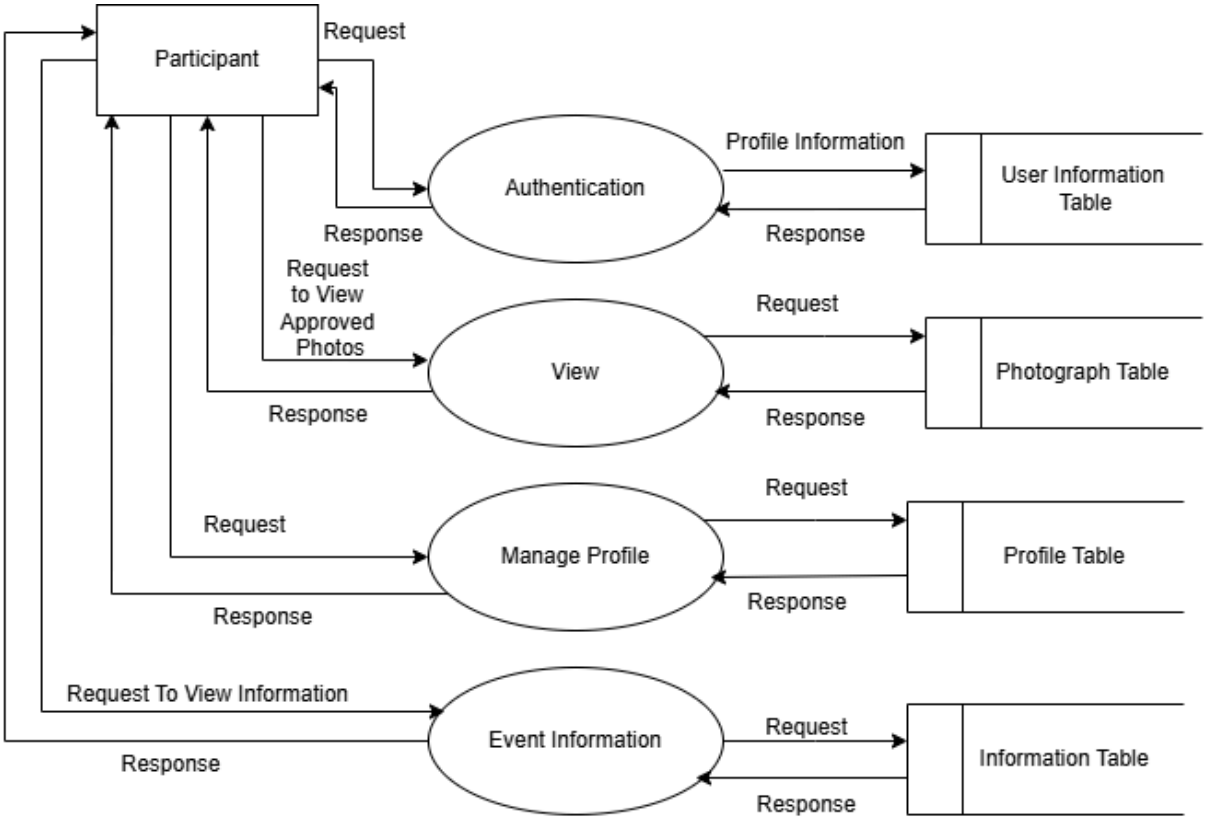


Figure 40: Level 1.2 Data Flow Diagram

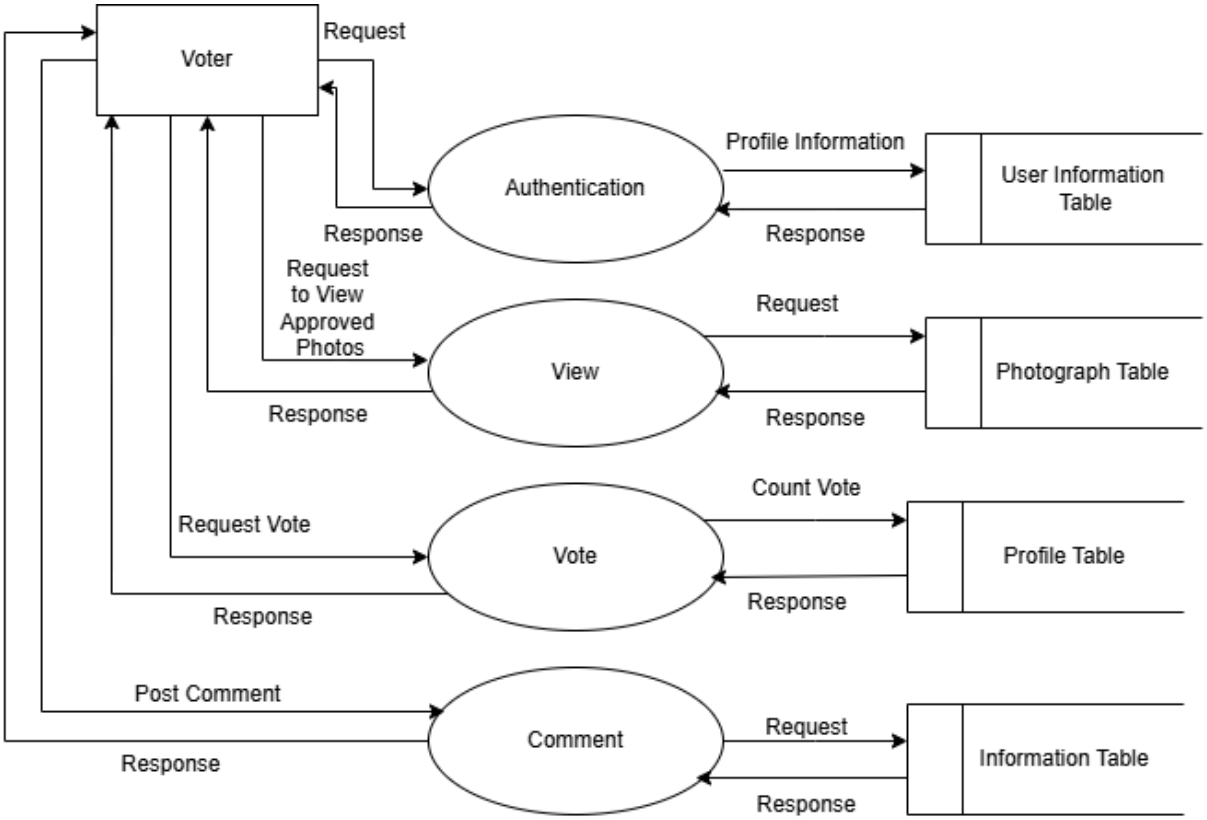


Figure 41: Level 1.3 Data Flow Diagram

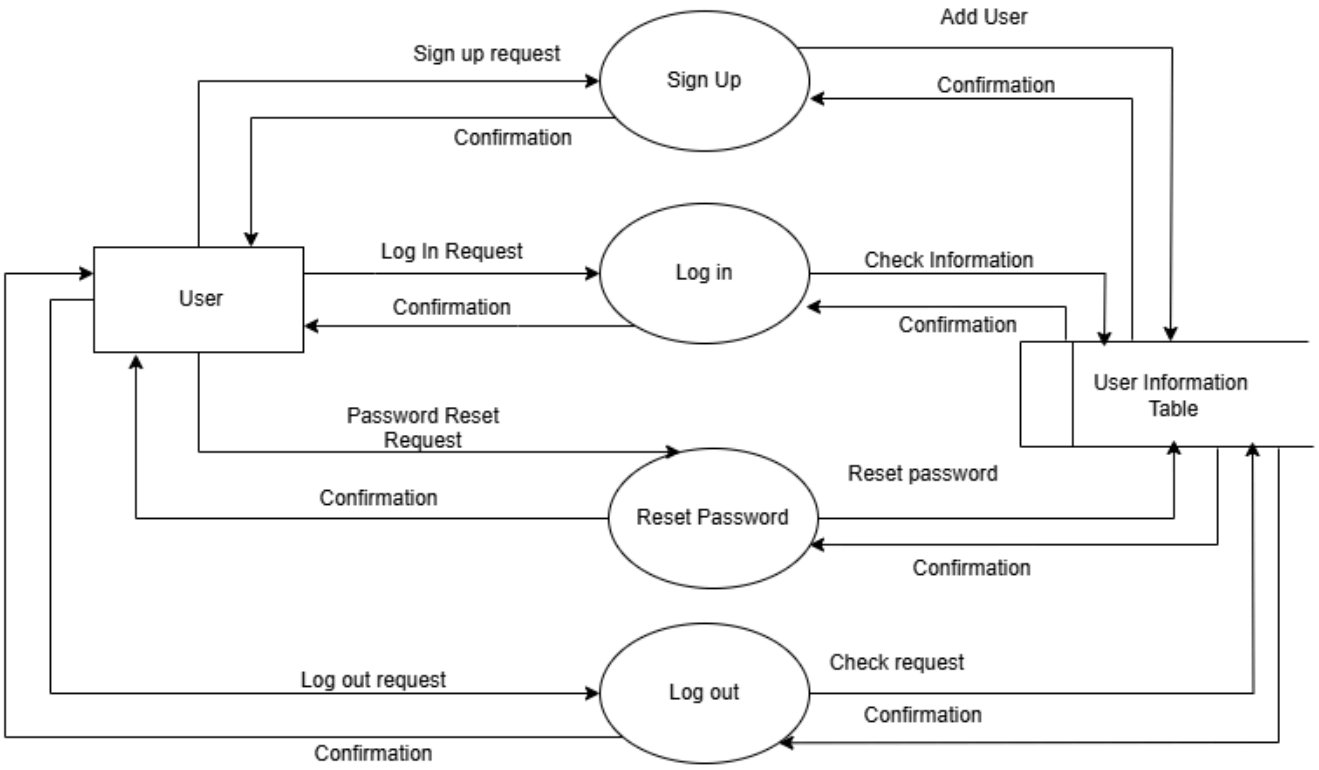


Figure 42: Level 2.1 Data Flow Diagram

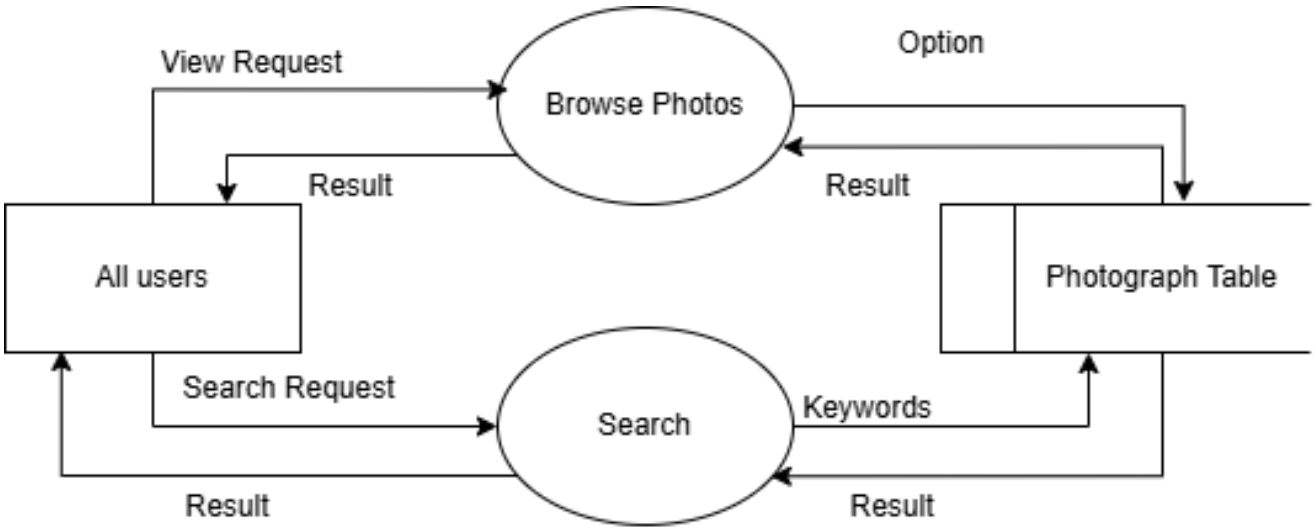


Figure 43: Level 2.2 Data Flow Diagram

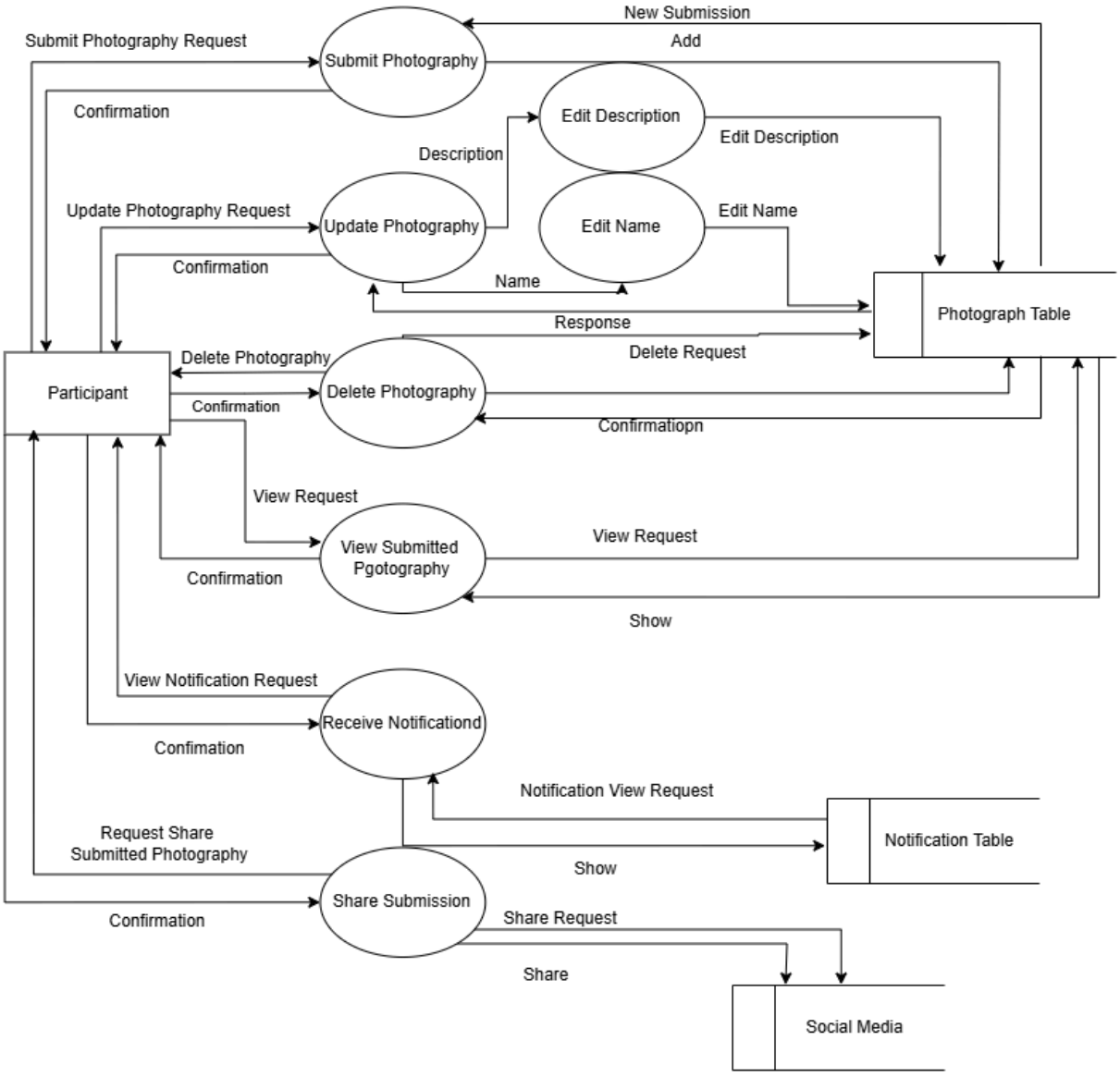


Figure 44: Level 2.3 Data Flow Diagram

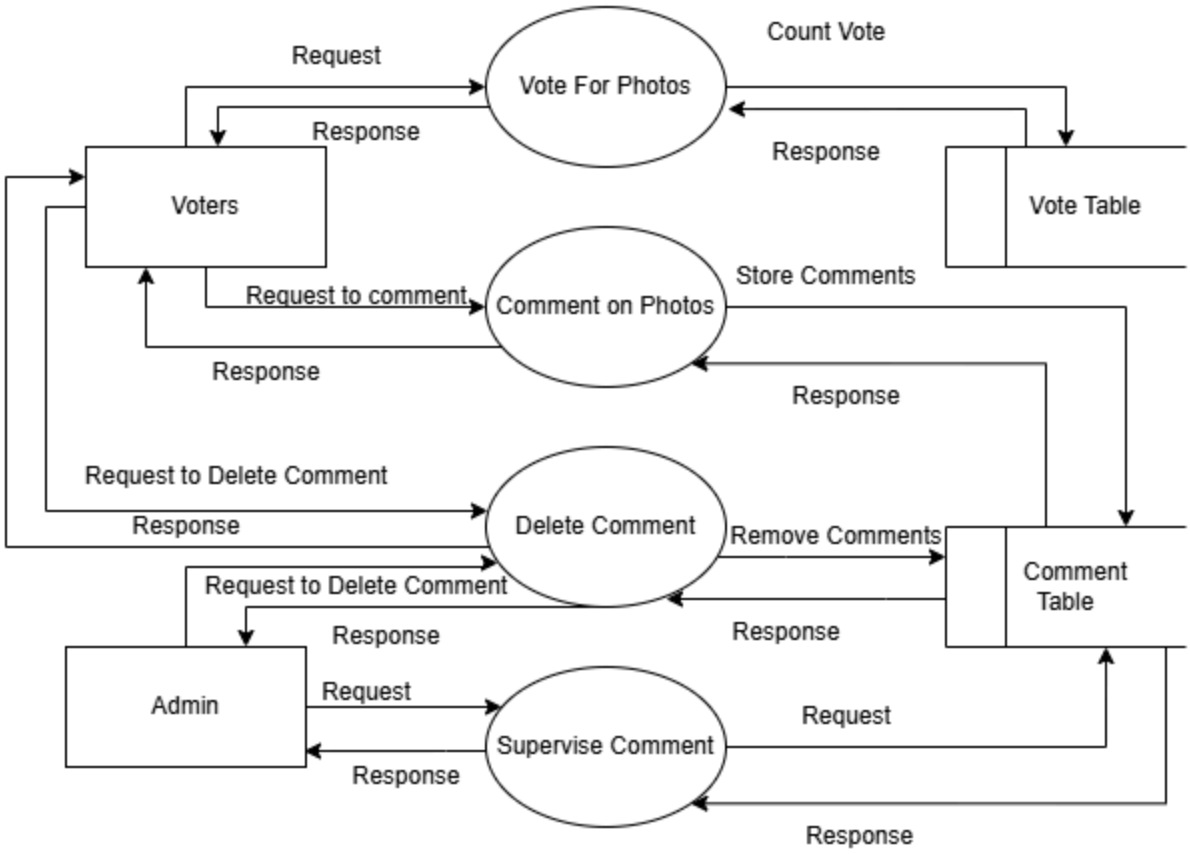


Figure 45: Level 2.4 Data Flow Diagram

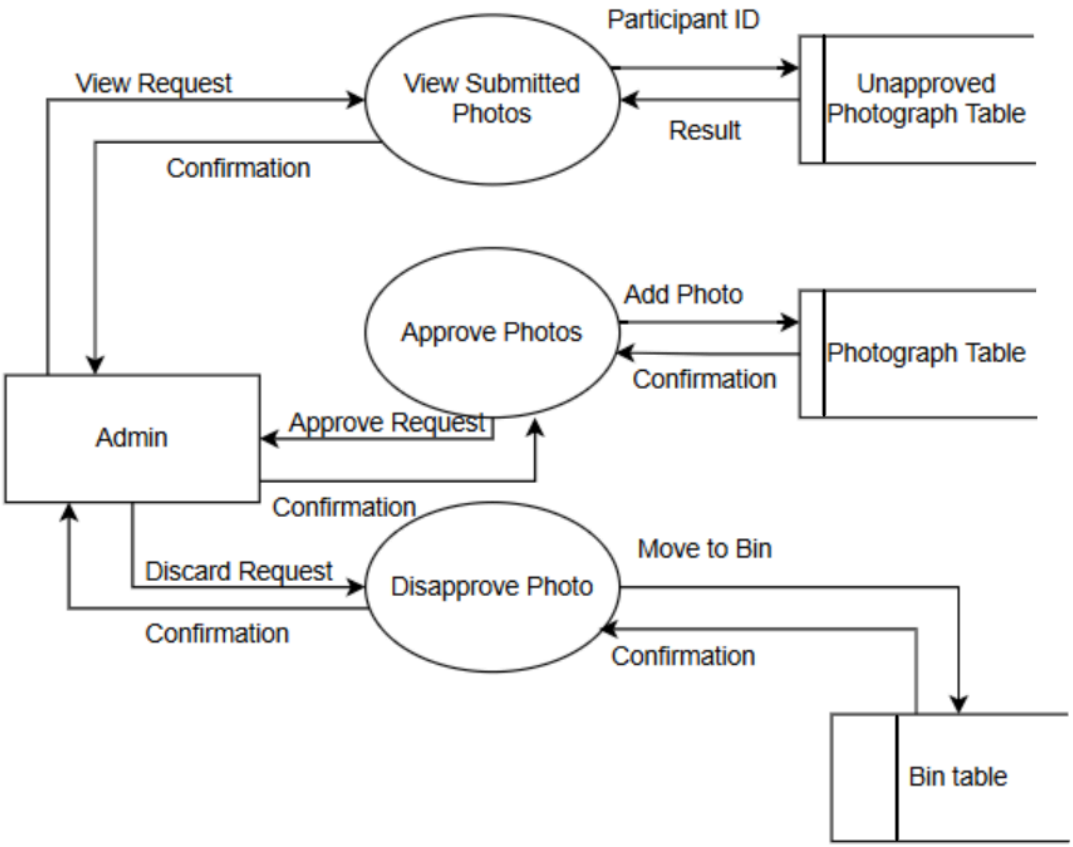


Figure 46: Level 2.5 Data Flow Diagram

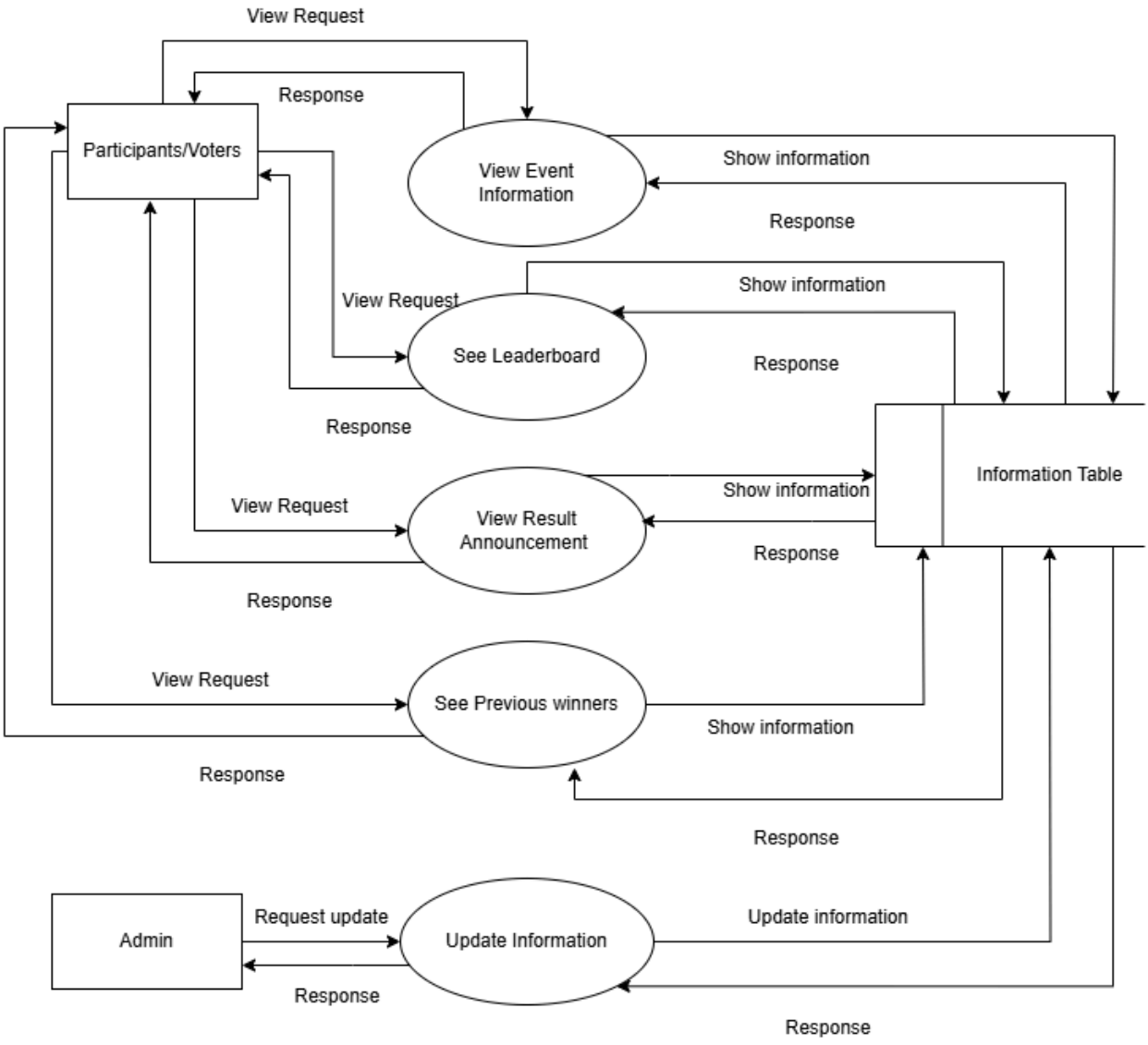


Figure 47: Level 2.6 Data Flow Diagram

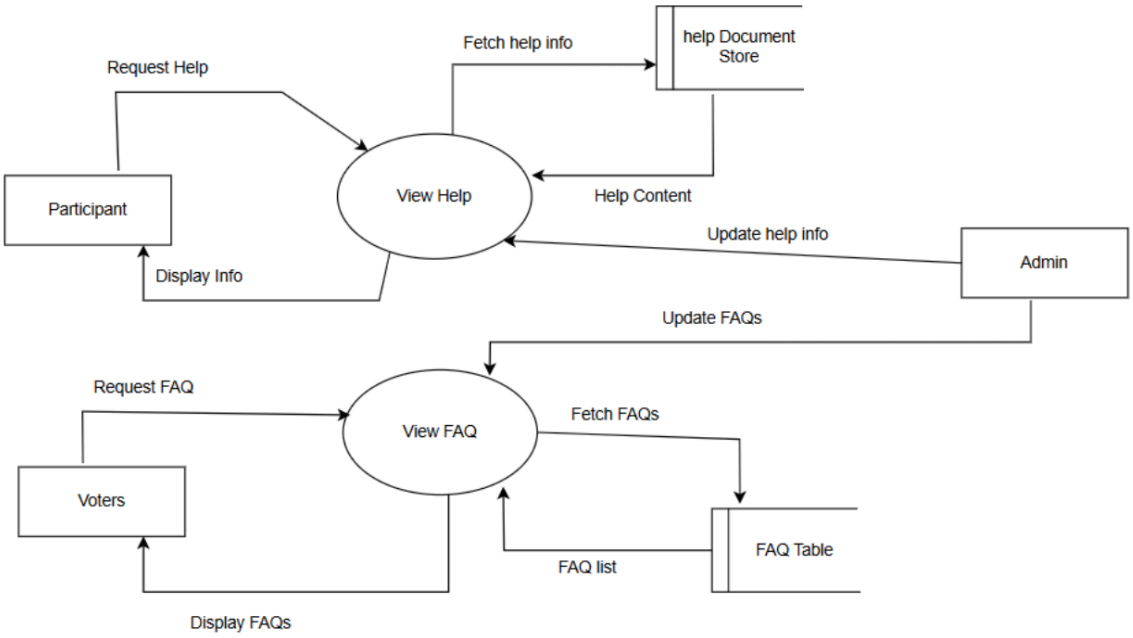


Figure 48: Level 2.7 Data Flow Diagram

7 Sequence Diagram

A sequence diagram on a website depicts user interactions and the flow of messages between components over time.

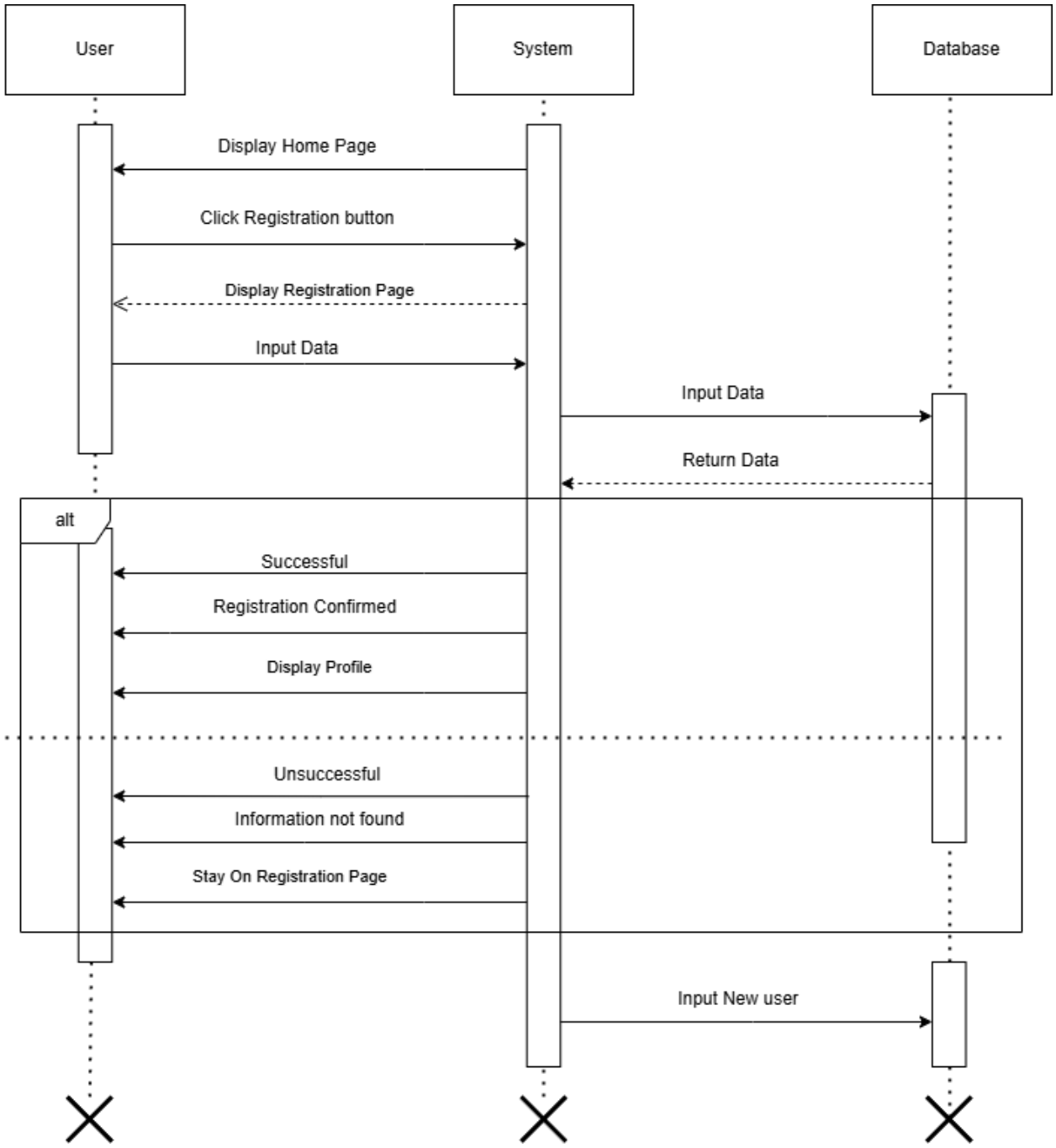


Figure 49: Sequence Diagram (Sign up)

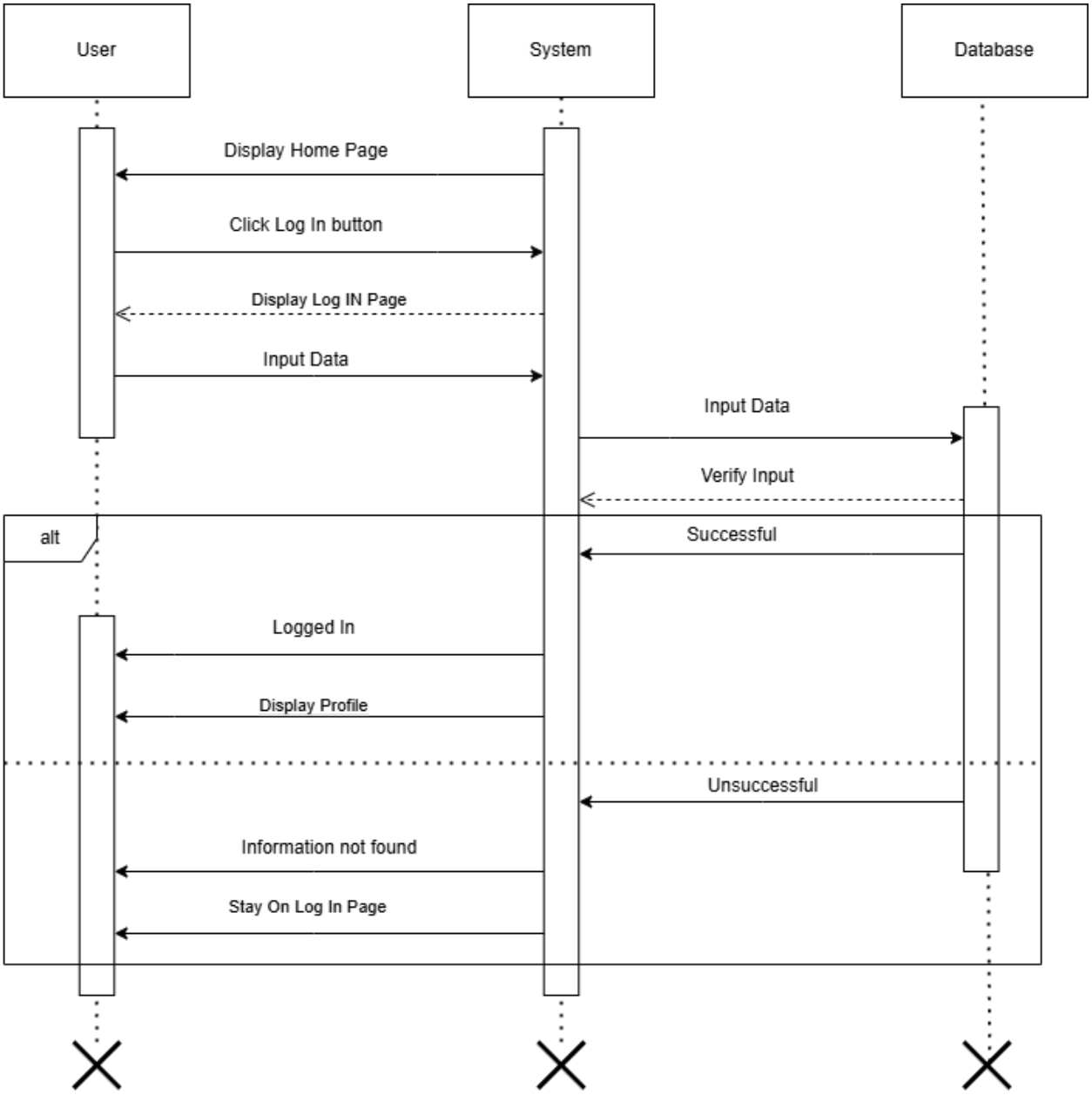


Figure 50: Sequence Diagram (log in)

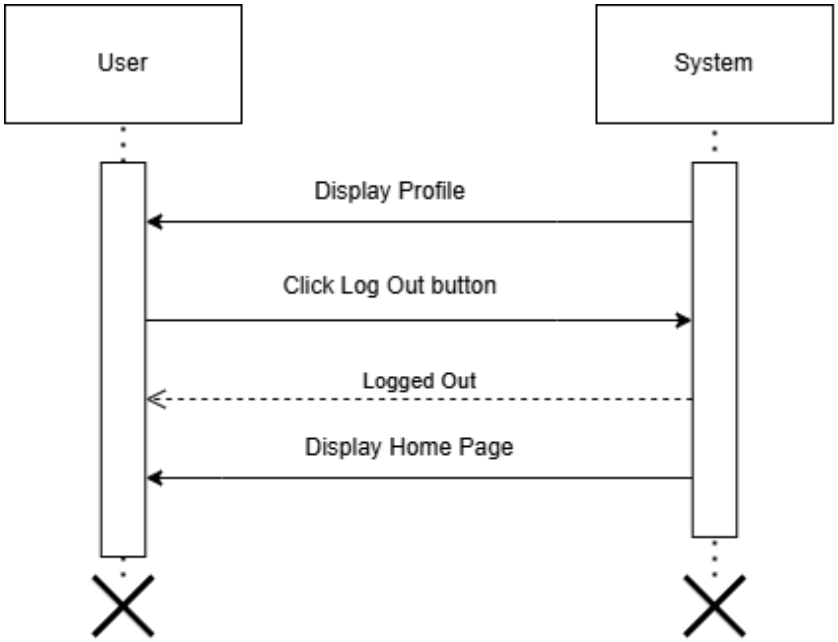


Figure 51: Sequence diagram (log out)

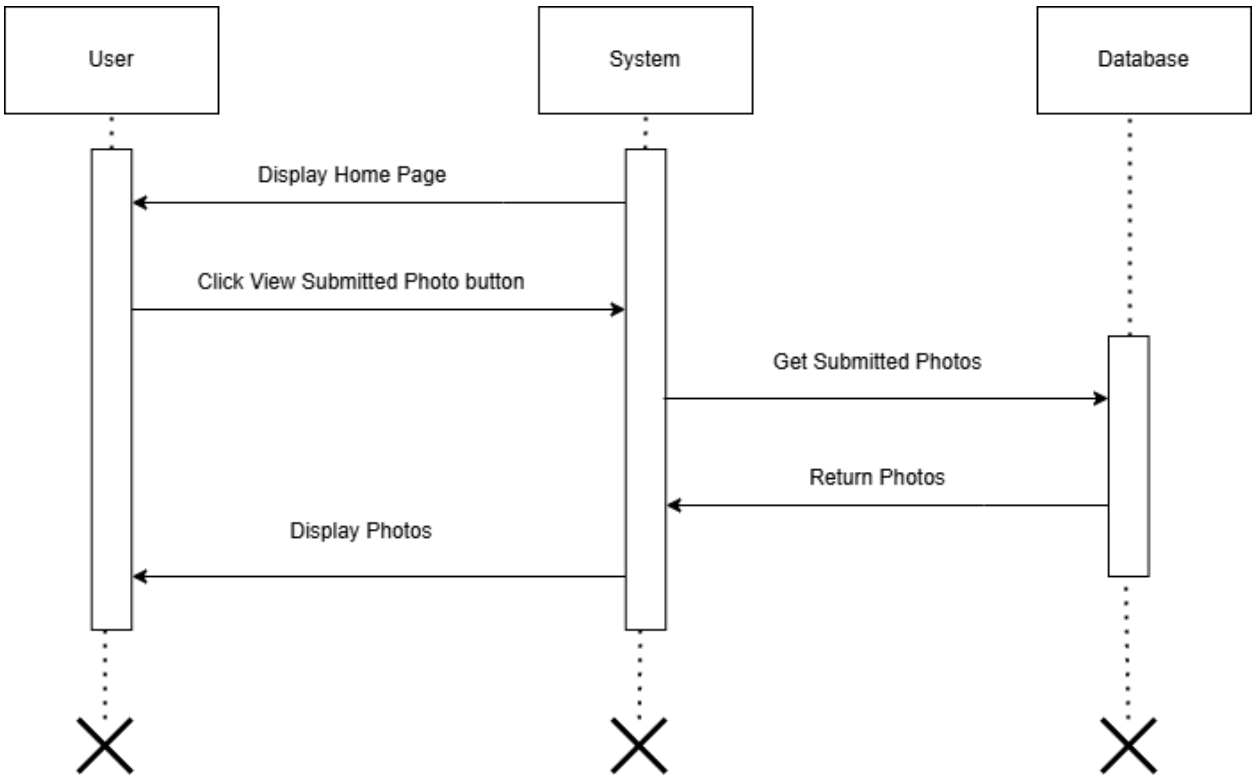


Figure 52: Sequence Diagram (Browse Photos)

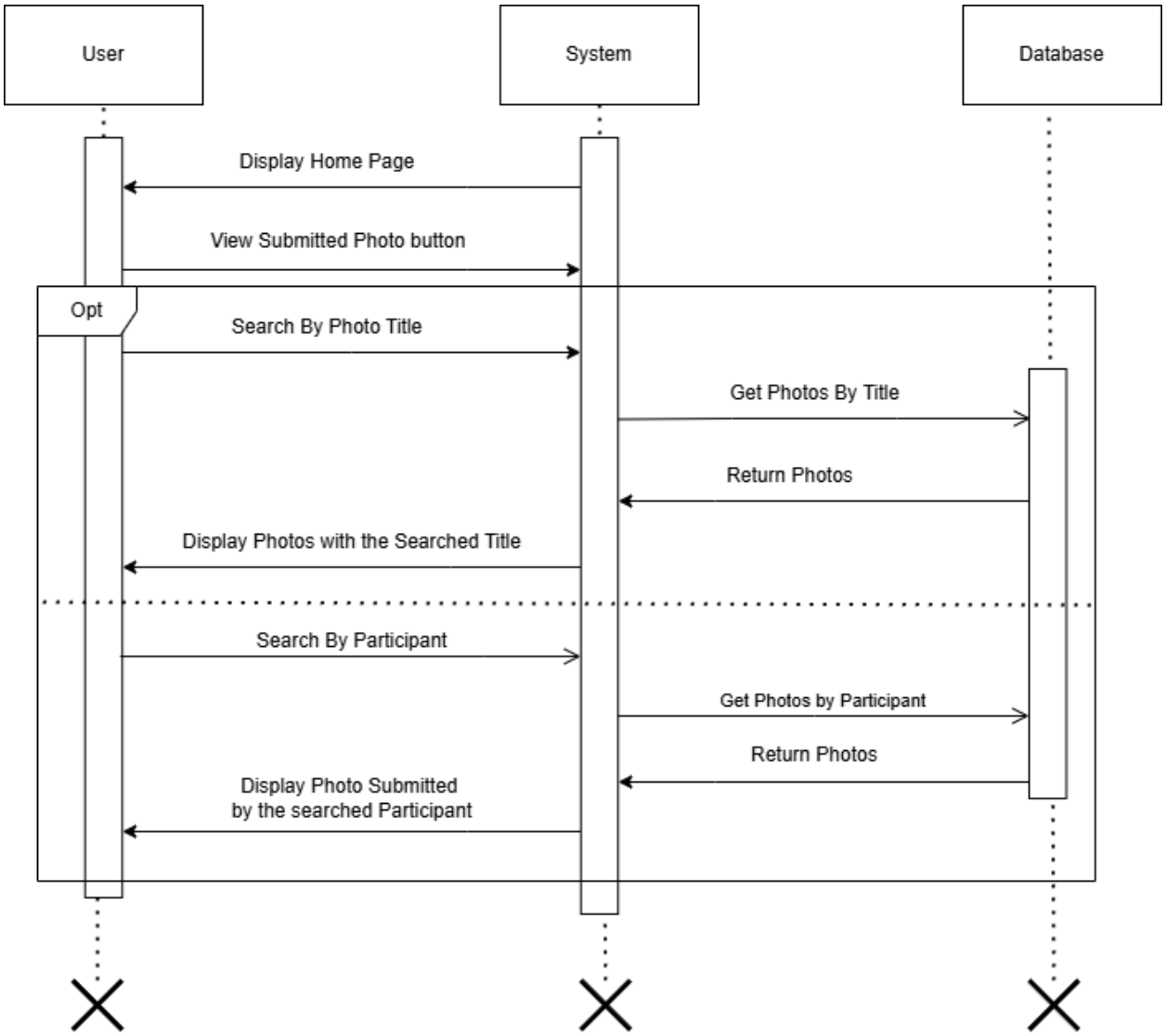


Figure 53: Sequence Diagram (Search Photo)

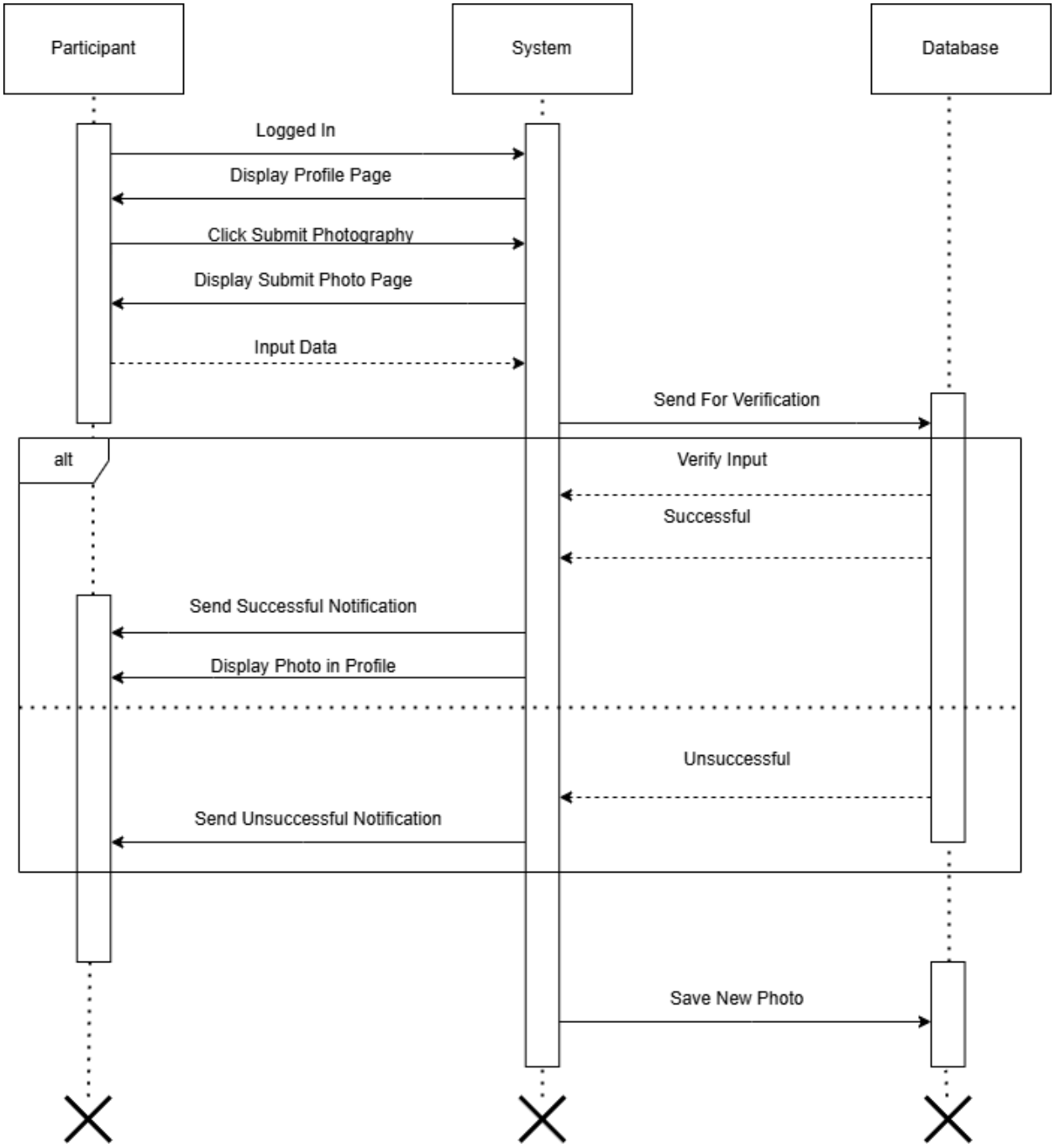


Figure 54: Sequence Diagram (Submit Photography)

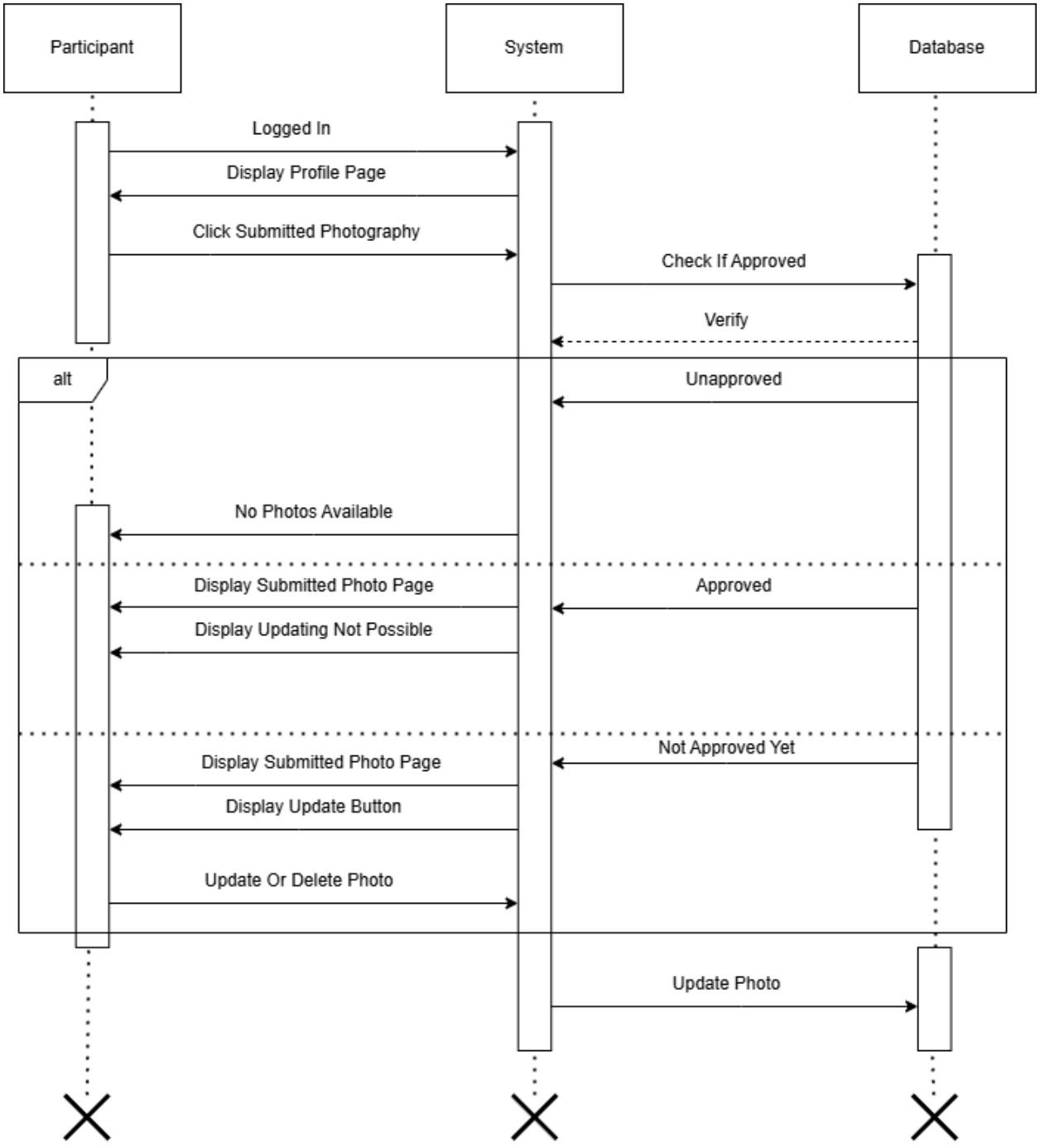


Figure 55: Sequence Diagram (Update and Delete Photo)

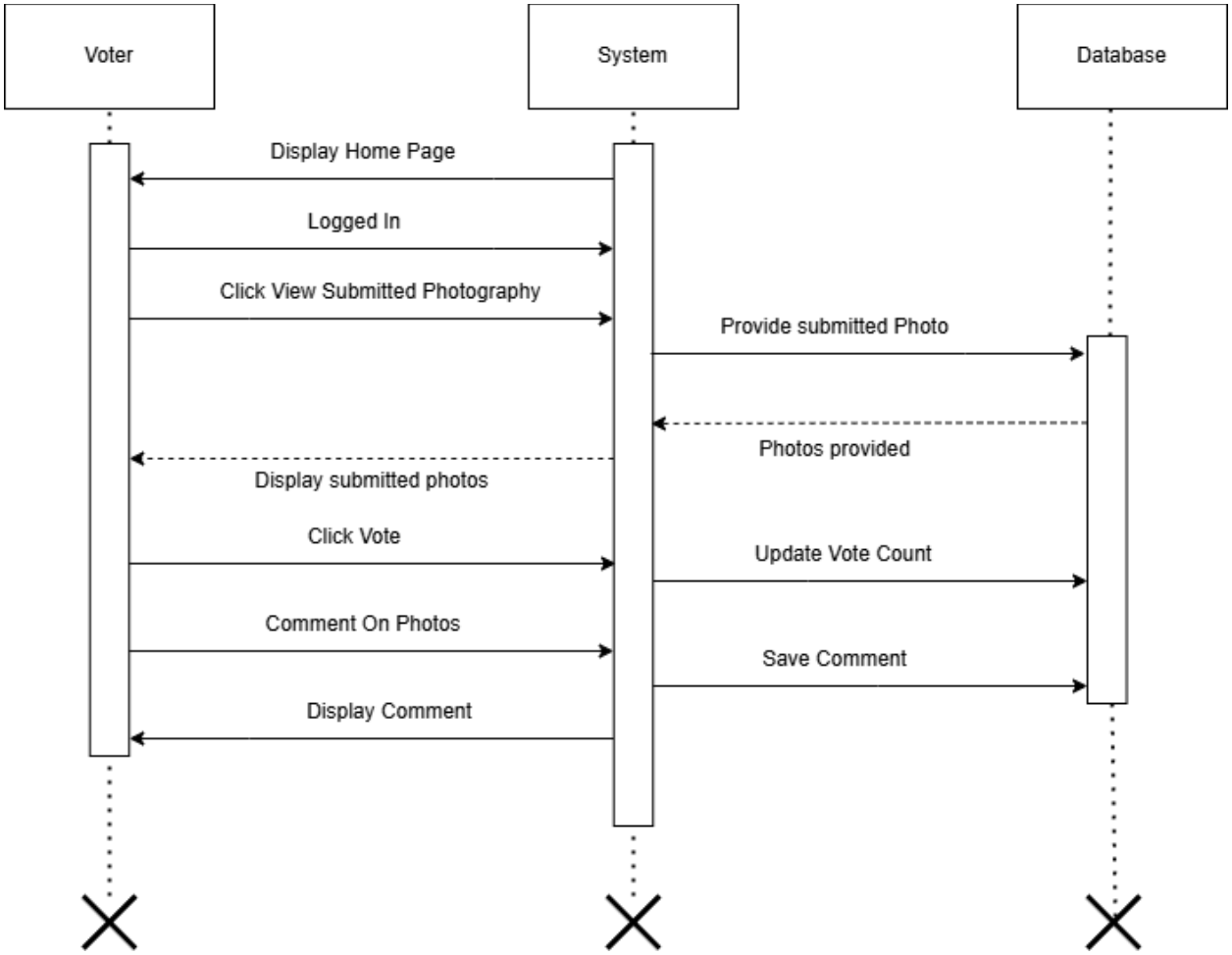


Figure 56: Sequence Diagram (Vote and Comment on photos)

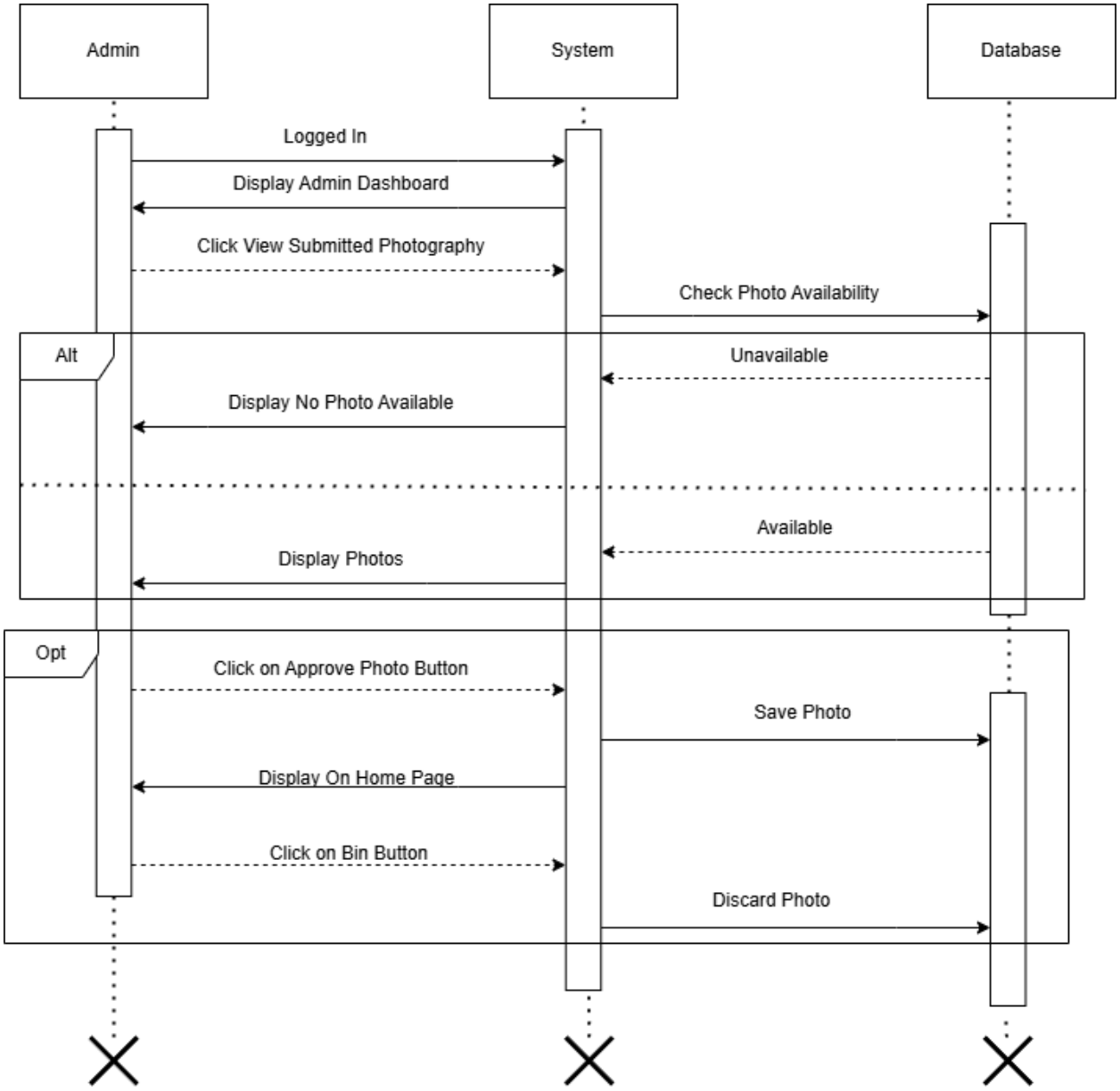


Figure 57: Sequence Diagram (Approve Photos)

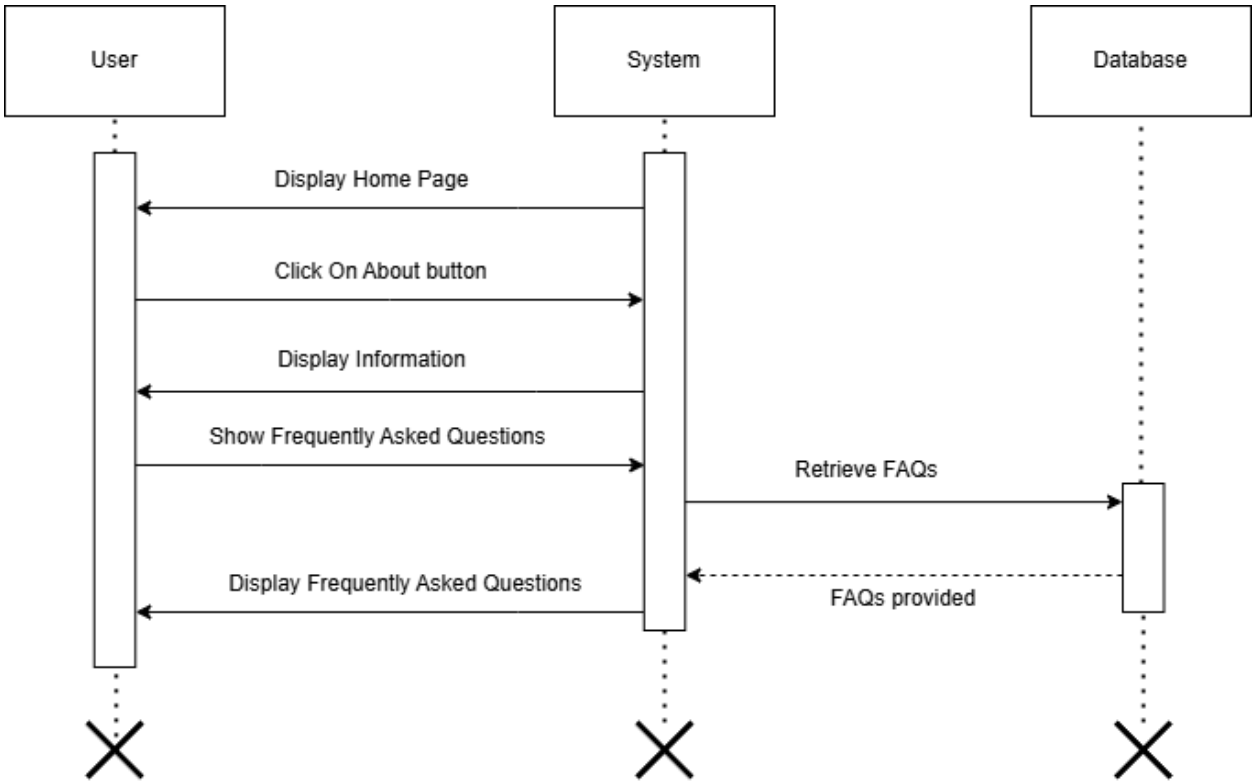


Figure 58: Sequence Diagram (View Information)

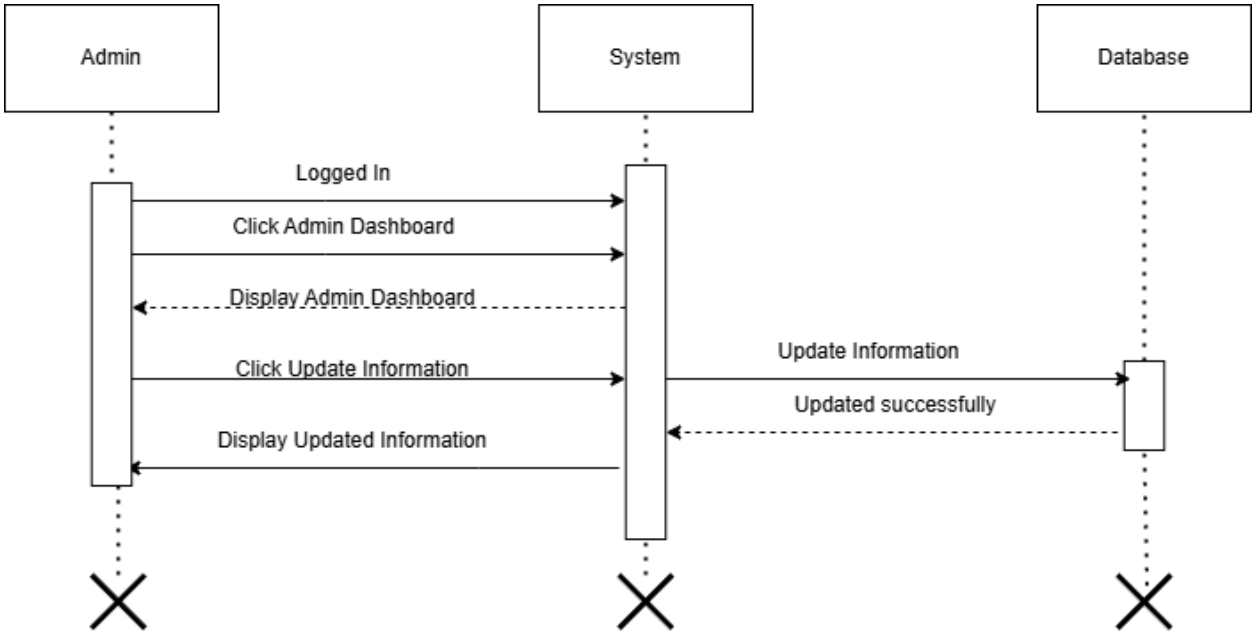


Figure 59: Sequence Diagram (Update Information)

8 State Diagram:

State diagram represents active states for each class of events (triggers).

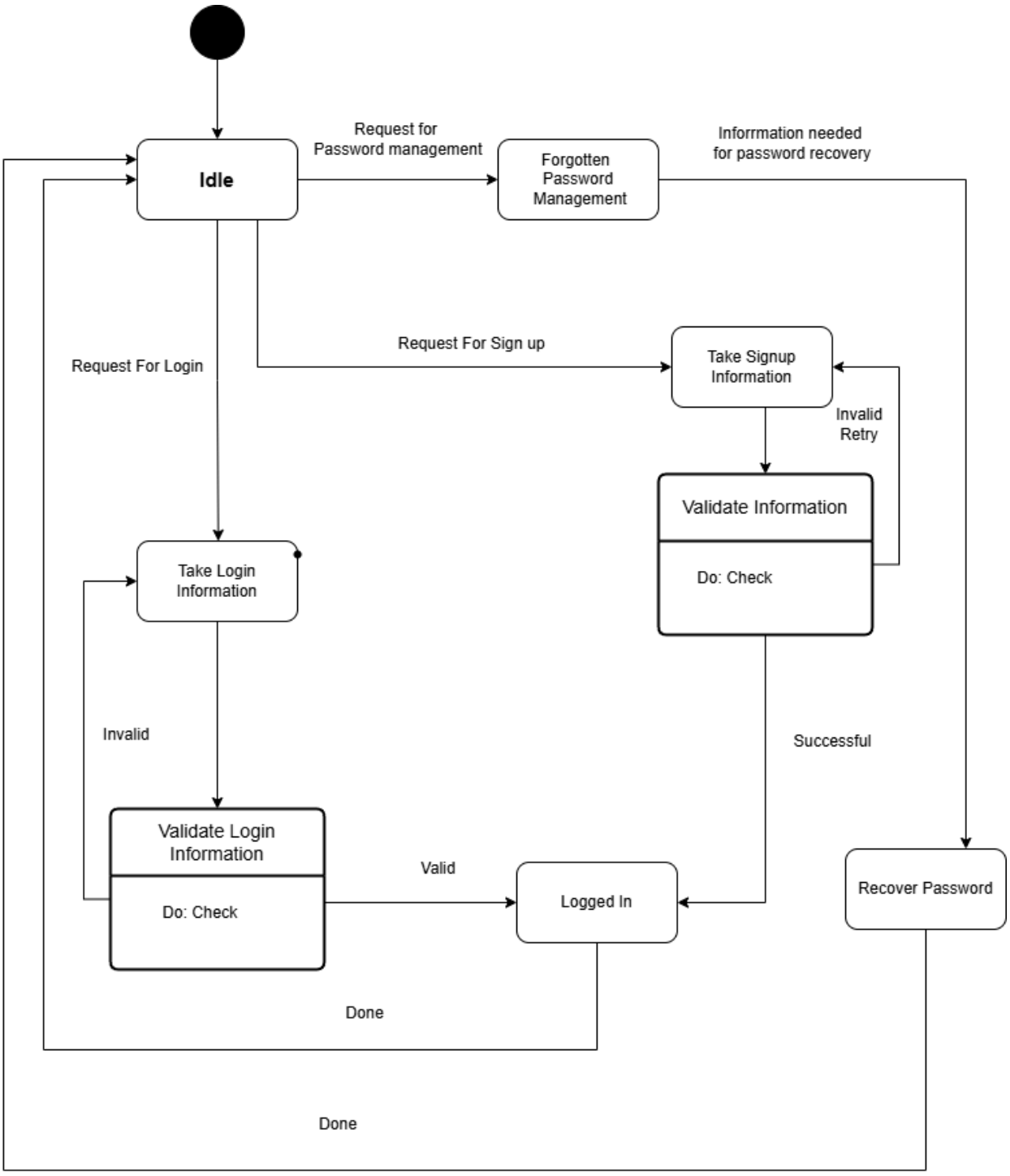


Figure 60: State Diagram (Account Class)

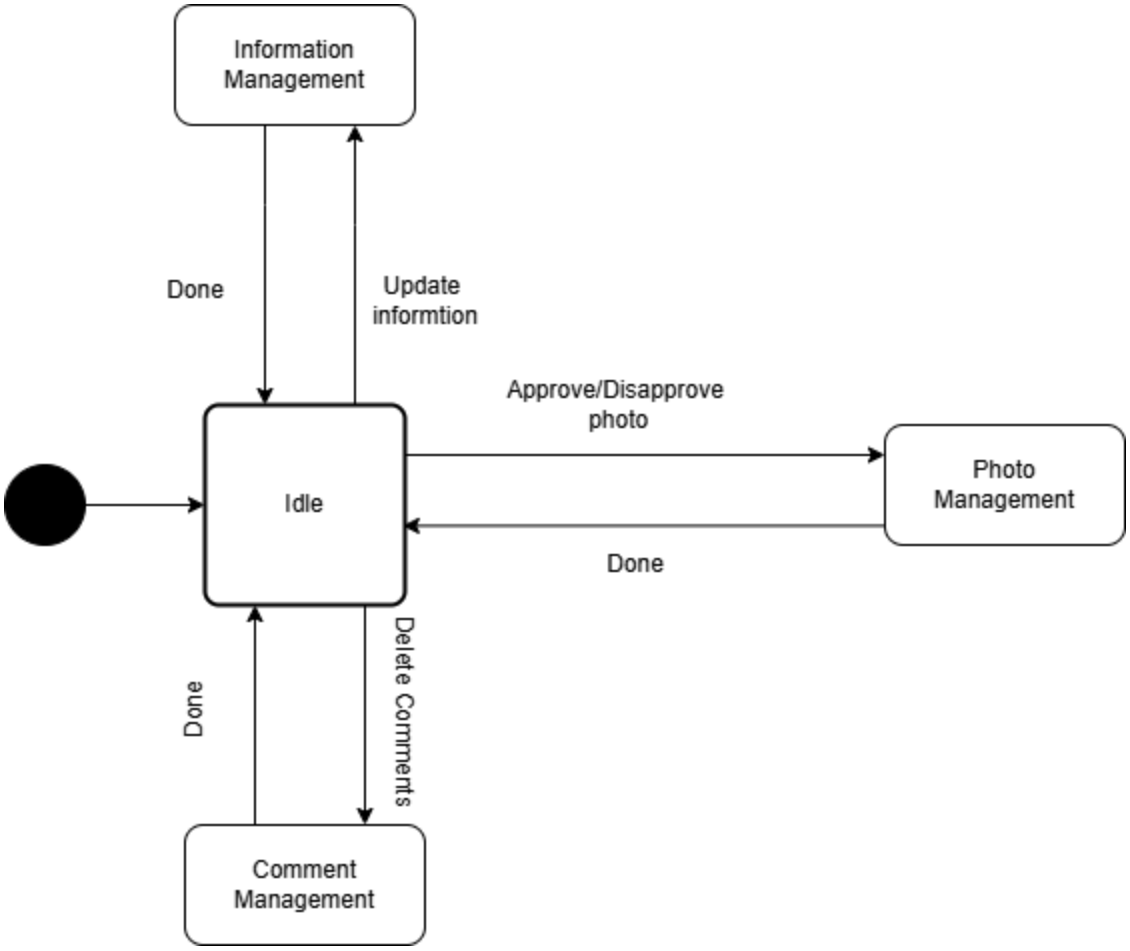


Figure 61: State Diagram (Admin Class)

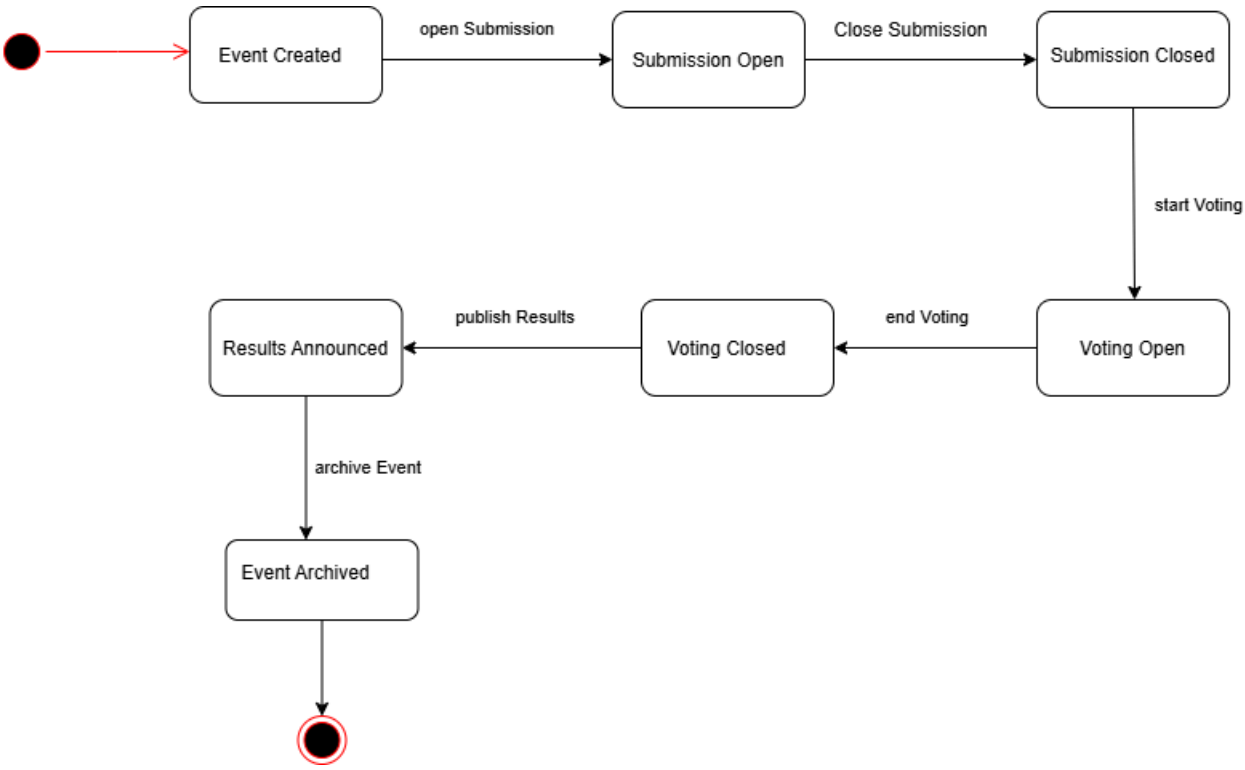


Figure 62: State Diagram (Event Class)

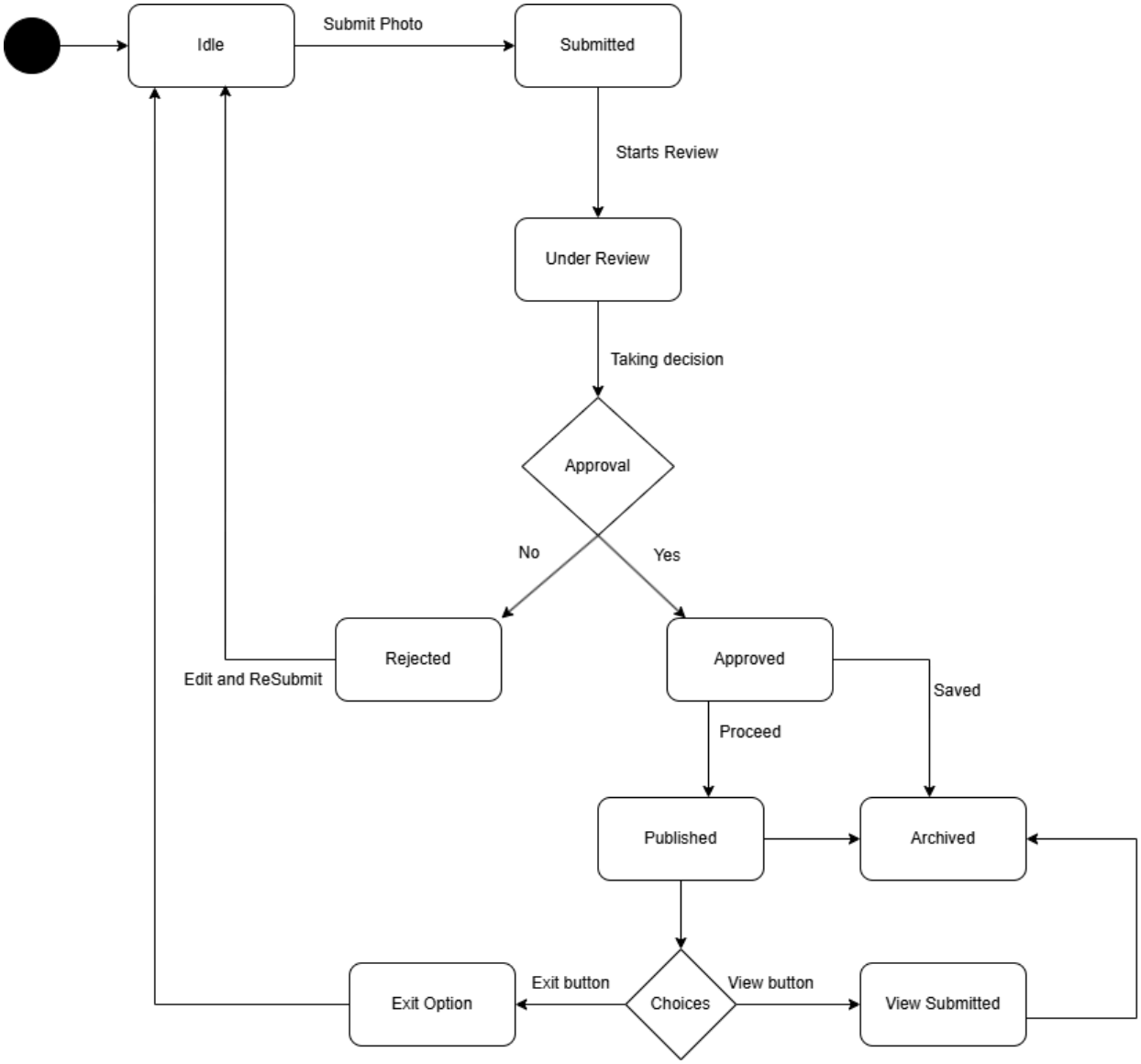


Figure 63: State Diagram (Photo Submission Class)

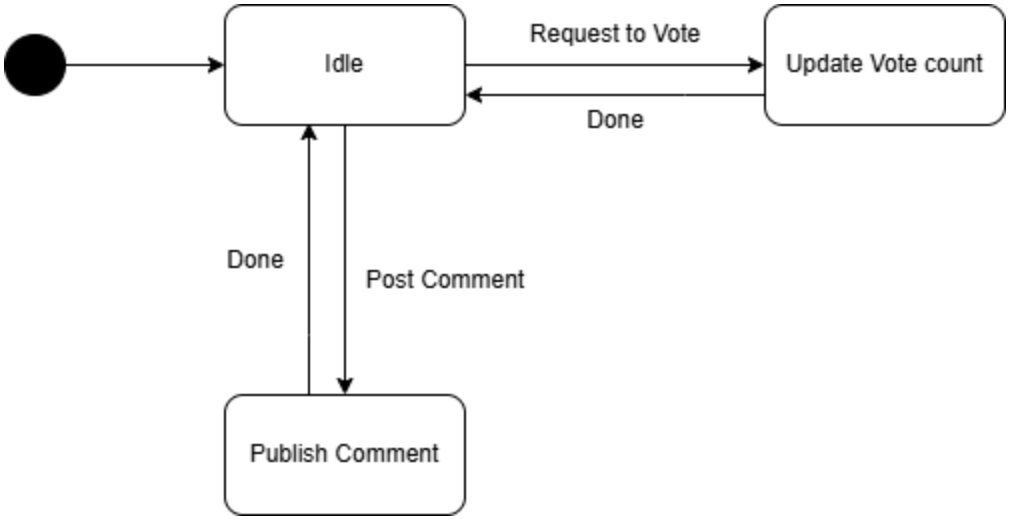


Figure 64: State Diagram (Vote Class)

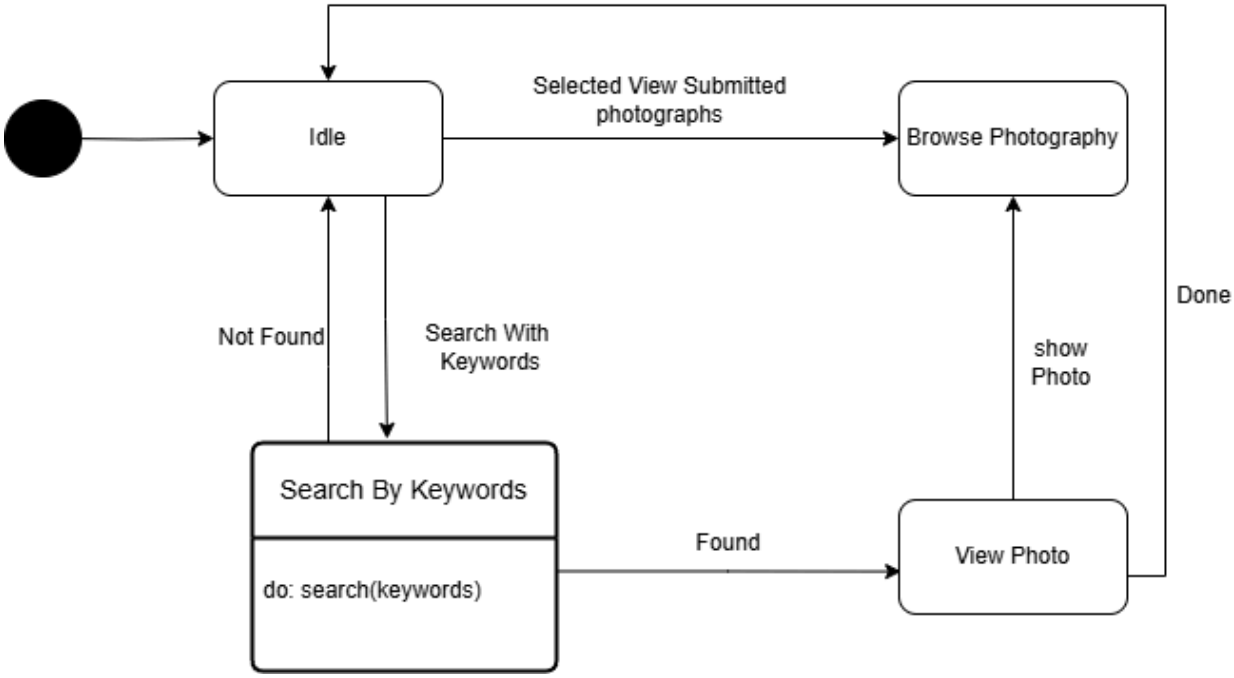


Figure 65: State Diagram (Search and Browse Class)

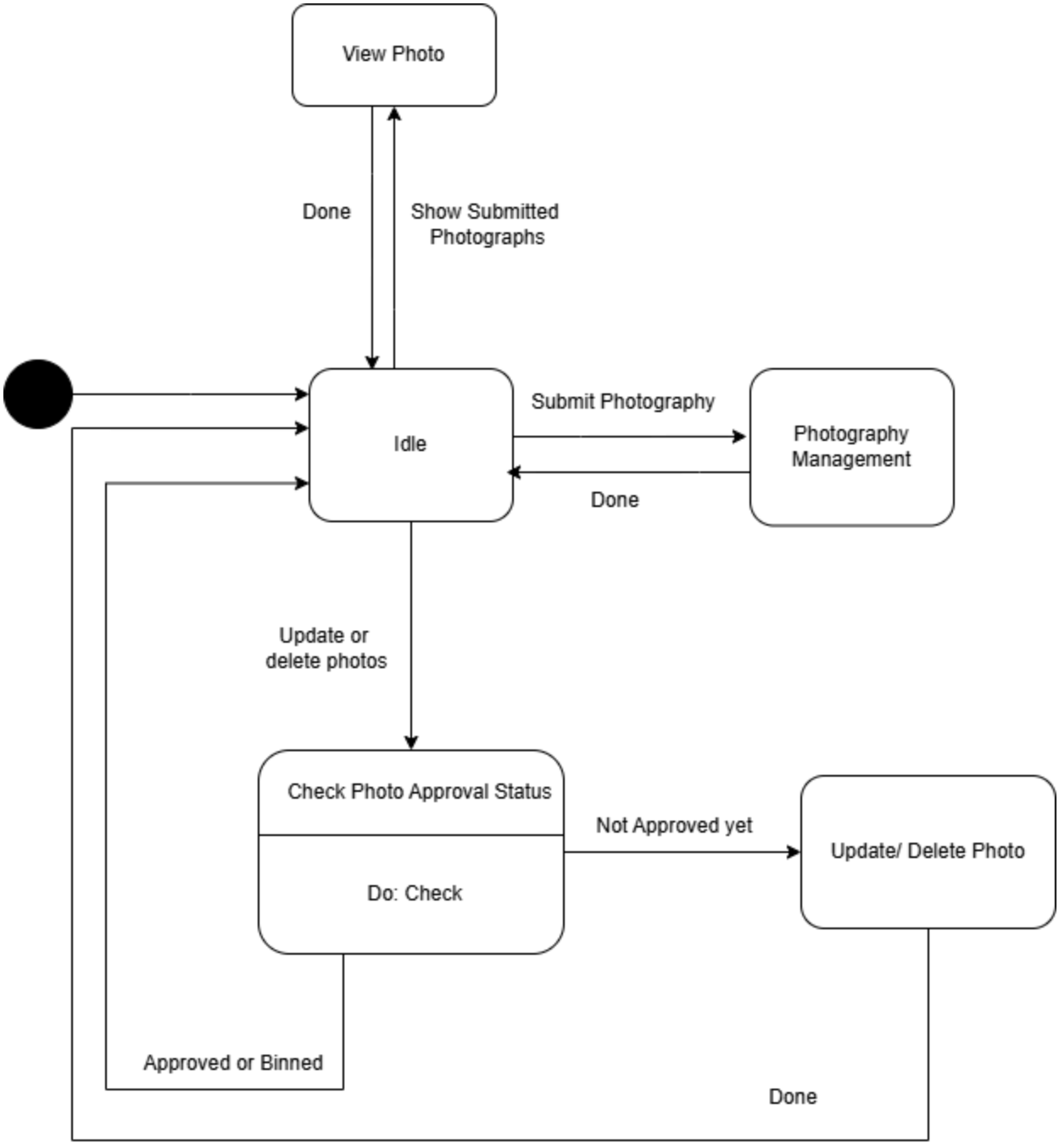


Figure 66: State Diagram (Participant Class)

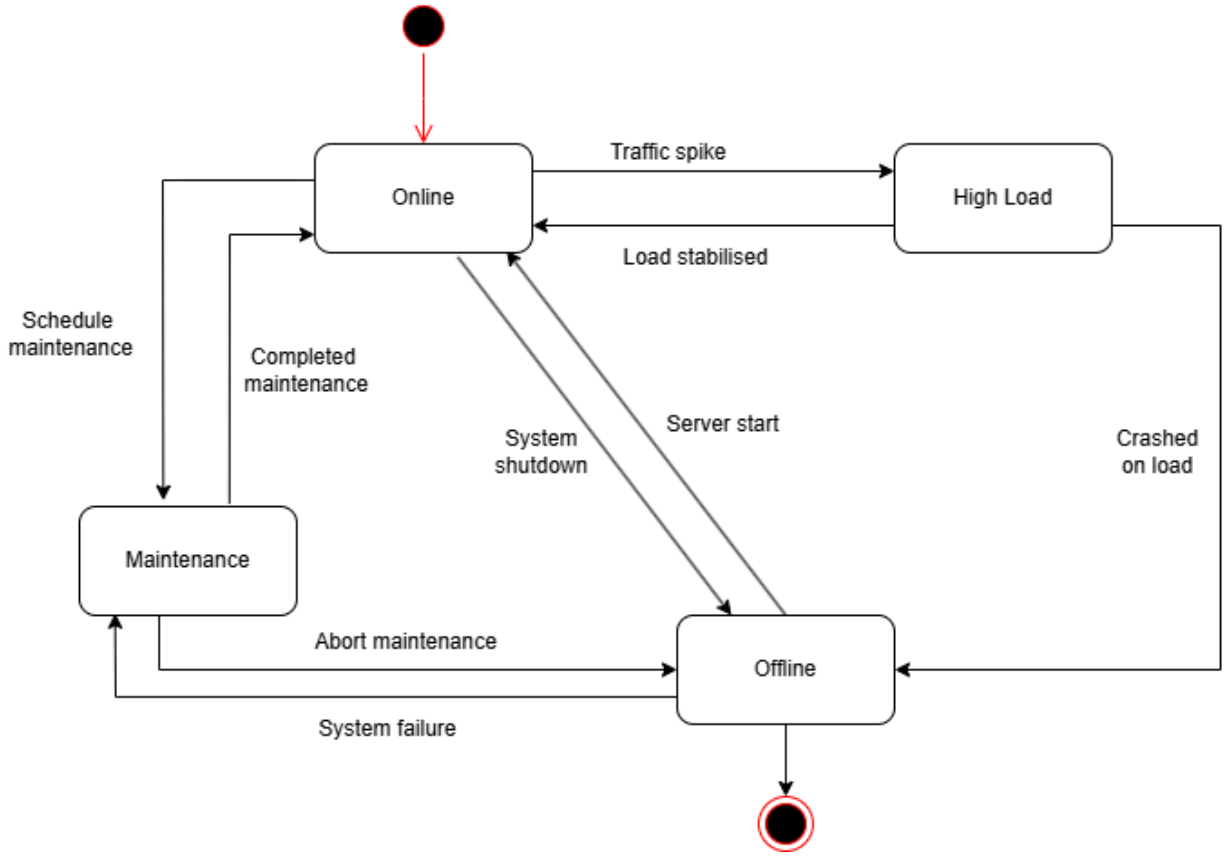


Figure 67: State Diagram (System Class)

9 Deployment Diagram

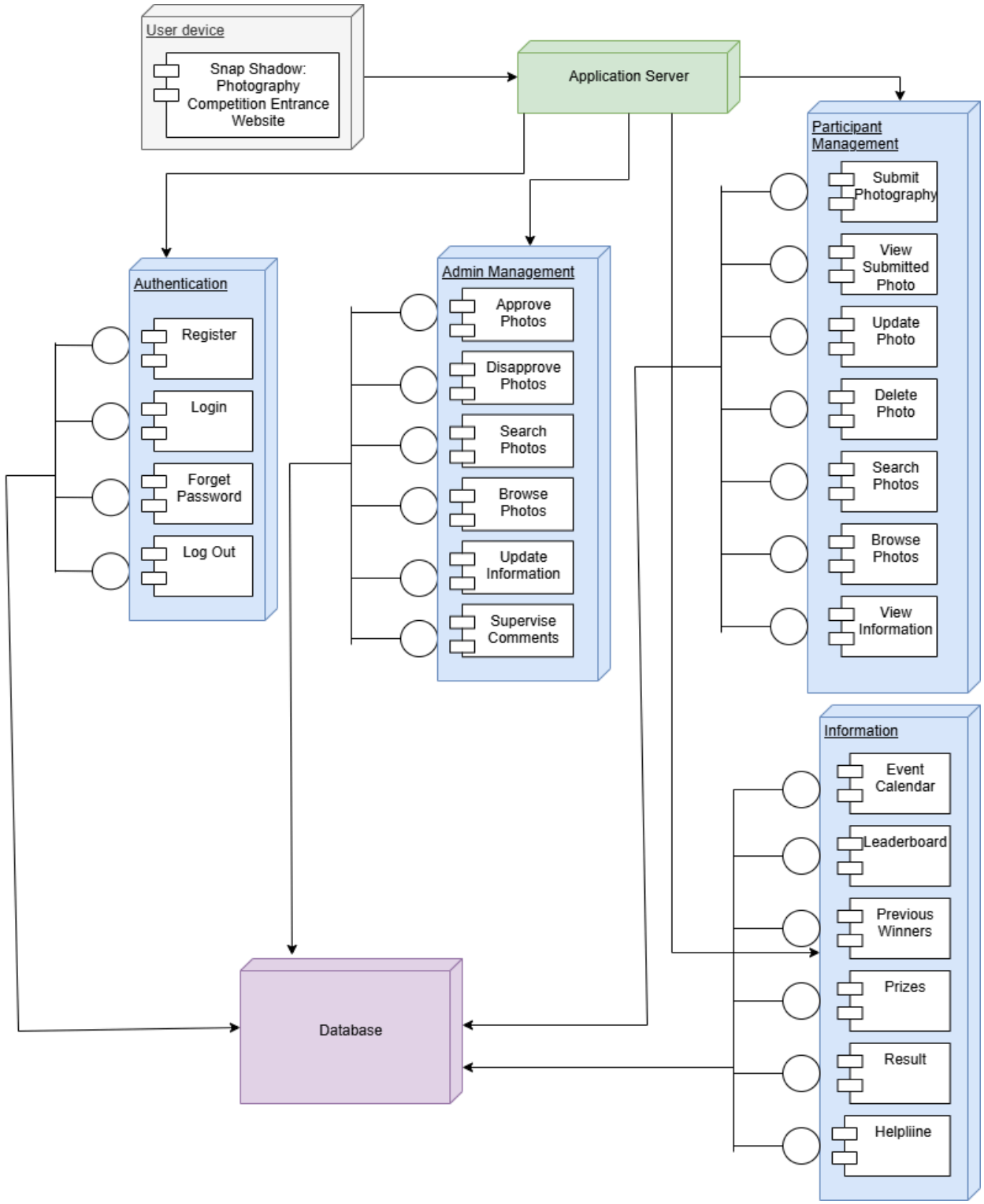


Figure 68: Deployment Diagram

9 Conclusion :

The Software Requirements Specification (SRS) for the "Snap Showdown: A Photography Competition Entrance Website" is a pivotal document that encapsulates the vision and objectives of the project. It serves as a detailed roadmap for stakeholders, providing clear insights into the functionalities, user needs, and technical requirements essential for the successful development of the platform.

Throughout this process, we have engaged in rigorous stakeholder identification and requirement elicitation, ensuring that diverse perspectives are considered. The application of methodologies has enabled us to prioritize features that not only meet basic expectations but also enhance user engagement and satisfaction.

Our collaborative approach, involving continuous feedback and iterative discussions, has refined the system's design and functionality. The SRS outlines critical use cases, system interactions, and design principles that will guide developers and designers in creating a seamless user experience.

In summary, this SRS lays the groundwork for a robust and user-centric photography competition website. It aims to facilitate an enriching experience for participants, administrators, and viewers alike.

Appendix

Meeting Schedule

1.

Date: 20.02.2025

Location: Online

Discussion: Preparing the steps for stakeholder identification

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089
Iqbal Hossain Safy	2022-1-60-038

2.

Date: 21.02.2025

Location: East West University

Discussion: First visit with the website owner

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089
Iqbal Hossain Safy	2022-1-60-038

3.

Date: 23.02.2025

Location: East West University

Discussion: Discussing visits with the website owner

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089
Iqbal Hossain Safy	2022-1-60-038

4.

Date: 28.02.2025

Location: Online

Discussion: Identifying intended audience

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089
Iqbal Hossain Safy	2022-1-60-038

5.

Date: 01.03.2025

Location: Online

Discussion: Identifying Viewpoints of Stakeholders

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089
Iqbal Hossain Safy	2022-1-60-038

6.

Date: 02.03.2025

Location: East West University

Discussion: Preparing usage scenario

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089
Iqbal Hossain Safy	2022-1-60-038

7.

Date: 03.03.2025

Location: Online

Discussion: Preparing usage scenario

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089

Iqbal Hossain Safy	2022-1-60-038

8..

Date: 04.03.2025

Location: East West University

Discussion: Preparing usage scenario

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089
Iqbal Hossain Safy	2022-1-60-038

9.

Date: 10.03.2025

Location: Online

Discussion: Preparing use case diagram

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089
Iqbal Hossain Safy	2022-1-60-038

10.

Date: 11.03.2025

Location: Online

Discussion: Preparing use case diagram

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089
Iqbal Hossain Safy	2022-1-60-038

11.

Date: 13.03.2025

Location: East West University

Discussion: Correcting use case diagram

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089
Iqbal Hossain Safy	2022-1-60-038

12.

Date: 15.03.2025

Location: Online

Discussion: Preparing activity diagram

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089
Iqbal Hossain Safy	2022-1-60-038

13.

Date: 17.03.2025

Location: Online

Discussion: Correcting activity diagram

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089
Iqbal Hossain Safy	2022-1-60-038

14.

Date: 19.03.2025

Location: Online

Discussion: Preparing swimlane diagram

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089

Iqbal Hossain Safy	2022-1-60-038

15.

Date: 10.04.2025

Location: Online

Discussion: Preparing the website homepage

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089
Iqbal Hossain Safy	2022-1-60-038

16.

Date: 15.04.2025

Location: Online

Discussion: Preparing the website homepage

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089
Iqbal Hossain Safy	2022-1-60-038

17.

Date: 20.04.2025

Location: Online

Discussion: Preparing UML diagram

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089
Iqbal Hossain Safy	2022-1-60-038

18.

Date: 22.04.2025

Location: Online

Discussion: Correcting UML diagram

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089
Iqbal Hossain Safy	2022-1-60-038

19.

Date: 25.04.2025

Location: Online

Discussion: Preparing the website

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089
Iqbal Hossain Safy	2022-1-60-038

20.

Date: 25.04.2025

Location: East West University

Discussion: Preparing data flow diagram

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089
Iqbal Hossain Safy	2022-1-60-038

21.

Date: 25.04.2025

Location: Online

Discussion: Correcting data flow diagram

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089
Iqbal Hossain Safy	2022-1-60-038

22.

Date: 01.05.2025

Location: Online

Discussion: Correcting UML and data flow diagram

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089
Iqbal Hossain Safy	2022-1-60-038

23.

Date: 05.05.2025

Location: Online

Discussion: Preparing the website

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089
Iqbal Hossain Safy	2022-1-60-038

24.

Date: 07.05.2025

Location: Online

Discussion: Preparing the sequence diagram

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089
Iqbal Hossain Safy	2022-1-60-038

25.

Date: 11.05.2025

Location: Online

Discussion: Correcting the sequence diagram

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089
Iqbal Hossain Safy	2022-1-60-038

26.

Date: 14.05.2025

Location: East West University

Discussion: Preparing State Machine Diagram

Members:

Fairuj Saima	2022-1-60-049

Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089
Iqbal Hossain Safy	2022-1-60-038

27.

Date: 16.05.2025

Location: Online

Discussion: Correcting State Machine Diagram

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089
Iqbal Hossain Safy	2022-1-60-038

28.

Date: 17.05.2025

Location: Online

Discussion: Preparing Deployment Diagram

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089

Iqbal Hossain Safy	2022-1-60-038

29.

Date: 18.05.2025-20.05.2025

Location: Online

Discussion: Preparing the website

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089
Iqbal Hossain Safy	2022-1-60-038

30.

Date: 21.05.2025

Location: Online

Discussion: Finalizing SRS file

Members:

Fairuj Saima	2022-1-60-049
Nafis Mahamud Sarker	2022-1-60-075
Sharafat Ahmed	2022-1-60-089
Iqbal Hossain Safy	2022-1-60-038